MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: Mansourah Massarah Gold Project
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION
49WF POTABLE WATER TREATMENT PLANT
FW PUMP HOUSE & ELECTRICAL BLDG DESIGN

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
FW	:	Fire Water
BLDG	:	Building

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1. INTENT

This document is to Prepare for design of **Fire Water Pump House & Electrical Building**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the structure.

2. EXECUTIVE SUMMARY

No index entries found.

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis & design of beam & column members have been carried out in STAAD. For foundation design, force values are directly taken from STAAD output & designed in excel sheet.

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4. DESIGN REQUIREMENTS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

4.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundation Design

Foundations shall be isolated type. The design calculations as determined by the standard in Civil, Structural & Architectural design criteria (Doc no. MD-216-1000-EG-ST-CRT-1001).

4.4 Grouting

Grout shall be 30 mm thick cementitious under steel column bases of strength 40 Mpa.

4.5 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation Top & Column	50
Exposed to weather or contact with ground	Beam	40

4.6 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

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4.7 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 750 kN/m^2 at 1.5m depth below EGL (ie. RL 917.00m) refer Table 10 - BH-21 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.8 Stability Against Overturning & Sliding

Overturning

Structures and building foundations, shall have the following minimum stability ratios against overturning:

- 1.5 for load combinations with wind loads
- 2 for other load combinations

Sliding

A minimum factor of safety against sliding shall be 1.5.

5. DESIGN LOADS

5.1 Dead Load

5.1.1 Roof Dead Load

- Provision for elecrical lighting ceiling = 0.05 kN/m^2
- Suspended steel channel system = 0.10 kN/m^2
- Acoustical Fibre Tile (16mm) = 0.05 kN/m^2

  Sub total = 0.20 kN/m^2
- Hollow Core slab (PCI 6th edition)

  (HCS-200 Topped) = 3.1 kN/m²

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Prestressed Hollowcore Floor

OVERALL STRUCTURAL DEPTH MM	SELF WT. KN/M²	NO. OF CORES IN UNIT	SPANS INDICATED BELOW ALLOW FOR CHARACTERISTIC SERVICE LOAD (LIVE LOAD) PLUS SELF WEIGHT PLUS 1.5KN/M² FOR FINISHES CHARACTERISTIC SERVICE LOADS KN/M²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.0
			Clear span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense Slab)	2.94	11	7.40	7.40	7.40	7.35	7.06	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

  100mm thk Polystyrene Foam
  0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²
- Water proofing membrane

  Bituminous, smooth surface = 0.07 kN/m²

- 90 mm thick average cement foam concrete screed Lightweight

  Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

- 75 mm Concrete screed above HCS = 21 x 0.075
  =1.6 kN/m²

Sub-total = 3.1 kN/m²
Total Load = 6.4 kN/m²

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UDL on edge ledger beams (RB1) = 6.4 x 6.5/2 = 20.8 kN/m

Moment on all the extrenal beams due to cantilever portion of slab & Screed
= (0.15x24+(0.05+0.2)/2x20)*0.55*0.55/2 = 0.93 knm/m

5.1.2 Ground Floor Wall Load

- 250 thk. Outer CMU (Partially Grouted Cell @ 800 c-c)

  16 x 0.25 x (4.2-0.5)m clear height = 14.8 kN/m
  16 x 0.25 x (6.7-0.5)m clear height = 24.8 kN/m

5.2 Live Load

5.2.1 Roof Live Load

- Live loads were taken from Design Criteria clause no 4.2.2.

  Roof not accessed for maintenance = 0.5 kN/m²
  Udl Due to Live Load on the Edge Ledger Beam (RB1) = 0.5 x 6.25/2 kN/m²kN/m
  = 1.56 kN/m²kN/m

  Moment on all the External Beams due to Live Load on

- Cantilever Portion of slab = 0.5 x 0.55 x 0.55/2 = 0.075 kN/m /m

5.2.2 Monorail Live Load

Load due to 1.5 Ton capacity monorail with hoist = 15 kN

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6 WIND LOAD CALCULATION

Velocity Pressure , qz evaluated at height z shall be calculated by the following equation:

$qz = 0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ kN/m^2 (Refer Cl. No 7.2.8 of SBC-301)

Basic Wind Speed	V =	177	km/h
Velocity pressure exposure coefficient	Kz =	1.04	As per Table 7.2-2(Page-7/44)- z = 0 to 5m of SBC-301
Exposure Categories.	=	D	As per Project Basis of Design
Topographic Factor	K_{zt}=	1.00	As per Cl. 6.4.3.2

(As per Pt no 4 of cl 6.4.3.1 of SBC-301 , $H/L_h < 0.2$, hence topographic effect not considered)

Wind Directionality Factor	K_d =	0.85	As per Table 6.4-1(Page-6/7)	
Importance factor	I =	1.00	As per Table 6.5-1(Page-6/9)	
Hence, for z = 10m	qz =	1.31	Kn/m2 =	0.131 Ton/m2

Design wind pressure on structure is ; p = qh (GCpf - GCpi) (Refer Eq. 7.2-15 of SBC-301)

GCpf = External pressure co-efficient (Refer Figure No. 7.2 - 6 of SBC-301)
GCpi = Internal pressure co-efficient (Refer Figure No. 7.2 - 1 of SBC-301)

GCpi = ±0.18

WIND ALONG X DIRECTION

GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)
Higher values are adopted for conservative design

Wind load for GCpi = + 0.18 =	1.31 x (0.61-0.18) =	0.56	Kn/m2	(Windward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.43-0.18) =	-0.80	Kn/m2	(Leeward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.45-0.18) =	-0.83	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 3 m =	3.0 x 0.56 =	1.68	Kn/m
Load on columns for span 4.0m =	4.0 x 0.56 =	2.24	Kn/m
Load on columns for span 4.5m =	4.5 x 0.56 =	2.52	Kn/m

Load on Leeward columns :

Load on columns for span 3 m =	3.0 x (-0.80) =	-2.40	Kn/m
Load on columns for span 4.0m =	4.0 x (-0.80) =	-3.20	Kn/m
Load on columns for span 4.5m =	4.5 x (-0.80) =	-3.60	Kn/m

Load on side wall columns :

Load on columns for span 3.25m =	3.25 x (-0.83) =	-2.70	Kn/m

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GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)
Higher values are adopted for conservative design

Wind load for GCpi = - 0.18 =	1.31 x (0.61+0.18) =	1.03	Kn/m2	(Windward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.43+0.18) =	-0.33	Kn/m2	(Leeward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.45+0.18) =	-0.35	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 3m =	3.0 x 1.03 =	3.09	Kn/m
Load on columns for span 4.0m =	4.0 x 1.03 =	4.12	Kn/m
Load on columns for span 4.5m =	4.5 x 1.03 =	4.64	Kn/m

Load on Leeward columns :

Load on columns for span 3.0 m =	3.0 x (-0.33) =	-0.99	Kn/m
Load on columns for span 4.0m =	4.0 x (-0.33) =	-1.32	Kn/m
Load on columns for span 4.9m =	4.5 x (-0.33) =	-1.49	Kn/m

Load on side wall columns :

Load on columns for span 3.25m =	3,25 x (-0.35) =	-1.14	Kn/m

WIND ALONG Z DIRECTION

GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)
Higher values are adopted for conservative design

Wind load for GCpi = + 0.18 =	1.31 x (0.61-0.18) =	0.56	Kn/m2	(Windward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.43-0.18) =	-0.80	Kn/m2	(Leeward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.45-0.18) =	-0.83	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 3.25m =	3.25x 0.56 =	1.82	Kn/m

Load on Leeward columns :

Load on columns for span 3.25m =	3.25 x (-0.80) =	-2.60	Kn/m

Load on side wall columns :

Load on columns for span 3.0 m =	3.0 x (-0.83) =	-2.49	Kn/m
Load on columns for span 4.0m =	4.0 x (-0.83) =	-3.32	Kn/m
Load on columns for span 4.5m =	4.5 x (-0.83) =	-3.74	Kn/m

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GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)

Higher values are adopted for conservative design

Wind load for GCpi = - 0.18 =	1.31 x (0.61+0.18) =	1.03	Kn/m2	(Windward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.43+0.18) =	-0.33	Kn/m2	(Leeward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.45+0.18) =	-0.35	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 3.25m =	3.25 x 1.03 =	3.35	Kn/m

Load on Leeward columns :

Load on columns for span 3.25m =	3.25 x (-0.33) =	-1.07	Kn/m

Load on side wall columns :

Load on columns for span 3 m =	3 x (-0.35) =	-1.05	Kn/m
Load on columns for span 4.0m =	4.0 x (-0.35) =	-1.40	Kn/m
Load on columns for span 4.5m =	4.5 x (-0.35) =	-1.58	Kn/m

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN OF FOUNDATION MKD. F1

(STAAD Node -108)

	Along-X	Along-Z
Size of column :	0.50m	0.50m
Size of fdn. raft :	2.20m	2.20m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Fluid Load	LL
Roof Live Load	LR
Wind Load	WLX1
Wind Load	WLX2
Wind Load	WLZ1
Wind Load	WLZ2
Temp Load	TL

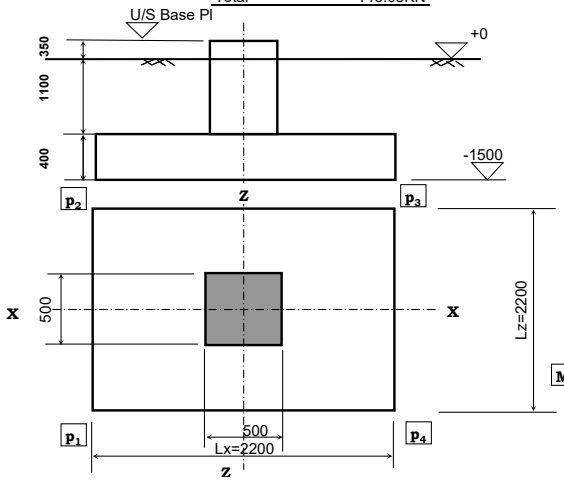
Design Input:

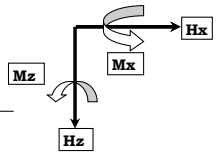
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%

Calculation of Weight of Foundation :

Weight of raft & Col.= (2.2*2.2*0.4*24)+(0.5*0.5*1.45*24) = 55.16KN
Weight of soil = ((2.2*2.2)-(0.5*0.5))*1.1*18 = 90.88KN
Total = 146.05KN

Effective depth of slab along X direction = 344mm = 0.344m
Effective depth of slab along Z direction = 332mm = 0.332m
Clear projection along Z direction = 0.85m
Clear projection along X direction = 0.85m





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F1

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0LL+1.0TL	276.0	44.8	0.0	-10.0	0.0	422.02	44.84	82.95	-10.04	-18.57
202	1.0DL+1.0LR	269.2	44.4	0.0	-13.5	0.0	415.21	44.41	82.16	-13.53	-25.03
203	1.0DL+0.75LL+0.75LR+0.75TL	275.9	44.7	0.0	-10.9	0.0	421.99	44.73	82.75	-10.85	-20.07
204	1.0DL+1.0WLX1	249.8	23.1	0.0	-11.9	0.0	395.86	23.10	42.74	-11.92	-22.05
205	1.0DL+1.0WLX2	283.3	68.3	0.0	-14.2	0.0	429.38	68.33	126.40	-14.20	-26.27
206	1.0DL+1.0WLZ1	268.4	36.3	0.0	-13.1	0.0	414.44	36.26	67.08	-13.14	-24.30
207	1.0DL+1.0WLZ2	262.2	41.0	0.0	-14.6	0.0	408.23	40.97	75.79	-14.62	-27.04
208	1.0DL+0.75LL+0.75LR+0.75WLX1	264.5	26.6	0.0	-12.0	0.0	410.58	26.60	49.21	-12.03	-22.25
209	1.0DL+0.75LL+0.75LR+0.75WLX2	289.7	60.5	0.0	-13.7	0.0	435.72	60.52	111.95	-13.74	-25.41
210	1.0DL+0.75LL+0.75LR+0.75WLZ1	278.5	36.5	0.0	-12.9	0.0	424.51	36.47	67.46	-12.94	-23.93
211	1.0DL+0.75LL+0.75LR+0.75WLZ2	273.8	40.0	0.0	-14.0	0.0	419.85	40.00	73.99	-14.05	-25.99
212	0.6DL+1.0WLX1	143.5	5.3	0.0	-6.5	0.0	289.54	5.34	9.87	-6.46	-11.95
213	0.6DL+1.0WLX2	177.0	50.6	0.0	-8.7	0.0	323.06	50.56	93.54	-8.74	-16.17
214	0.6DL+1.0WLZ1	162.1	18.5	0.0	-7.7	0.0	308.12	18.50	34.22	-7.67	-14.20
215	0.6DL+1.0WLZ2	155.9	23.2	0.0	-9.2	0.0	301.91	23.20	42.93	-9.16	-16.94
Ultimate Conditions											
101	1.4DL	372.1	62.2	0.0	-19.1	0.0	518.18	62.17	115.01	-19.11	-35.36
102	1.4DL+1.7LL+0.5LR	392.4	58.0	0.0	-18.5	0.0	538.45	58.02	107.34	-18.52	-34.26
103	1.2DL+1.6LR+0.8WLX1	311.5	36.2	0.0	-14.8	0.0	457.58	36.25	67.06	-14.80	-27.38
104	1.2DL+1.6LR+0.8WLX2	338.4	72.4	0.0	-16.6	0.0	484.40	72.43	133.99	-16.62	-30.75
105	1.2DL+1.6LR+0.8WLZ1	326.4	46.8	0.0	-15.8	0.0	472.45	46.78	86.53	-15.77	-29.18
106	1.2DL+1.6LR+0.8WLZ2	321.4	50.5	0.0	-17.0	0.0	467.47	50.54	93.50	-16.96	-31.37
107	1.2DL+1.6LR+1.0LL	335.3	50.9	0.0	-15.9	0.0	481.31	50.85	94.08	-15.87	-29.36
108	1.2DL+1.0LL+1.6WLX1+0.5LR	306.0	16.8	0.0	-13.2	0.0	452.04	16.76	31.00	-13.24	-24.48
109	1.2DL+1.0LL+1.6WLX1+0.5LR	359.6	89.1	0.0	-16.9	0.0	505.68	89.12	164.86	-16.89	-31.24
110	1.2DL+1.0LL+1.6WLX1+0.5LR	335.7	37.8	0.0	-15.2	0.0	481.77	37.81	69.95	-15.18	-28.08
111	1.2DL+1.0LL+1.6WLX1+0.5LR	325.8	45.3	0.0	-17.5	0.0	471.83	45.34	83.88	-17.55	-32.47
112	1.2DL+1.0LL	329.9	50.8	0.0	-16.1	0.0	475.95	50.85	94.07	-16.07	-29.73
113	0.9DL+1.6WLX1	213.6	5.9	0.0	-9.5	0.0	359.68	5.88	10.87	-9.52	-17.60
114	0.9DL+1.6WLX2	267.3	78.2	0.0	-13.2	0.0	413.32	78.23	144.73	-13.16	-24.35
115	0.9DL+1.6WLZ1	243.4	26.9	0.0	-11.5	0.0	389.42	26.93	49.82	-11.46	-21.20
116	0.9DL+1.6WLZ2	233.4	34.5	0.0	-13.8	0.0	379.47	34.46	63.75	-13.83	-25.58
117	1.4DL+ 1.4TL+1.7LL+0.5LR	391.3	62.0	0.0	-13.9	0.0	537.36	62.04	114.77	-13.90	-25.72

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	1.0DL+1.0LL+1.0TL	422.0	44.8	82.9	-10.0	-18.6	5.6	25.0	3.8	16.8
202	1.0DL+1.0LR	415.2	44.4	82.2	-13.5	-25.0	5.6	18.2	3.7	12.3
203	1.0DL+0.75LL+0.75LR+0.75TL	422.0	44.7	82.8	-10.9	-20.1	5.6	23.1	3.8	15.6
204	1.0DL+1.0WLX1	395.9	23.1	42.7	-11.9	-22.1	10.2	19.7	6.9	13.3
205	1.0DL+1.0WLX2	429.4	68.3	126.4	-14.2	-26.3	3.7	18.0	2.5	12.1
206	1.0DL+1.0WLZ1	414.4	36.3	67.1	-13.1	-24.3	6.8	18.8	4.6	12.6
207	1.0DL+1.0WLZ2	408.2	41.0	75.8	-14.6	-27.0	5.9	16.6	4.0	11.2
208	1.0DL+0.75LL+0.75LR+0.75WLX1	410.6	26.6	49.2	-12.0	-22.2	9.2	20.3	6.2	13.7
209	1.0DL+0.75LL+0.75LR+0.75WLX2	435.7	60.5	112.0	-13.7	-25.4	4.3	18.9	2.9	12.7
210	1.0DL+0.75LL+0.75LR+0.75WLZ1	424.5	36.5	67.5	-12.9	-23.9	6.9	19.5	4.7	13.1
211	1.0DL+0.75LL+0.75LR+0.75WLZ2	419.9	40.0	74.0	-14.0	-26.0	6.2	17.8	4.2	12.0
212	0.6DL+1.0WLX1	289.5	5.3	9.9	-6.5	-11.9	32.3	26.7	21.7	17.9
213	0.6DL+1.0WLX2	323.1	50.6	93.5	-8.7	-16.2	3.8	22.0	2.6	14.8
214	0.6DL+1.0WLZ1	308.1	18.5	34.2	-7.7	-14.2	9.9	23.9	6.7	16.1
215	0.6DL+1.0WLZ2	301.9	23.2	42.9	-9.2	-16.9	7.7	19.6	5.2	13.2
						Min.=	3.7	16.6	2.5	11.2
							OK	OK	OK	OK

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DESIGN OF FOUNDATION MKD. F1

Load Combination & Calculation of Base pressure

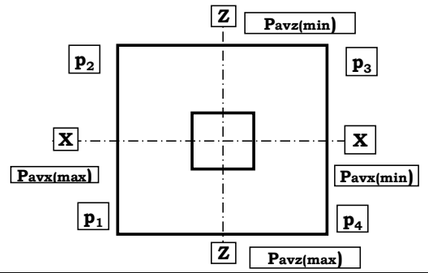
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 500 kN/m^2 For WL/SL case = 625 kN/m^2

Gross SBC of soil for DL, LL= 527 kN/m^2 For WL/SL case = 652 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-gh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	1.0DL+1.0LL+1.0TL	422.02	82.95	-18.57	87.19	46.74	10.47	144.40	123.47	50.92	29.99	117.40	133.94	40.46	97.66	76.73	**O.K**
202	1.0DL+1.0LR	415.21	82.16	-25.03	85.79	46.30	14.10	146.18	117.98	53.59	25.39	119.18	132.08	39.49	99.89	71.69	**O.K**
203	1.0DL+0.75LL+0.75LR+0.75TL	421.99	82.75	-20.07	87.19	46.63	11.31	145.13	122.51	51.87	29.25	118.13	133.82	40.56	98.50	75.88	**O.K**
204	1.0DL+1.0WLX1	395.86	42.74	-22.05	81.79	24.08	12.43	118.30	93.44	70.13	45.28	91.30	105.87	57.71	94.22	69.36	**O.K**
205	1.0DL+1.0WLX2	429.38	126.40	-26.27	88.72	71.23	14.80	174.74	145.14	32.29	2.69	147.74	159.94	17.49	103.52	73.92	**O.K**
206	1.0DL+1.0WLZ1	414.44	67.08	-24.30	85.63	37.80	13.69	137.12	109.73	61.52	34.14	110.12	123.43	47.83	99.32	71.94	**O.K**
207	1.0DL+1.0WLZ2	408.23	75.79	-27.04	84.34	42.70	15.24	142.29	111.81	56.88	26.40	115.29	127.05	41.64	99.59	69.11	**O.K**
208	1.0DL+0.75LL+0.75LR+0.75WLX1	410.58	49.21	-22.25	84.83	27.73	12.54	125.09	100.02	69.64	44.57	98.09	112.56	57.11	97.37	72.30	**O.K**
209	1.0DL+0.75LL+0.75LR+0.75WLX2	435.72	111.95	-25.41	90.02	63.08	14.32	167.43	138.79	41.26	12.62	140.43	153.11	26.94	104.35	75.71	**O.K**
210	1.0DL+0.75LL+0.75LR+0.75WLZ1	424.51	67.46	-23.93	87.71	38.01	13.49	139.21	112.24	63.18	36.21	112.21	125.73	49.70	101.20	74.23	**O.K**
211	1.0DL+0.75LL+0.75LR+0.75WLZ2	419.85	73.99	-25.99	86.75	41.69	14.64	143.08	113.80	59.69	30.41	116.08	128.44	45.05	101.39	72.11	**O.K**
212	0.6DL+1.0WLX1	289.54	9.87	-11.95	59.82	5.56	6.73	72.12	58.65	60.99	47.52	45.12	65.39	54.26	66.56	53.09	**O.K**
213	0.6DL+1.0WLX2	323.06	93.54	-16.17	66.75	52.71	9.11	128.57	110.35	23.15	4.93	101.57	119.46	14.04	75.86	57.64	**O.K**
214	0.6DL+1.0WLZ1	308.12	34.22	-14.20	63.66	19.28	8.00	90.94	74.94	52.38	36.38	63.94	82.94	44.38	71.66	55.66	**O.K**
215	0.6DL+1.0WLZ2	301.91	42.93	-16.94	62.38	24.19	9.54	96.11	77.02	47.73	28.65	69.11	86.57	38.19	71.92	52.84	**O.K**

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Ultimate Conditions

101 1.4DL	518.18	115.01	-35.36	107.06	64.81	19.93	191.80	151.95	62.18	22.33	164.80	171.88	42.26	126.99	87.14	
102 1.4DL+1.7LL+0.5LR	538.45	107.34	-34.26	111.25	60.48	19.30	191.04	152.43	70.07	31.46	164.04	171.74	50.77	130.56	91.95	
103 1.2DL+1.6LR+0.8WLX1	457.58	67.06	-27.38	94.54	37.79	15.43	147.76	116.90	72.18	41.33	120.76	132.33	56.76	109.97	79.12	
104 1.2DL+1.6LR+0.8WLX2	484.40	133.99	-30.75	100.08	75.50	17.33	192.91	158.26	41.91	7.25	165.91	175.59	24.58	117.41	82.76	
105 1.2DL+1.6LR+0.8WLZ1	472.45	86.53	-29.18	97.61	48.76	16.44	162.82	129.93	65.29	32.41	135.82	146.38	48.85	114.06	81.17	
106 1.2DL+1.6LR+0.8WLZ2	467.47	93.50	-31.37	96.59	52.69	17.68	166.95	131.60	61.58	26.22	139.95	149.28	43.90	114.27	78.91	
107 1.2DL+1.6LR+1.0LL	481.31	94.08	-29.36	99.45	53.01	16.55	169.00	135.91	62.98	29.89	142.00	152.46	46.44	115.99	82.90	
108 1.2DL+1.0LL+1.6WLX1+0.5LR	452.04	31.00	-24.48	93.40	17.47	13.80	124.66	97.07	89.72	62.13	97.66	110.87	75.93	107.19	79.60	
109 1.2DL+1.0LL+1.6WLX1+0.5LR	505.68	164.86	-31.24	104.48	92.90	17.60	214.98	179.78	29.18	-6.02	187.98	197.38	11.58	122.08	86.88	
110 1.2DL+1.0LL+1.6WLX1+0.5LR	481.77	69.95	-28.08	99.54	39.42	15.82	154.78	123.13	75.95	44.30	127.78	138.96	60.13	115.37	83.72	
111 1.2DL+1.0LL+1.6WLX1+0.5LR	471.83	83.88	-32.47	97.48	47.27	18.29	163.05	126.46	68.51	31.92	136.05	144.76	50.22	115.78	79.19	
112 1.2DL+1.0LL	475.95	94.07	-29.73	98.34	53.00	16.75	168.09	134.59	62.08	28.58	141.09	151.34	45.33	115.09	81.59	
113 0.9DL+1.6WLX1	359.68	10.87	-17.60	74.31	6.12	9.92	90.36	70.52	78.11	58.27	63.36	80.44	68.19	84.24	64.40	
114 0.9DL+1.6WLX2	413.32	144.73	-24.35	85.40	81.55	13.72	180.67	153.23	17.56	-9.88	153.67	166.95	3.84	99.12	71.68	
115 0.9DL+1.6WLZ1	389.42	49.82	-21.20	80.46	28.07	11.95	120.48	96.58	64.33	40.44	93.48	108.53	52.39	92.41	68.51	
116 0.9DL+1.6WLZ2	379.47	63.75	-25.58	78.40	35.92	14.42	128.74	99.91	56.90	28.06	101.74	114.33	42.48	92.82	63.99	
117 1.4DL+ 1.4TL+1.7LL+0.5LR	537.36	114.77	-25.72	111.03	64.67	14.49	190.19	161.21	60.84	31.86	163.19	175.70	46.35	125.52	96.54	
Max :	436	126	-12													
Min :	290	10	-27													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) =	**117.40 kN/m^2**
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) =	**127.40 kN/m^2**
Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) =	**147.74 kN/m^2**
Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) =	**157.74 kN/m^2**

Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0LR

Maximum pressure, p_1 (pmax) =	85.79+46.30+14.10=	146.18 kN/m^2	Maximum average Pressure along X, $p_{avx(max)}$ =	(146.18+117.98)/2=	132.08 kN/m^2
p_2=	85.79+46.30-14.10=	117.98 kN/m^2	Minimum average Pressure along X, $p_{avx(minx)}$ =	(25.39+53.59)/2=	39.49 kN/m^2
p_3=	85.79-(46.30+14.10)=	25.39 kN/m^2	Maximum average Pressure along Y, $p_{avy(max)}$ =	(146.18+53.59)/2=	99.89 kN/m^2
p_4=	85.79-46.30+14.10=	53.59 kN/m^2	Minimum average Pressure along Y, $p_{avy(minx)}$ =	(25.39+117.98)/2=	71.68 kN/m^2
Maxm pressure excluding weight of soil (p_{max}-gh)=	146.18-1.5*18=	119.18 kN/m^2			

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DESIGN OF FOUNDATION MKD. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.5+2*0.5*0.332)+(0.5+2*0.5*0.332))*2

= 3.328 m

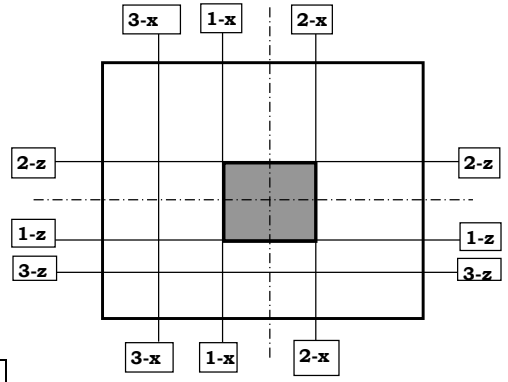
Punching Area= ((0.5+2*0.5*0.332)*(0.5+2*0.5*0.332))

= 0.69 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)

= 29.40 kN/m^2

Area of footing, A_f= 4.84 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.$F_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL	121.80	92.34	142.07	45.44	10.68	64.55	322.29
102	1.4DL+1.7LL+0.5LR	125.00	97.51	143.92	45.79	13.35	64.98	339.67
103	1.2DL+1.6LR+0.8WLX1	103.13	85.96	114.95	33.67	13.40	47.69	270.34
104	1.2DL+1.6LR+0.8WLX2	117.25	82.92	140.86	45.79	5.28	65.18	293.33
105	1.2DL+1.6LR+0.8WLZ1	108.70	86.53	123.95	37.72	11.56	53.52	283.08
106	1.2DL+1.6LR+0.8WLZ2	108.57	84.62	125.04	38.40	10.14	54.53	278.82
107	1.2DL+1.6LR+1.0LL	111.50	87.40	128.08	39.52	11.09	56.10	290.69
108	1.2DL+1.0LL+1.6WLX1+0.5LR	97.37	89.43	102.83	27.81	18.43	39.19	265.58
109	1.2DL+1.0LL+1.6WLX1+0.5LR	125.59	83.37	154.65	52.04	2.21	74.19	311.58
110	1.2DL+1.0LL+1.6WLX1+0.5LR	108.50	90.59	120.83	35.91	14.77	50.85	291.08
111	1.2DL+1.0LL+1.6WLX1+0.5LR	108.23	86.75	123.02	37.28	11.92	52.87	282.55
112	1.2DL+1.0LL	110.38	86.29	126.96	39.12	10.69	55.53	286.09
113	0.9DL+1.6WLX1	75.71	72.92	77.62	17.87	14.58	25.11	186.40
114	0.9DL+1.6WLX2	103.93	66.86	129.43	42.10	-1.64	60.11	232.39
115	0.9DL+1.6WLZ1	86.84	74.08	95.62	25.97	10.92	36.77	211.89
116	0.9DL+1.6WLZ2	86.57	70.24	97.80	27.34	8.07	38.79	203.36
117	1.4DL+ 1.4TL+1.7LL+0.5LR	125.72	96.33	145.95	46.83	12.14	66.50	338.74

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p_{1x} =	171.88-(171.88-42.26)*0.85/2.2 =	121.80	kN/m^2	Overhang =	0.85 m
p_{2x} =	42.26+(171.88-42.26)*0.85/2.2 =	92.34	kN/m^2	Effective depth =	0.344 m
p_{3x} =	171.88-(171.88-42.26)*(0.85-0.344)/2.2 =	142.07	kN/m^2		
BM_{1x} =	(121.80-29.4)*0.85*0.85/2+0.5*(171.88-121.80)*0.85*0.85*0.67 =	45.44	kN-m/m		
BM_{2x} =	(92.34-29.4)*0.85*0.85/2-0.5*(-42.26+92.34)*0.85*0.85*0.67 =	10.68	kN-m/m		
SF_{3x} =	(142.07-29.4)*(0.85-0.344)+0.5*(171.88-142.07)*(0.85-0.344) =	64.55	kN/m		
$S.F_{(punching)}$ =	(4.84-0.69)*(107.06-29.4) =	322.29	kN		

Location	Mux (kN-m)	f Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	r=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	52.04	0.44	0.59	0.001	485.07	619.20	12fT	182.65	300	175	646.27

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (f Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 74.19 kN/m

Shear stress , t_v= (74.19*1000)/(1000*344)

= 0.22 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 339.67 kN

Strength of Concrete in shear Vc1 = 3386.46 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2295.63 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2257.64 kN **Table 22.6.5.2_ (c)**

Design Vc= 2257.64 kN

Allowable shear stress, (f Vc)/bd = 1.48 N/mm^2

Design punching shear stress, t = (339.67*1000)/(3328*344)

= 0.30 N/mm^2 **No shear reinforcement reqd.**

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p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4DL	111.59	102.54	117.61	33.42	22.69	48.12
102	1.4DL+1.7LL+0.5LR	115.64	106.87	121.47	34.77	24.37	50.05
103	1.2DL+1.6LR+0.8WLX1	98.05	91.04	102.71	27.68	19.38	39.85
104	1.2DL+1.6LR+0.8WLX2	104.02	96.15	109.25	30.20	20.87	43.48
105	1.2DL+1.6LR+0.8WLZ1	101.35	93.88	106.32	29.07	20.22	41.85
106	1.2DL+1.6LR+0.8WLZ2	100.61	92.57	105.94	29.03	19.51	41.81
107	1.2DL+1.6LR+1.0LL	103.21	95.68	108.20	29.76	20.85	42.84
108	1.2DL+1.0LL+1.6WLX1+0.5LR	96.53	90.26	100.69	26.83	19.41	38.61
109	1.2DL+1.0LL+1.6WLX1+0.5LR	108.48	100.48	113.79	31.86	22.39	45.86
110	1.2DL+1.0LL+1.6WLX1+0.5LR	103.14	95.95	107.92	29.60	21.08	42.60
111	1.2DL+1.0LL+1.6WLX1+0.5LR	101.64	93.33	107.16	29.52	19.67	42.51
112	1.2DL+1.0LL	102.15	94.53	107.20	29.41	20.40	42.34
113	0.9DL+1.6WLX1	76.57	72.07	79.57	18.90	13.56	27.20
114	0.9DL+1.6WLX2	88.52	82.28	92.66	23.92	16.54	34.44
115	0.9DL+1.6WLZ1	83.18	77.74	86.78	21.66	15.23	31.18
116	0.9DL+1.6WLZ2	81.68	75.13	86.03	21.58	13.82	31.09
117	1.4DL+ 1.4TL+1.7LL+0.5LR	114.32	107.74	118.70	33.39	25.59	48.02

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Sample calculation of BM & SF

Load case : 101 1.4DL

p_{1z} = 126.99-(126.99-87.14)*0.85/2.2= 111.59 kN/m^2

p_{2z} = 87.14+(126.99-87.14)*0.85/2.2= 102.54 kN/m^2

p_{3z} = 126.99-(126.99-87.14)*(0.85-0.332)/2.2= 117.61 kN/m^2

Overhang = 0.85 m

Effective depth = 0.332 m

BM_{1z} = (111.59-29.4)*0.85*0.85/2+0.5*(126.99-111.59)*0.85*0.85*0.67 = 33.42 kN-m/m

BM_{2z} = (102.54-29.4)*0.85*0.85/2-0.5*(-87.14+102.54)*0.85*0.85*0.67 = 22.69 kN-m/m

SF_{3z} = (117.61-29.4)*(0.85-0.332)+0.5*(126.99-117.61)*(0.85-0.332) = 48.12 kN/m

Location	Mux (kN-m)	f Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	r=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	34.77	0.315	0.421	0.0010	346.92	597.60	12fT	189.25	300	175	646.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (f Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 50.05 kN/m

Shear stress , t_v= (50.05*1000)/(1000*332)

= 0.15 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	**12fT**	**175**	c/c
	At Top	**12fT**	**175**	c/c
Along Z Direction :				
	At Bottom	**12fT**	**175**	c/c
	At Top	**12fT**	**175**	c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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ATTACHMENT 1 STAAD INPUT

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 30.10.2020
ENGINEER NAME BJB
CHECKER NAME SJM
APPROVED NAME AG
JOB CLIENT MAA'DEN
JOB REV A01
JOB NAME Fire Water Pump House
JOB NO O19045
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 6.50001 0 0; 4 0 4.19961 0; 5 6.50001 4.19961 0; 7 0 0 4.00001;
10 0 4.19961 4.00001; 23 0 -2 0; 24 6.50001 -2 0; 26 0 -2 4.00001;
79 6.5 0 4.00001; 80 6.5 4.19961 4.00001; 81 6.5 -2 4.00001; 82 0 0 8.50001;
83 0 4.19961 8.50001; 84 0 -2 8.50001; 85 6.5 0 8.50001;
86 6.5 4.19961 8.50001; 87 6.5 -2 8.50001; 88 0 0 13; 89 0 6.69961 13;
90 0 -2 13; 91 6.5 0 13; 92 6.5 6.69961 13; 93 6.5 -2 13; 94 0 0 16;
95 0 6.69961 16; 96 0 -2 16; 97 6.5 0 16; 98 6.5 6.69961 16; 99 6.5 -2 16;
100 0 0 20.5; 101 0 6.69961 20.5; 102 0 -2 20.5; 103 6.5 0 20.5;
104 6.5 6.69961 20.5; 105 6.5 -2 20.5; 106 0 0 25; 107 0 6.69961 25;
108 0 -2 25; 109 6.5 0 25; 110 6.5 6.69961 25; 111 6.5 -2 25; 112 0 4.19961 13;
113 6.5 4.19961 13; 114 4.75 6.69961 13; 115 4.75 6.69961 16;
116 6.5 6.69961 22.75; 117 0 6.69961 22.75; 118 6.5 6.69961 18.25;
119 0 6.69961 18.25;
MEMBER INCIDENCES
1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24;
34 7 26; 583 79 80; 584 79 2; 585 79 81; 587 5 80; 590 10 80; 591 82 83;
592 82 84; 593 85 86; 594 85 87; 596 83 86; 597 88 112; 598 88 90; 599 91 113;
600 91 93; 601 88 91; 602 89 114; 603 94 95; 604 94 96; 605 97 98; 606 97 99;
608 95 115; 609 100 101; 610 100 102; 611 103 104; 612 103 105; 614 101 104;
615 106 107; 616 106 108; 617 109 110; 618 109 111; 619 106 109; 620 107 110;
621 10 83; 622 80 86; 623 112 89; 624 83 112; 625 113 92; 626 86 113;
627 92 98; 628 89 95; 629 95 119; 630 101 117; 631 98 118; 632 104 116;
633 7 82; 634 82 88; 635 88 94; 636 94 100; 637 100 106; 638 79 85; 639 85 91;
640 91 97; 641 97 103; 642 103 109; 643 112 113; 644 114 92; 645 115 98;
646 114 115; 647 116 110; 648 117 107; 650 118 104; 651 119 101;
START GROUP DEFINITION
MEMBER
_COLUMN_C1 3 4 8 31 32 34 583 585 591 TO 594 597 TO 600 603 TO 606 -
609 TO 612 615 TO 618 623 625
_RB_1 11 587
_RB_2 590 596 602 608 614 620 644 645
_GB_1 25 30 584 601 619
_RB_4 1
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
CONSTANTS
MATERIAL CONCRETE ALL
MEMBER PROPERTY AMERICAN
3 4 8 31 32 34 583 585 591 TO 594 597 TO 600 603 TO 606 609 TO 612 -
615 TO 618 623 625 PRIS YD 0.5 ZD 0.5
1 11 587 590 596 602 608 614 620 TO 622 624 626 TO 632 643 TO 645 647 648 -
650 651 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
25 30 584 601 619 633 TO 642 PRIS YD 0.6 ZD 0.25
MEMBER PROPERTY
646 PRIS YD 0.45 ZD 0.25
SUPPORTS
23 24 26 81 84 87 90 93 96 99 102 105 108 111 FIXED
*******************************************************************************************************
****
LOAD 1 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
25 30 584 633 634 638 639 UNI GY -14.8
11 587 621 622 624 626 TO 632 647 648 650 651 UNI GY -20.8
11 UMOM GZ 1
587 UMOM GZ -1
1 UMOM GX -1
620 UMOM GX -1
601 619 635 TO 637 640 TO 642 UNI GY -24.8
*******************************************************************************************************
****
```

```
MEMBER LOAD
614 620 627 646 CON GY -15
608 CON GY -15 3.25
* LOAD DUE TO MONORAIL
************************************************************************************************************
****
LOAD 3 LOADTYPE Roof Live  TITLE LR
MEMBER LOAD
11 587 621 622 624 626 TO 632 647 648 650 651 UNI GY -1.625
11 621 624 628 UMOM GZ 0.075
587 622 626 UMOM GZ -0.075
1 UMOM GX -1.625
620 CMOM GX 1.625
629 630 CMOM GZ 0.075 2.25
631 632 CMOM GZ -0.075 2.25
************************************************************************************************************
****
LOAD 4 LOADTYPE Wind  TITLE WIND ALONG X DIR. ( CPI = + 0.18) WLX1
MEMBER LOAD
603 UNI GX 1.68
605 UNI GX 2.4
615 617 UNI GZ 2.7
3 4 UNI GZ -2.7
3 8 UNI GX 2.24
591 597 609 615 UNI GX 2.5
4 583 UNI GX 3.2
593 599 611 617 625 UNI GX 3.6
************************************************************************************************************
****
LOAD 5 LOADTYPE Wind  TITLE WIND ALONG X DIR. ( CPI = - 0.18) WLX2
MEMBER LOAD
603 UNI GX -0.99
605 UNI GX -3.09
615 617 UNI GZ -1.14
3 4 UNI GZ 1.14
3 8 UNI GX -1.32
591 597 609 615 UNI GX -1.62
4 583 UNI GX -4.12
593 599 611 617 625 UNI GX -4.64
************************************************************************************************************
****
LOAD 6 LOADTYPE Wind  TITLE WIND ALONG Z DIR. ( CPI = + 0.18) WLZ1
MEMBER LOAD
3 4 UNI GZ 1.82
615 617 UNI GZ 2.6
603 UNI GX -2.49
605 UNI GX 2.49
8 UNI GX -3.32
4 583 UNI GX 3.32
593 599 611 617 625 UNI GX 3.74
591 597 609 615 UNI GX -3.74
************************************************************************************************************
****
LOAD 7 LOADTYPE Wind  TITLE WIND ALONG Z DIR. ( CPI = - 0.18) WLZ2
MEMBER LOAD
3 4 UNI GZ -1.07
615 617 UNI GZ -3.35
603 UNI GX -1.05
605 UNI GX 1.05
8 UNI GX -1.4
4 583 UNI GX 1.4
593 599 611 617 625 UNI GX 1.58
591 597 609 615 UNI GX -1.58
************************************************************************************************************
****
LOAD 8 LOADTYPE Temperature  TITLE TEMPERATURE LOAD ( TL )
TEMPERATURE LOAD
3 4 8 583 591 593 597 599 603 605 609 611 615 617 623 625 TEMP 25
1 11 587 590 596 602 608 614 620 TO 622 624 626 TO 632 643 TO 648 650 -
651 TEMP 0 25
3 4 8 583 591 593 597 599 603 605 609 611 615 617 623 625 TEMP 0 0 25
************************************************************************************************************
****
* FACTORED LOAD COMBINATIONS
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.7LL+0.5LR
1 1.4 2 1.7 3 0.5
LOAD COMB 103 1.2DL+1.6LR+0.8WLX1
1 1.2 3 1.6 4 0.8
LOAD COMB 104 1.2DL+1.6LR+0.8WLX2
1 1.2 3 1.6 5 0.8
LOAD COMB 105 1.2DL+1.6LR+0.8WLZ1
1 1.2 3 1.6 6 0.8
```

```
1 1.2 2 1.0 3 0.5 5 1.6
LOAD COMB 110 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 6 1.6
LOAD COMB 111 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 7 1.6
LOAD COMB 112 1.2DL+1.0LL
1 1.2 2 1.0
LOAD COMB 113 0.9DL+1.6WLX1
1 0.9 4 1.6
LOAD COMB 114 0.9DL+1.6WLX2
1 0.9 5 1.6
LOAD COMB 115 0.9DL+1.6WLZ1
1 0.9 6 1.6
LOAD COMB 116 0.9DL+1.6WLZ2
1 0.9 7 1.6
LOAD COMB 117 1.4DL+ 1.4TL+1.7LL+0.5LR
1 1.4 2 1.7 3 0.5 8 1.4
*******************************************************************************************************************
****
* UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
LOAD COMB 201 1.0DL+1.0LL+1.0TL
1 1.0 2 1.0 8 1.0
LOAD COMB 202 1.0DL+1.0LR
1 1.0 3 1.0
LOAD COMB 203 1.0DL+0.75LL+0.75LR+0.75TL
1 1.0 2 0.75 3 0.75 8 0.75
LOAD COMB 204 1.0DL+1.0WLX1
1 1.0 4 1.0
LOAD COMB 205 1.0DL+1.0WLX2
1 1.0 5 1.0
LOAD COMB 206 1.0DL+1.0WLZ1
1 1.0 6 1.0
LOAD COMB 207 1.0DL+1.0WLZ2
1 1.0 7 1.0
LOAD COMB 208 1.0DL+0.75LL+0.75LR+0.75WLX1
1 1.0 2 0.75 3 0.75 4 0.75
LOAD COMB 209 1.0DL+0.75LL+0.75LR+0.75WLX2
1 1.0 2 0.75 3 0.75 5 0.75
LOAD COMB 210 1.0DL+0.75LL+0.75LR+0.75WLZ1
1 1.0 2 0.75 3 0.75 6 0.75
LOAD COMB 211 1.0DL+0.75LL+0.75LR+0.75WLZ2
1 1.0 2 0.75 3 0.75 7 0.75
LOAD COMB 212 0.6DL+1.0WLX1
1 0.6 4 1.0
LOAD COMB 213 0.6DL+1.0WLX2
1 0.6 5 1.0
LOAD COMB 214 0.6DL+1.0WLZ1
1 0.6 6 1.0
LOAD COMB 215 0.6DL+1.0WLZ2
1 0.6 7 1.0
*******************************************************************************************************************
****
PERFORM ANALYSIS
LOAD LIST 101 TO 117
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 MEMB 25 30 584 601 619 633 TO 642
CLT 0.04 ALL
MAXMAIN 20 ALL
SFACE 0.5 MEMB 1 11 25 30 584 587 590 596 601 602 608 614 619 TO 622 624 626 -
627 TO 648 650 651
MINMAIN 12 ALL
MINSEC 10 ALL
DESIGN BEAM 1 11 25 30 584 587 590 596 601 602 608 614 619 TO 622 624 626 -
627 TO 648 650 651
DESIGN COLUMN 3 4 8 31 32 34 583 585 591 TO 594 597 TO 600 603 TO 606 609 -
610 TO 612 615 TO 618 623 625
CONCRETE TAKE
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 2 STAAD OUTPUT

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    OCT 15, 2020                *
*           Time=    16: 1: 9                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\MAADEN\Potable Water Treatment Plant\FWPH\STAAD\FWPH-Elec.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 30.10.2020
    4. ENGINEER NAME BJB
    5. CHECKER NAME SJM
    6. APPROVED NAME AG
    7. JOB CLIENT MAA'DEN
    8. JOB REV A01
    9. JOB NAME FIRE WATER PUMP HOUSE
   10. JOB NO O19045
   11. END JOB INFORMATION
   12. INPUT WIDTH 79
   13. UNIT METER KN
   14. JOINT COORDINATES
   15. 1 0 0 0; 2 6.50001 0 0; 4 0 4.19961 0; 5 6.50001 4.19961 0; 7 0 0 4.00001
   16. 10 0 4.19961 4.00001; 23 0 -2 0; 24 6.50001 -2 0; 26 0 -2 4.00001
   17. 79 6.5 0 4.00001; 80 6.5 4.19961 4.00001; 81 6.5 -2 4.00001; 82 0 0 8.50001
   18. 83 0 4.19961 8.50001; 84 0 -2 8.50001; 85 6.5 0 8.50001
   19. 86 6.5 4.19961 8.50001; 87 6.5 -2 8.50001; 88 0 0 13; 89 0 6.69961 13
   20. 90 0 -2 13; 91 6.5 0 13; 92 6.5 6.69961 13; 93 6.5 -2 13; 94 0 0 16
   21. 95 0 6.69961 16; 96 0 -2 16; 97 6.5 0 16; 98 6.5 6.69961 16; 99 6.5 -2 16
   22. 100 0 0 20.5; 101 0 6.69961 20.5; 102 0 -2 20.5; 103 6.5 0 20.5
   23. 104 6.5 6.69961 20.5; 105 6.5 -2 20.5; 106 0 0 25; 107 0 6.69961 25
   24. 108 0 -2 25; 109 6.5 0 25; 110 6.5 6.69961 25; 111 6.5 -2 25; 112 0 4.19961 13
   25. 113 6.5 4.19961 13; 114 4.75 6.69961 13; 115 4.75 6.69961 16
   26. 116 6.5 6.69961 22.75; 117 0 6.69961 22.75; 118 6.5 6.69961 18.25
   27. 119 0 6.69961 18.25
   28. MEMBER INCIDENCES
   29. 1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24
   30. 34 7 26; 583 79 80; 584 79 2; 585 79 81; 587 5 80; 590 10 80; 591 82 83
   31. 592 82 84; 593 85 86; 594 85 87; 596 83 86; 597 88 112; 598 88 90; 599 91 113
   32. 600 91 93; 601 88 91; 602 89 114; 603 94 95; 604 94 96; 605 97 98; 606 97 99
   33. 608 95 115; 609 100 101; 610 100 102; 611 103 104; 612 103 105; 614 101 104
   34. 615 106 107; 616 106 108; 617 109 110; 618 109 111; 619 106 109; 620 107 110
   35. 621 10 83; 622 80 86; 623 112 89; 624 83 112; 625 113 92; 626 86 113
   36. 627 92 98; 628 89 95; 629 95 119; 630 101 117; 631 98 118; 632 104 116
   37. 633 7 82; 634 82 88; 635 88 94; 636 94 100; 637 100 106; 638 79 85; 639 85 91
   38. 640 91 97; 641 97 103; 642 103 109; 643 112 113; 644 114 92; 645 115 98
```

```
39. 646 114 115; 647 116 110; 648 117 107; 650 118 104; 651 119 101
40. START GROUP DEFINITION
41. MEMBER
42. _COLUMN_C1 3 4 8 31 32 34 583 585 591 TO 594 597 TO 600 603 TO 606 -
43. 609 TO 612 615 TO 618 623 625
44. _RB_1 11 587
45. _RB_2 590 596 602 608 614 620 644 645
46. _GB_1 25 30 584 601 619
47. _RB_4 1
48. END GROUP DEFINITION
49. DEFINE MATERIAL START
50. ISOTROPIC CONCRETE
51. E 2.17184E+007
52. POISSON 0.17
53. DENSITY 23.6158
54. ALPHA 5E-006
55. DAMP 0.05
56. TYPE CONCRETE
57. STRENGTH FCU 27578.9
58. END DEFINE MATERIAL
59. CONSTANTS
60. MATERIAL CONCRETE ALL
61. MEMBER PROPERTY AMERICAN
62. 3 4 8 31 32 34 583 585 591 TO 594 597 TO 600 603 TO 606 609 TO 612 -
63. 615 TO 618 623 625 PRIS YD 0.5 ZD 0.5
64. 1 11 587 590 596 602 608 614 620 TO 622 624 626 TO 632 643 TO 645 647 648 -
65. 650 651 PRIS YD 0.5 ZD 0.25
66. MEMBER PROPERTY AMERICAN
67. 25 30 584 601 619 633 TO 642 PRIS YD 0.6 ZD 0.25
68. MEMBER PROPERTY
69. 646 PRIS YD 0.45 ZD 0.25
70. SUPPORTS
71. 23 24 26 81 84 87 90 93 96 99 102 105 108 111 FIXED
72. ******************************************************************************
73. LOAD 1 LOADTYPE DEAD  TITLE DL
74. SELFWEIGHT Y -1
75. MEMBER LOAD
76. 25 30 584 633 634 638 639 UNI GY -14.8
77. 11 587 621 622 624 626 TO 632 647 648 650 651 UNI GY -20.8
78. 11 UMOM GZ 1
79. 587 UMOM GZ -1.
80. 1 UMOM GX -1.
81. 620 UMOM GX -1.
82. 601 619 635 TO 637 640 TO 642 UNI GY -24.8
83. ******************************************************************************
84. LOAD 2 LOADTYPE LIVE  TITLE LL
85. MEMBER LOAD
86. 614 620 627 646 CON GY -15
87. 608 CON GY -15 3.25
88. * LOAD DUE TO MONORAIL
89. ******************************************************************************
90. LOAD 3 LOADTYPE ROOF LIVE  TITLE LR
91. MEMBER LOAD
92. 11 587 621 622 624 626 TO 632 647 648 650 651 UNI GY -1.625
93. 11 621 624 628 UMOM GZ 0.075
94. 587 622 626 UMOM GZ -0.075
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 95. 1 UMOM GX -1.625
 96. 620 CMOM GX 1.625
 97. 629 630 CMOM GZ 0.075 2.25
 98. 631 632 CMOM GZ -0.075 2.25
 99. ********************************************************************************
100. LOAD 4 LOADTYPE WIND  TITLE WIND ALONG X DIR. ( CPI = + 0.18) WLX1
101. MEMBER LOAD
102. 603 UNI GX 1.68
103. 605 UNI GX 2.4
104. 615 617 UNI GZ 2.7
105. 3 4 UNI GZ -2.7
106. 3 8 UNI GX 2.24
107. 591 597 609 615 UNI GX 2.5
108. 4 583 UNI GX 3.2
109. 593 599 611 617 625 UNI GX 3.6
110. ********************************************************************************
111. LOAD 5 LOADTYPE WIND  TITLE WIND ALONG X DIR. ( CPI = - 0.18) WLX2
112. MEMBER LOAD
113. 603 UNI GX -0.99
114. 605 UNI GX -3.09
115. 615 617 UNI GZ -1.14
116. 3 4 UNI GZ 1.14
117. 3 8 UNI GX -1.32
118. 591 597 609 615 UNI GX -1.62
119. 4 583 UNI GX -4.12
120. 593 599 611 617 625 UNI GX -4.64
121. ********************************************************************************
122. LOAD 6 LOADTYPE WIND  TITLE WIND ALONG Z DIR. ( CPI = + 0.18) WLZ1
123. MEMBER LOAD
124. 3 4 UNI GZ 1.82
125. 615 617 UNI GZ 2.6
126. 603 UNI GX -2.49
127. 605 UNI GX 2.49
128. 8 UNI GX -3.32
129. 4 583 UNI GX 3.32
130. 593 599 611 617 625 UNI GX 3.74
131. 591 597 609 615 UNI GX -3.74
132. ********************************************************************************
133. LOAD 7 LOADTYPE WIND  TITLE WIND ALONG Z DIR. ( CPI = - 0.18) WLZ2
134. MEMBER LOAD
135. 3 4 UNI GZ -1.07
136. 615 617 UNI GZ -3.35
137. 603 UNI GX -1.05
138. 605 UNI GX 1.05
139. 8 UNI GX -1.4
140. 4 583 UNI GX 1.4
141. 593 599 611 617 625 UNI GX 1.58
142. 591 597 609 615 UNI GX -1.58
143. ********************************************************************************
144. LOAD 8 LOADTYPE TEMPERATURE  TITLE TEMPERATURE LOAD ( TL )
145. TEMPERATURE LOAD
146. 3 4 8 583 591 593 597 599 603 605 609 611 615 617 623 625 TEMP 25
147. 1 11 587 590 596 602 608 614 620 TO 622 624 626 TO 632 643 TO 648 650 -
148. 651 TEMP 0 25
149. 3 4 8 583 591 593 597 599 603 605 609 611 615 617 623 625 TEMP 0 0 25
150. ********************************************************************************
```

```
151. * FACTORED LOAD COMBINATIONS
152. LOAD COMB 101 1.4DL
153. 1 1.4
154. LOAD COMB 102 1.4DL+1.7LL+0.5LR
155. 1 1.4 2 1.7 3 0.5
156. LOAD COMB 103 1.2DL+1.6LR+0.8WLX1
157. 1 1.2 3 1.6 4 0.8
158. LOAD COMB 104 1.2DL+1.6LR+0.8WLX2
159. 1 1.2 3 1.6 5 0.8
160. LOAD COMB 105 1.2DL+1.6LR+0.8WLZ1
161. 1 1.2 3 1.6 6 0.8
162. LOAD COMB 106 1.2DL+1.6LR+0.8WLZ2
163. 1 1.2 3 1.6 7 0.8
164. LOAD COMB 107 1.2DL+1.6LR+1.0LL
165. 1 1.2 2 1.0 3 1.6
166. LOAD COMB 108 1.2DL+1.0LL+1.6WLX1+0.5LR
167. 1 1.2 2 1.0 3 0.5 4 1.6
168. LOAD COMB 109 1.2DL+1.0LL+1.6WLX1+0.5LR
169. 1 1.2 2 1.0 3 0.5 5 1.6
170. LOAD COMB 110 1.2DL+1.0LL+1.6WLX1+0.5LR
171. 1 1.2 2 1.0 3 0.5 6 1.6
172. LOAD COMB 111 1.2DL+1.0LL+1.6WLX1+0.5LR
173. 1 1.2 2 1.0 3 0.5 7 1.6
174. LOAD COMB 112 1.2DL+1.0LL
175. 1 1.2 2 1.0
176. LOAD COMB 113 0.9DL+1.6WLX1
177. 1 0.9 4 1.6
178. LOAD COMB 114 0.9DL+1.6WLX2
179. 1 0.9 5 1.6
180. LOAD COMB 115 0.9DL+1.6WLZ1
181. 1 0.9 6 1.6
182. LOAD COMB 116 0.9DL+1.6WLZ2
183. 1 0.9 7 1.6
184. LOAD COMB 117 1.4DL+ 1.4TL+1.7LL+0.5LR
185. 1 1.4 2 1.7 3 0.5 8 1.4
186. ********************************************************************************
187. * UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
188. LOAD COMB 201 1.0DL+1.0LL+1.0TL
189. 1 1.0 2 1.0 8 1.0
190. LOAD COMB 202 1.0DL+1.0LR
191. 1 1.0 3 1.0
192. LOAD COMB 203 1.0DL+0.75LL+0.75LR+0.75TL
193. 1 1.0 2 0.75 3 0.75 8 0.75
194. LOAD COMB 204 1.0DL+1.0WLX1
195. 1 1.0 4 1.0
196. LOAD COMB 205 1.0DL+1.0WLX2
197. 1 1.0 5 1.0
198. LOAD COMB 206 1.0DL+1.0WLZ1
199. 1 1.0 6 1.0
200. LOAD COMB 207 1.0DL+1.0WLZ2
201. 1 1.0 7 1.0
202. LOAD COMB 208 1.0DL+0.75LL+0.75LR+0.75WLX1
203. 1 1.0 2 0.75 3 0.75 4 0.75
204. LOAD COMB 209 1.0DL+0.75LL+0.75LR+0.75WLX2
205. 1 1.0 2 0.75 3 0.75 5 0.75
206. LOAD COMB 210 1.0DL+0.75LL+0.75LR+0.75WLZ1
```

```
207. 1 1.0 2 0.75 3 0.75 6 0.75
208. LOAD COMB 211 1.0DL+0.75LL+0.75LR+0.75WLZ2
209. 1 1.0 2 0.75 3 0.75 7 0.75
210. LOAD COMB 212 0.6DL+1.0WLX1
211. 1 0.6 4 1.0
212. LOAD COMB 213 0.6DL+1.0WLX2
213. 1 0.6 5 1.0
214. LOAD COMB 214 0.6DL+1.0WLZ1
215. 1 0.6 6 1.0
216. LOAD COMB 215 0.6DL+1.0WLZ2
217. 1 0.6 7 1.0
218. ********************************************************************************
219. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**50**	**NUMBER OF MEMBERS**	**72**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**14**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 29/ 8/ 42 DOF
TOTAL PRIMARY LOAD CASES = 8, TOTAL DEGREES OF FREEDOM = 216
TOTAL LOAD COMBINATION CASES = 32 SO FAR.
SIZE OF STIFFNESS MATRIX = 10 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.2/ 323681.9 MB

```
220. LOAD LIST 101 TO 117
221. START CONCRETE DESIGN
222. CODE ACI
223. TRACK 2 ALL
224. FC 35000 ALL
225. CLB 0.04 ALL
226. CLS 0.04 MEMB 25 30 584 601 619 633 TO 642
227. CLT 0.04 ALL
228. MAXMAIN 20 ALL
229. SFACE 0.5 MEMB 1 11 25 30 584 587 590 596 601 602 608 614 619 TO 622 624 626 -
230. 627 TO 648 650 651
231. MINMAIN 12 ALL
232. MINSEC 10 ALL
233. DESIGN BEAM 1 11 25 30 584 587 590 596 601 602 608 614 619 TO 622 624 626 -
234. 627 TO 648 650 651
```

ACI 318-11 BEAM NO. 1 DESIGN RESULTS

LEN - 6500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	0.	6500.	YES	YES
	CRITICAL POS MOMENT= 23.59 KN-MET AT 3250.MM, LOAD 117					
	REQD STEEL= 192.MM2, RHO=0.0017, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 142. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 25447.0 cm^4					
2	444.	3 - 12MM	0.	2561.	YES	NO
	CRITICAL NEG MOMENT= 29.17 KN-MET AT 0.MM, LOAD 109					
	REQD STEEL= 237.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					
3	444.	3 - 12MM	3668.	6500.	NO	YES
	CRITICAL NEG MOMENT= 28.25 KN-MET AT 6500.MM, LOAD 108					
	REQD STEEL= 230.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					

REQUIRED REINF. STEEL SUMMARY :

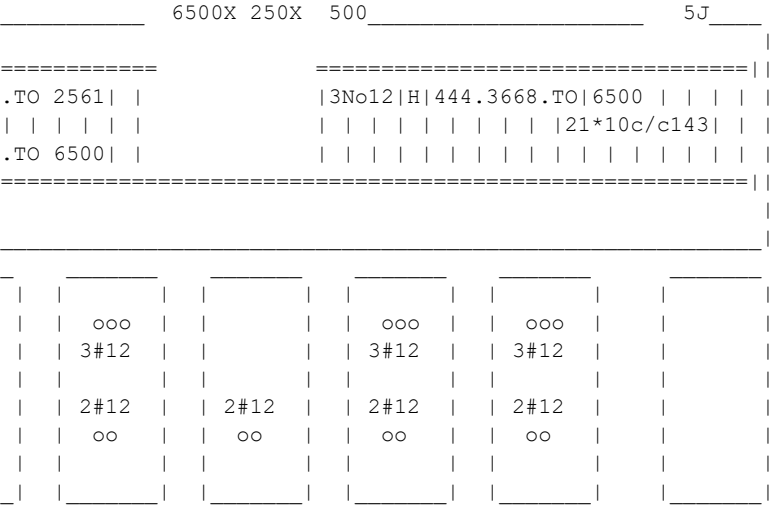
SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	39./ 182.	6./ 29.	113/ 109
542.	52./ 129.	8./ 21.	117/ 109
1083.	86./ 84.	14./ 14.	117/ 109
1625.	113./ 47.	18./ 8.	117/ 114
2167.	132./ 18.	21./ 3.	117/ 114
2708.	143./ 0.	23./ 0.	117/ 0
3250.	147./ 0.	24./ 0.	117/ 0
3792.	143./ 0.	23./ 0.	117/ 0

4333.	132./	13.	21./	2.	117/ 113
4875.	113./	41.	18./	7.	117/ 113
5417.	86./	78.	14./	13.	117/ 108
5958.	52./	124.	8./	20.	117/ 108
6500.	34./	176.	6./	28.	114/ 108

B E A M N O. 1 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.76 KNS Vc= 118.38 KNS Vs= 0.00 KNS
Tu= 8.83 KN-MET Tc= 3.9 KN-MET Ts= 11.8 KN-MET LOAD 103
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM C/C FOR 2310. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.86 SQ.CM.

AT END SUPPORT - Vu= 7.53 KNS Vc= 193.13 KNS Vs= 0.00 KNS
Tu= 10.72 KN-MET Tc= 3.9 KN-MET Ts= 14.3 KN-MET LOAD 104
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM C/C FOR 2810. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.47 SQ.CM.



ACI 318-11 BEAM NO. 11 DESIGN RESULTS

LEN - 4000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	154.	4000.	NO	YES
	CRITICAL POS MOMENT= 41.94 KN-MET AT 2333.MM, LOAD 117					
	REQD STEEL= 345.MM2, RHO=0.0031, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
2	442.	2 - 16MM	0.	1192.	YES	NO
	CRITICAL NEG MOMENT= 42.15 KN-MET AT 0.MM, LOAD 117					
	REQD STEEL= 346.MM2, RHO=0.0031, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
3	442.	2 - 16MM	2641.	4000.	NO	YES
	CRITICAL NEG MOMENT= 51.52 KN-MET AT 4000.MM, LOAD 102					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					

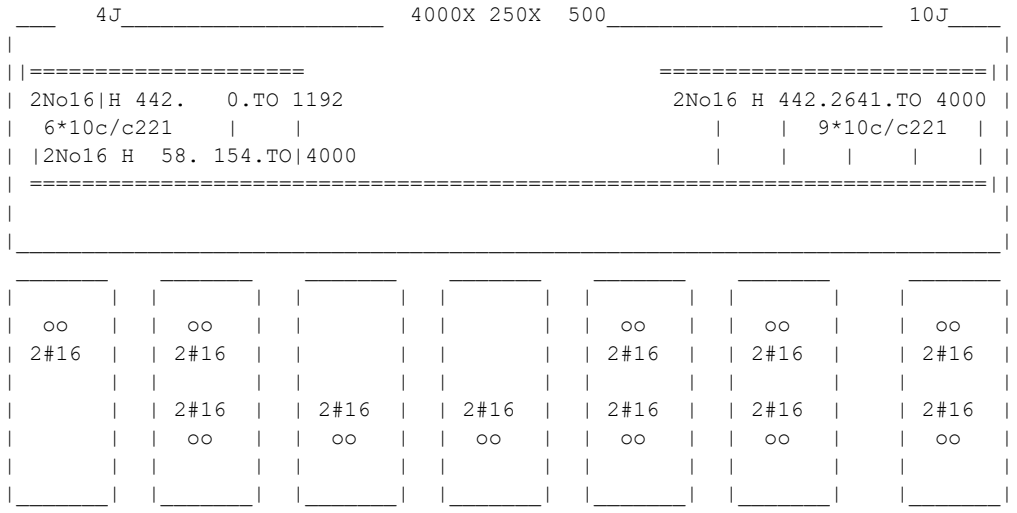
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 264.	0./ 42.	0/ 117
333.	0./ 117.	0./ 19.	0/ 117
667.	30./ 13.	5./ 2.	113/ 111
1000.	103./ 0.	17./ 0.	117/ 0
1333.	178./ 0.	29./ 0.	117/ 0
1667.	230./ 0.	37./ 0.	117/ 0
2000.	259./ 0.	41./ 0.	117/ 0
2333.	263./ 0.	42./ 0.	117/ 0
2667.	243./ 0.	39./ 0.	117/ 0
3000.	199./ 0.	32./ 0.	117/ 0
3333.	132./ 72.	21./ 12.	117/ 110
3667.	42./ 182.	7./ 29.	117/ 102
4000.	0./ 324.	0./ 52.	0/ 102

B E A M N O. 11 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 43.76 KNS Vc= 111.93 KNS Vs= 0.00 KNS
Tu= 1.11 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 1060. MM

AT END SUPPORT - Vu= 56.57 KNS Vc= 110.23 KNS Vs= 0.00 KNS
Tu= 2.76 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 1560. MM



ACI 318-11 BEAM NO. 25 DESIGN RESULTS

LEN - 6500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	447.	5782.	NO	NO

CRITICAL POS MOMENT= 54.43 KN-MET AT 3250.MM, LOAD 117
REQD STEEL= 364.MM2, RHO=0.0027, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 64443.6 cm^4

```
  2        542.     3 - 16MM         0.         2151.        YES  NO
|-------------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     99.06 KN-MET  AT     0.MM, LOAD  109|
|   REQD STEEL=   504.MM2, RHO=0.0037, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./   67. MMS         |
|   REQD. DEVELOPMENT LENGTH =   483. MMS                        |
|-------------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    91121.9 cm^4

  3        542.     3 - 16MM      4079.         6500.        NO   YES
|-------------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     99.74 KN-MET  AT  6500.MM, LOAD  108|
|   REQD STEEL=   507.MM2, RHO=0.0037, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./   67. MMS         |
|   REQD. DEVELOPMENT LENGTH =   483. MMS                        |
|-------------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    91121.9 cm^4

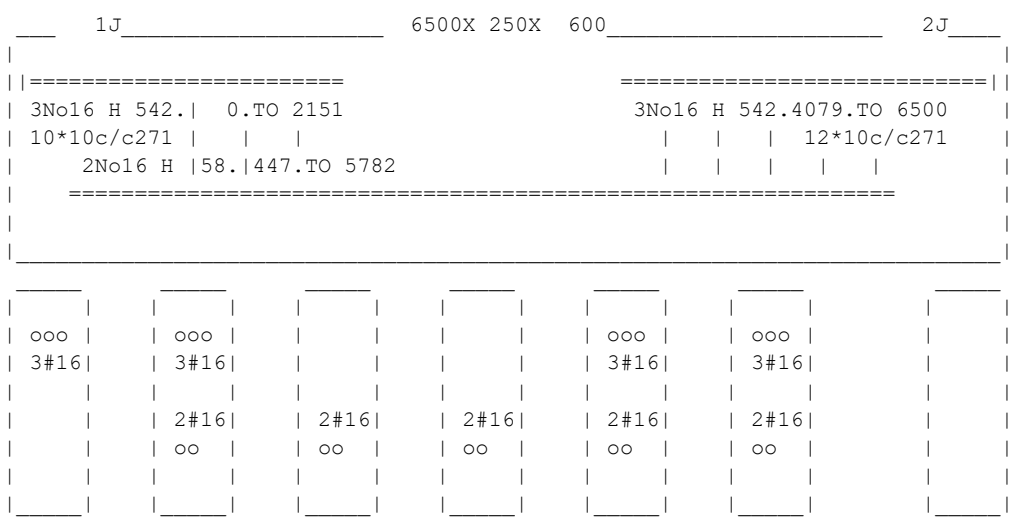

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     510.           0./       99.        0/  109
  542.           0./     299.           0./       59.        0/  109
 1083.          75./     125.          15./       25.      113/  109
 1625.         158./       7.          31./        1.      108/  114
 2167.         216./       0.          43./        0.      108/    0
 2708.         257./       0.          51./        0.      117/    0
 3250.         277./       0.          54./        0.      117/    0
 3792.         257./       0.          51./        0.      117/    0
 4333.         219./       0.          43./        0.      109/    0
 4875.         162./      11.          32./        2.      109/  113
 5417.          79./     128.          16./       25.      114/  108
 5958.           0./     302.           0./       59.        0/  108
 6500.           0./     514.           0./      100.        0/  108
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B E A M N O. 25 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   57.65 KNS  Vc=  136.01 KNS  Vs=    0.00 KNS
 Tu=    0.08 KN-MET  Tc=    5.1 KN-MET  Ts=    0.0 KN-MET  LOAD   109
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 2210. MM
```

AT END SUPPORT - Vu= 69.59 KNS Vc= 133.29 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2710. MM



ACI 318-11 BEAM NO. 30 DESIGN RESULTS

LEN - 4000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	299.	3867.	NO	NO

CRITICAL POS MOMENT= 22.68 KN-MET AT 2333.MM, LOAD 117
REQD STEEL= 150.MM2, RHO=0.0011, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

Cracked Moment of Inertia Iz at above location = 39023.5 cm^4

2	542.	2 - 16MM	0.	1526.	YES	NO

CRITICAL NEG MOMENT= 54.47 KN-MET AT 0.MM, LOAD 117
REQD STEEL= 364.MM2, RHO=0.0027, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 64443.6 cm^4

3	544.	2 - 12MM	2773.	4000.	NO	YES

CRITICAL NEG MOMENT= 33.80 KN-MET AT 4000.MM, LOAD 111
REQD STEEL= 224.MM2, RHO=0.0016, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

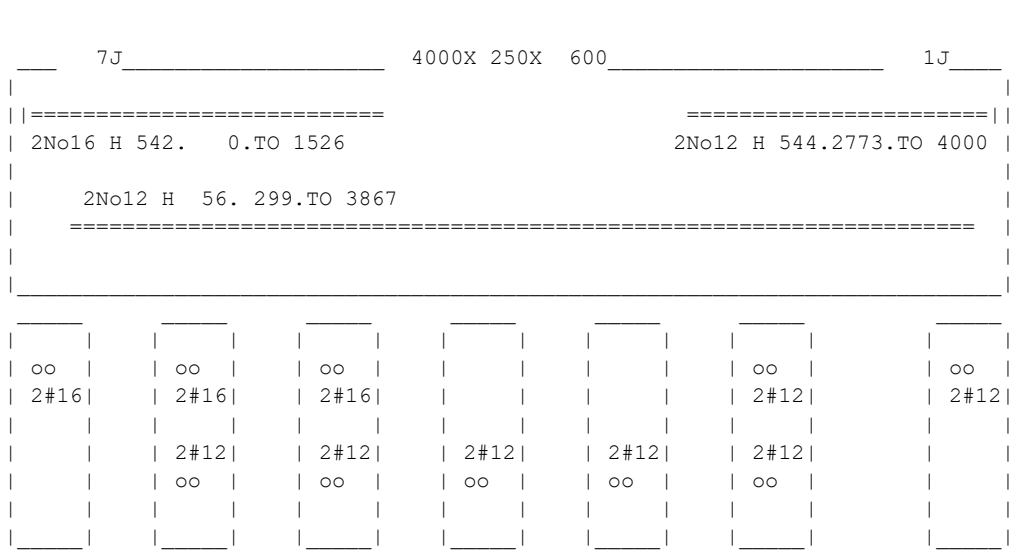
Cracked Moment of Inertia Iz at above location = 39023.5 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 277.	0./ 54.	0/ 117
333.	0./ 177.	0./ 35.	0/ 117
667.	1./ 91.	0./ 18.	116/ 117
1000.	32./ 22.	6./ 4.	111/ 117
1333.	59./ 0.	12./ 0.	111/ 0
1667.	81./ 0.	16./ 0.	102/ 0
2000.	102./ 0.	20./ 0.	117/ 0
2333.	114./ 0.	23./ 0.	117/ 0
2667.	112./ 0.	22./ 0.	117/ 0
3000.	96./ 0.	19./ 0.	117/ 116
3333.	65./ 39.	13./ 8.	117/ 111
3667.	20./ 99.	4./ 20.	117/ 111
4000.	0./ 171.	0./ 34.	0/ 111

B E A M N O. 30 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 26.64 KNS Vc= 133.15 KNS Vs= 0.00 KNS
Tu= 1.65 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 35.50 KNS Vc= 133.15 KNS Vs= 0.00 KNS
Tu= 1.65 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 584 DESIGN RESULTS

LEN - 4000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	633.	3867.	NO	NO
	CRITICAL POS MOMENT= 22.61 KN-MET AT 2333.MM, LOAD 117					
	REQD STEEL= 149.MM2, RHO=0.0011, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 39023.5 cm^4					
2	542.	2 - 16MM	0.	1526.	YES	NO
	CRITICAL NEG MOMENT= 54.15 KN-MET AT 0.MM, LOAD 117					
	REQD STEEL= 362.MM2, RHO=0.0027, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 64443.6 cm^4					
3	544.	2 - 12MM	2773.	4000.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     33.66 KN-MET  AT  4000.MM, LOAD  111|
|   REQD STEEL=   223.MM2, RHO=0.0016, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                        |
|--------------------------------------------------------------|

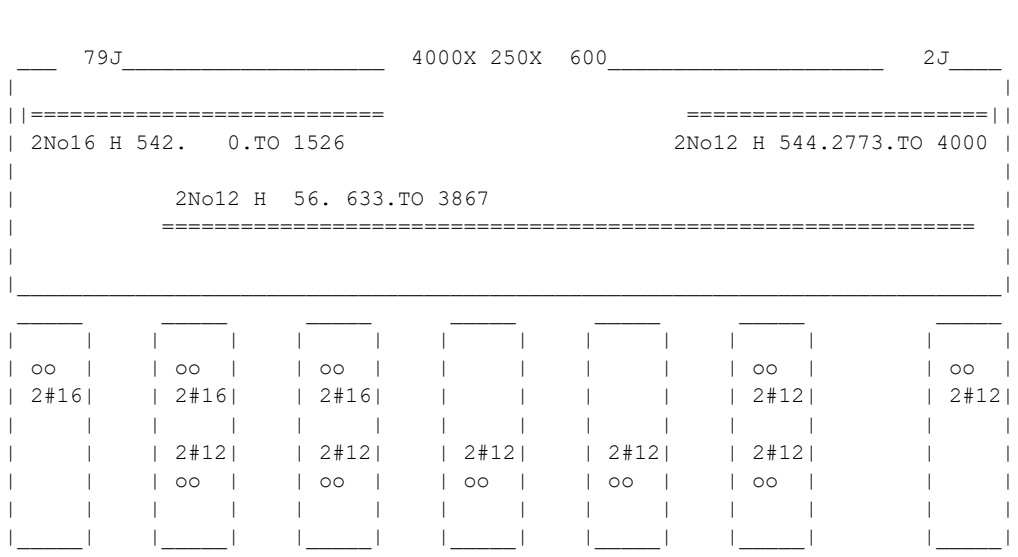
  Cracked Moment of Inertia Iz at above location =    39023.5 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     275.           0./       54.        0/  117
  333.           0./     175.           0./       35.        0/  117
  667.           0./      90.           0./       18.        0/  117
 1000.          32./      21.           6./        4.      111/  117
 1333.          59./       0.          12./        0.      111/    0
 1667.          81./       0.          16./        0.      102/    0
 2000.         102./       0.          20./        0.      117/    0
 2333.         114./       0.          23./        0.      117/    0
 2667.         112./       0.          22./        0.      117/    0
 3000.          95./       0.          19./        0.      117/    0
 3333.          64./      38.          13./        8.      117/  111
 3667.          19./      98.           4./       19.      117/  111
 4000.           0./     170.           0./       34.        0/  111
```

B E A M N O. 584 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   26.63 KNS  Vc=  133.15 KNS  Vs=    0.00 KNS
 Tu=    1.64 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   35.51 KNS  Vc=  133.15 KNS  Vs=    0.00 KNS
 Tu=    1.64 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 587 DESIGN RESULTS

LEN - 4000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	153.	4000.	NO	YES
	CRITICAL POS MOMENT= 41.97 KN-MET AT 2333.MM, LOAD 117					
	REQD STEEL= 345.MM2, RHO=0.0031, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
2	442.	2 - 16MM	0.	1192.	YES	NO
	CRITICAL NEG MOMENT= 42.51 KN-MET AT 0.MM, LOAD 117					
	REQD STEEL= 349.MM2, RHO=0.0032, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
3	442.	2 - 16MM	2641.	4000.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     51.15 KN-MET  AT  4000.MM, LOAD  102|
|   REQD STEEL=   394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4

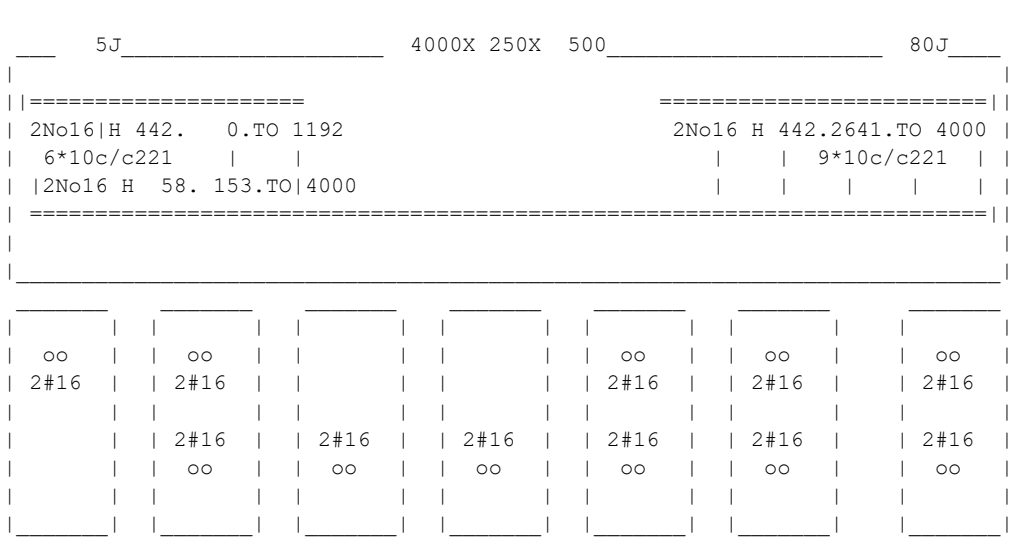

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     267.           0./       43.        0/  117
  333.           0./     119.           0./       19.        0/  117
  667.          23./      13.           4./        2.      115/  111
 1000.         102./       0.          16./        0.      117/    0
 1333.         178./       0.          28./        0.      117/    0
 1667.         230./       0.          37./        0.      117/    0
 2000.         259./       0.          41./        0.      117/    0
 2333.         263./       0.          42./        0.      117/    0
 2667.         244./       0.          39./        0.      117/    0
 3000.         200./       0.          32./        0.      117/    0
 3333.         133./      74.          21./       12.      117/  110
 3667.          43./     180.           7./       29.      117/  102
 4000.           0./     322.           0./       51.        0/  102
```

B E A M N O. 587 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   43.93 KNS  Vc=  112.15 KNS  Vs=    0.00 KNS
Tu=    1.11 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1060. MM

AT END   SUPPORT - Vu=   56.56 KNS  Vc=  110.25 KNS  Vs=    0.00 KNS
Tu=    2.76 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1560. MM
```



ACI 318-11 BEAM NO. 590 DESIGN RESULTS

LEN - 6500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	0.	6500.	YES	YES
	CRITICAL POS MOMENT= 25.33 KN-MET AT 3250.MM, LOAD 117					
	REQD STEEL= 206.MM2, RHO=0.0019, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 142. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 25447.0 cm^4					
2	444.	3 - 12MM	0.	2561.	YES	NO
	CRITICAL NEG MOMENT= 30.36 KN-MET AT 0.MM, LOAD 109					
	REQD STEEL= 247.MM2, RHO=0.0022, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					
3	444.	3 - 12MM	3668.	6500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     28.53 KN-MET  AT  6500.MM, LOAD  108|
|   REQD STEEL=   232.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./   71. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                        |
|--------------------------------------------------------------|

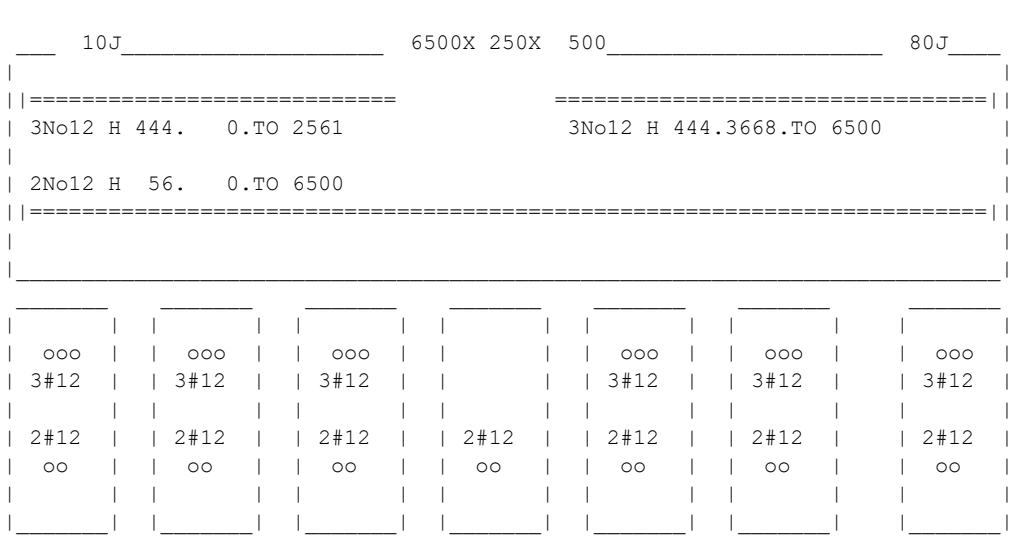
  Cracked Moment of Inertia Iz at above location =    36333.1 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.          66./     189.          11./       30.      113/  109
  542.          74./     134.          12./       22.      113/  109
 1083.          97./      86.          16./       14.      117/  109
 1625.         123./      49.          20./        8.      117/  114
 2167.         142./      18.          23./        3.      117/  114
 2708.         154./       0.          25./        0.      117/    0
 3250.         158./       0.          25./        0.      117/    0
 3792.         154./       0.          25./        0.      117/    0
 4333.         142./       7.          23./        1.      117/  113
 4875.         124./      38.          20./        6.      117/  113
 5417.          97./      75.          16./       12.      117/  108
 5958.          65./     123.          10./       20.      114/  108
 6500.          57./     178.           9./       29.      114/  108
```

B E A M N O. 590 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    9.55 KNS  Vc=  115.06 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   11.61 KNS  Vc=  106.39 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 596 DESIGN RESULTS

LEN - 6500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	0.	6500.	YES	YES

CRITICAL POS MOMENT= 25.82 KN-MET AT 3250.MM, LOAD 117
REQD STEEL= 210.MM2, RHO=0.0019, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 142. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

Cracked Moment of Inertia Iz at above location = 25447.0 cm^4

2	444.	3 - 12MM	0.	2561.	YES	NO

CRITICAL NEG MOMENT= 35.60 KN-MET AT 0.MM, LOAD 109
REQD STEEL= 290.MM2, RHO=0.0026, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

Cracked Moment of Inertia Iz at above location = 36333.1 cm^4

3	444.	3 - 12MM	3668.	6500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     32.74 KN-MET  AT  6500.MM, LOAD  108|
|   REQD STEEL=   267.MM2, RHO=0.0024, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./   71. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                        |
|--------------------------------------------------------------|

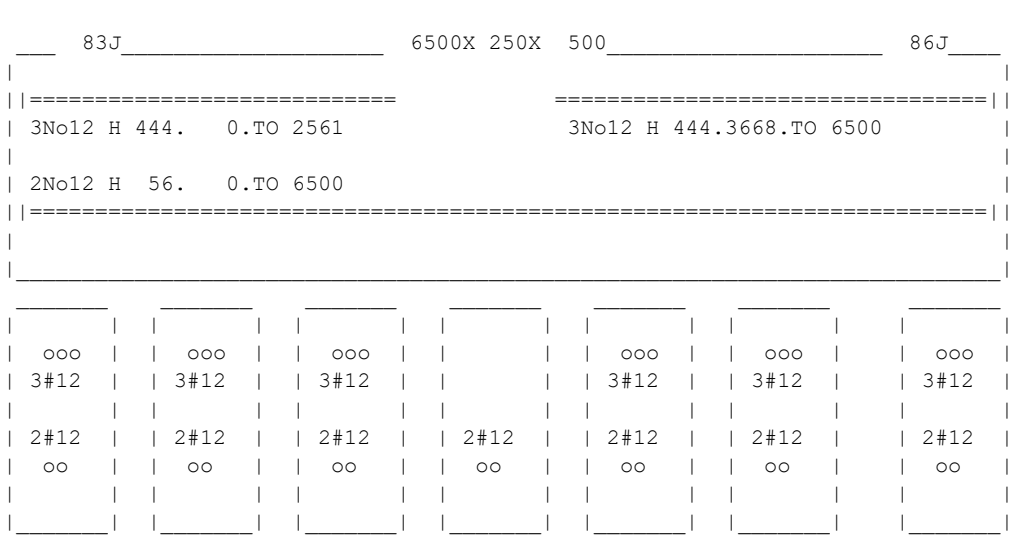
  Cracked Moment of Inertia Iz at above location =    36333.1 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.          97./     223.          16./       36.      113/  109
  542.         100./     161.          16./       26.      113/  109
 1083.          99./     107.          16./       17.      117/  109
 1625.         126./      65.          20./       10.      117/  114
 2167.         145./      28.          23./        4.      117/  114
 2708.         157./       0.          25./        0.      117/    0
 3250.         161./       0.          26./        0.      117/    0
 3792.         157./       0.          25./        0.      117/    0
 4333.         146./      15.          23./        2.      117/  113
 4875.         127./      51.          20./        8.      117/  113
 5417.         101./      91.          16./       15.      117/  113
 5958.          95./     144.          15./       23.      114/  108
 6500.          93./     204.          15./       33.      114/  108


      B E A M  N O.  596 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=    9.56 KNS  Vc=  122.92 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   11.60 KNS  Vc=  106.72 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 601 DESIGN RESULTS

LEN - 6500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	3 - 16MM	0.	6282.	YES	NO

CRITICAL POS MOMENT= 82.93 KN-MET AT 3250.MM, LOAD 117
REQD STEEL= 483.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS
REQD. DEVELOPMENT LENGTH = 483. MMS

Cracked Moment of Inertia Iz at above location = 91121.9 cm^4

2	540.	3 - 20MM	0.	2383.	YES	NO

CRITICAL NEG MOMENT= 166.84 KN-MET AT 0.MM, LOAD 109
REQD STEEL= 869.MM2, RHO=0.0064, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 65. MMS
REQD. DEVELOPMENT LENGTH = 777. MMS

Cracked Moment of Inertia Iz at above location = 130398.8 cm^4

3	540.	3 - 20MM	3846.	6500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=    163.54 KN-MET  AT  6500.MM, LOAD  108|
|   REQD STEEL=   851.MM2, RHO=0.0063, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   45./   65. MMS         |
|   REQD. DEVELOPMENT LENGTH =   777. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =   130398.8 cm^4

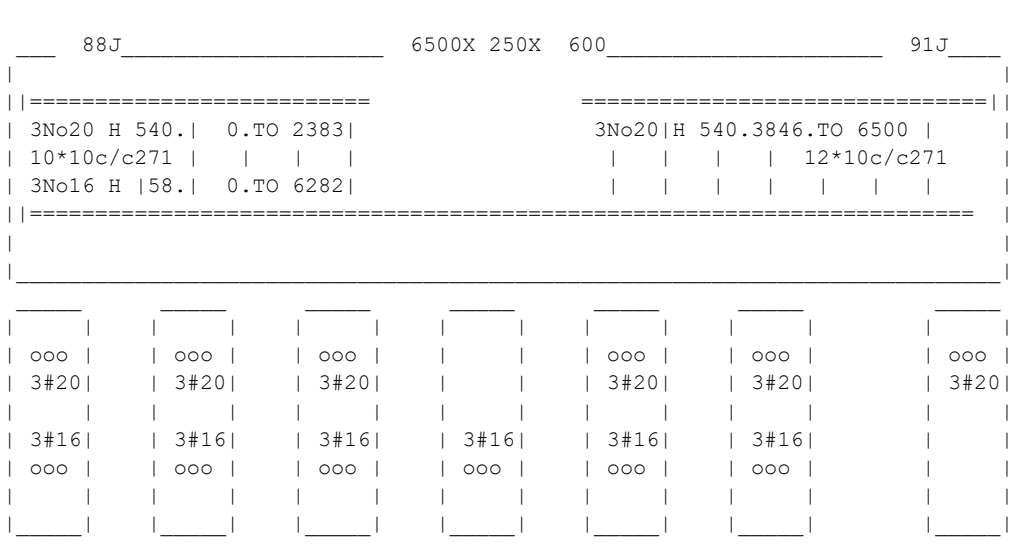

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )         (SQ.  MM )                (KNS-MET )

    0.           0./     876.           0./      167.        0/  109
  542.          24./     527.           5./      102.      113/  109
 1083.         149./     241.          29./       48.      113/  109
 1625.         269./      46.          53./        9.      108/  114
 2167.         351./       0.          69./        0.      108/    0
 2708.         394./       0.          77./        0.      117/    0
 3250.         425./       0.          83./        0.      117/    0
 3792.         395./       0.          77./        0.      117/    0
 4333.         362./       0.          71./        0.      109/    0
 4875.         283./      41.          56./        8.      109/  113
 5417.         165./     231.          33./       46.      114/  108
 5958.          44./     513.           9./      100.      114/  108
 6500.           0./     858.           0./      164.        0/  108
```

B E A M N O. 601 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   87.73 KNS  Vc=  198.96 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 2210. MM

AT END   SUPPORT - Vu=  107.49 KNS  Vc=  135.50 KNS  Vs=    7.82 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 2710. MM
```



ACI 318-11 BEAM NO. 602 DESIGN RESULTS

LEN - 4750. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	0.	4750.	YES	YES

CRITICAL POS MOMENT= 37.47 KN-MET AT 4354.MM, LOAD 117
REQD STEEL= 306.MM2, RHO=0.0028, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

Cracked Moment of Inertia Iz at above location = 36333.1 cm^4

2	444.	3 - 12MM	0.	2769.	YES	NO

CRITICAL NEG MOMENT= 40.50 KN-MET AT 0.MM, LOAD 109
REQD STEEL= 331.MM2, RHO=0.0030, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

Cracked Moment of Inertia Iz at above location = 36333.1 cm^4

3	444.	2 - 12MM	3762.	4750.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      4.71 KN-MET  AT  4750.MM, LOAD  113|
|   REQD STEEL=    38.MM2, RHO=0.0003, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./  142. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                       |
|--------------------------------------------------------------|

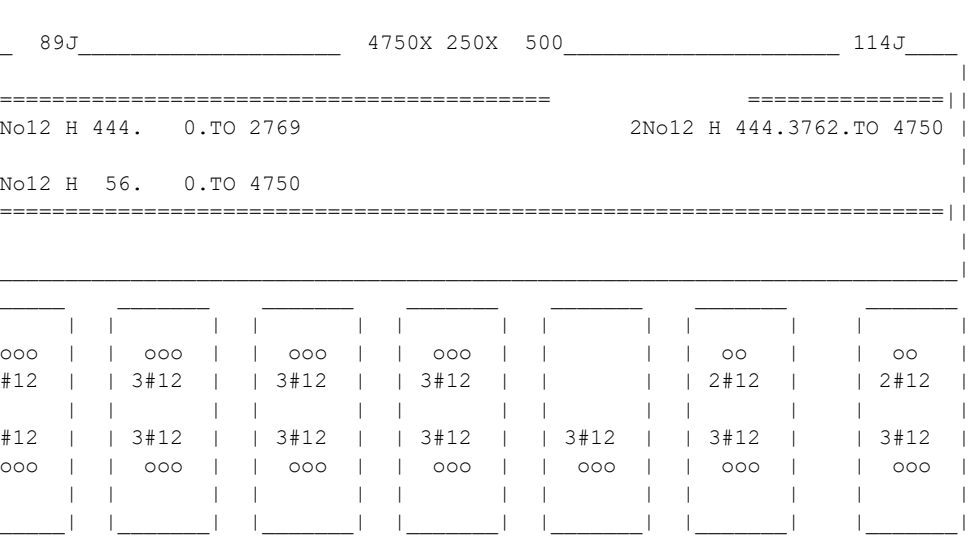
  Cracked Moment of Inertia Iz at above location =    25447.0 cm^4
```

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	75./	254.	12./	40.	113/	109
396.	80./	201.	13./	32.	113/	109
792.	83./	151.	13./	24.	113/	109
1188.	90./	106.	15./	17.	117/	109
1583.	122./	64.	20./	10.	117/	109
1979.	150./	35.	24./	6.	117/	114
2375.	174./	9.	28./	1.	117/	114
2771.	195./	0.	31./	0.	117/	0
3167.	211./	0.	34./	0.	117/	0
3563.	223./	0.	36./	0.	117/	0
3958.	231./	0.	37./	0.	117/	0
4354.	234./	6.	37./	1.	117/	113
4750.	234./	29.	37./	5.	117/	113

B E A M N O. 602 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   10.87 KNS  Vc=  115.64 KNS  Vs=    0.00 KNS
 Tu=    0.25 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    3.06 KNS  Vc=  104.44 KNS  Vs=    0.00 KNS
 Tu=    0.25 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 608 DESIGN RESULTS

LEN - 4750. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	4750.	YES	YES
	CRITICAL POS MOMENT= 55.21 KN-MET AT 3167.MM, LOAD 117					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
2	442.	2 - 16MM	0.	2901.	YES	NO
	CRITICAL NEG MOMENT= 52.54 KN-MET AT 0.MM, LOAD 109					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
3	444.	2 - 12MM	3762.	4750.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      4.16 KN-MET  AT  4750.MM, LOAD  113|
|   REQD STEEL=    34.MM2, RHO=0.0003, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./  142. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                       |
|--------------------------------------------------------------|

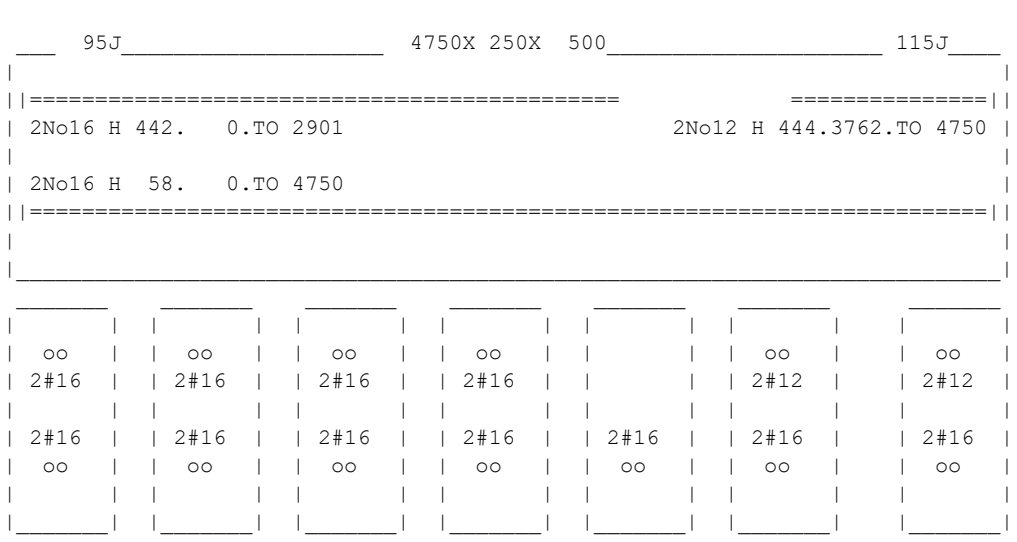
  Cracked Moment of Inertia Iz at above location =    25447.0 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.         107./     331.          17./       53.      113/  109
  396.         110./     257.          18./       41.      113/  109
  792.         111./     187.          18./       30.      113/  109
 1188.         108./     122.          17./       20.      113/  109
 1583.         148./      81.          24./       13.      117/  114
 1979.         204./      49.          33./        8.      117/  114
 2375.         256./      19.          41./        3.      117/  114
 2771.         304./       0.          48./        0.      117/    0
 3167.         348./       0.          55./        0.      117/    0
 3563.         337./       0.          53./        0.      117/    0
 3958.         308./       0.          49./        0.      117/    0
 4354.         275./       1.          44./        0.      117/  113
 4750.         238./      26.          38./        4.      117/  113
```

B E A M N O. 608 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   10.88 KNS  Vc=  135.28 KNS  Vs=    0.00 KNS
 Tu=    0.43 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    3.05 KNS  Vc=  103.95 KNS  Vs=    0.00 KNS
 Tu=    0.43 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 614 DESIGN RESULTS

LEN - 6500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	6500.	YES	YES

CRITICAL POS MOMENT= 51.40 KN-MET AT 3250.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	3234.	YES	NO

CRITICAL NEG MOMENT= 57.00 KN-MET AT 0.MM, LOAD 109
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

3	442.	2 - 16MM	3537.	6500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     50.60 KN-MET  AT  6500.MM, LOAD  108|
|   REQD STEEL=   394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

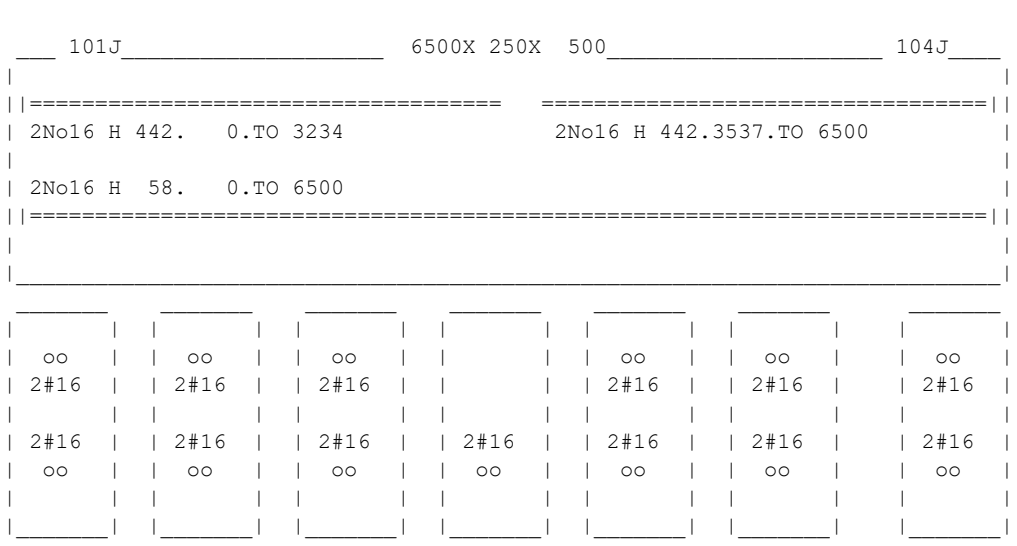
  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.         171./     360.          27./       57.      113/  109
  542.         163./     260.          26./       41.      113/  109
 1083.         150./     167.          24./       27.      113/  109
 1625.         157./     108.          25./       17.      108/  114
 2167.         219./      60.          35./       10.      117/  114
 2708.         275./      17.          44./        3.      117/  114
 3250.         324./       0.          51./        0.      117/    0
 3792.         276./       0.          44./        0.      117/    0
 4333.         221./      28.          35./        5.      117/  113
 4875.         158./      75.          25./       12.      117/  113
 5417.         125./     131.          20./       21.      114/  108
 5958.         140./     221.          22./       35.      114/  108
 6500.         149./     318.          24./       51.      114/  108
```

B E A M N O. 614 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    9.59 KNS  Vc=  190.43 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   11.57 KNS  Vc=  109.68 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 619 DESIGN RESULTS

LEN - 6500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	3 - 16MM	0.	6282.	YES	NO

CRITICAL POS MOMENT= 84.48 KN-MET AT 3250.MM, LOAD 101
REQD STEEL= 483.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS
REQD. DEVELOPMENT LENGTH = 483. MMS

Cracked Moment of Inertia Iz at above location = 91121.9 cm^4

2	540.	3 - 20MM	0.	2383.	YES	NO

CRITICAL NEG MOMENT= 177.72 KN-MET AT 0.MM, LOAD 109
REQD STEEL= 928.MM2, RHO=0.0069, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 65. MMS
REQD. DEVELOPMENT LENGTH = 777. MMS

Cracked Moment of Inertia Iz at above location = 130398.8 cm^4

3	540.	3 - 20MM	3846.	6500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=    177.85 KN-MET  AT  6500.MM, LOAD  108|
|   REQD STEEL=   929.MM2, RHO=0.0069, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   45./   65. MMS         |
|   REQD. DEVELOPMENT LENGTH =   777. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =   130398.8 cm^4

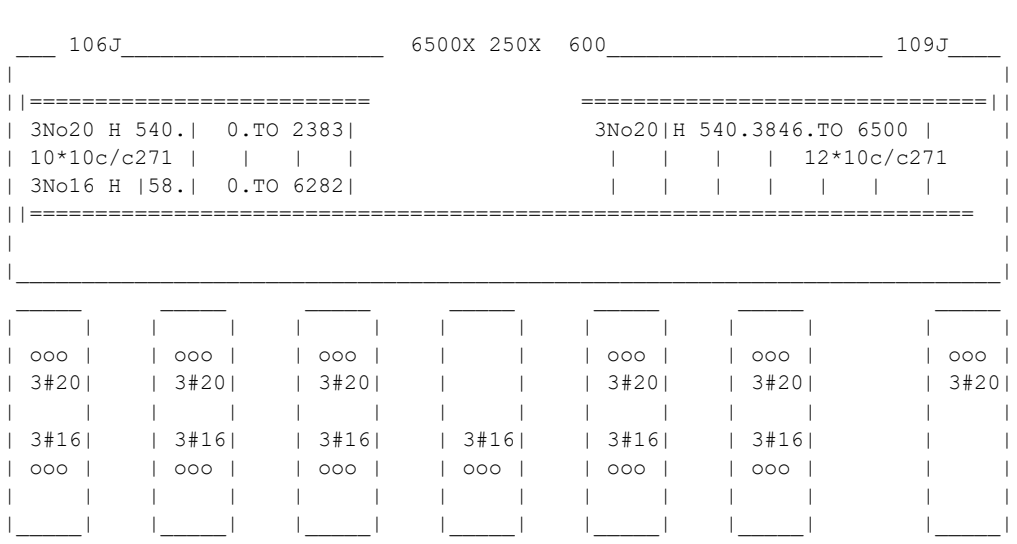

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     937.           0./      178.        0/  109
  542.          91./     573.          18./      111.      113/  109
 1083.         204./     275.          40./       54.      113/  109
 1625.         311./      68.          61./       13.      108/  114
 2167.         381./       0.          74./        0.      108/    0
 2708.         403./       0.          79./        0.      101/    0
 3250.         433./       0.          84./        0.      101/    0
 3792.         412./       0.          80./        0.      109/    0
 4333.         396./       0.          77./        0.      109/    0
 4875.         328./      74.          64./       15.      109/  113
 5417.         222./     279.          44./       55.      114/  108
 5958.         111./     576.          22./      111.      114/  108
 6500.           0./     937.           0./      178.        0/  108
```

B E A M N O. 619 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   86.16 KNS  Vc=  144.31 KNS  Vs=    0.00 KNS
Tu=    0.04 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD   104
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 2210. MM

AT END   SUPPORT - Vu=  107.52 KNS  Vc=  136.48 KNS  Vs=    6.88 KNS
Tu=    0.00 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 2710. MM
```



ACI 318-11 BEAM NO. 620 DESIGN RESULTS

LEN - 6500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	6500.	YES	YES

CRITICAL POS MOMENT= 47.70 KN-MET AT 3250.MM, LOAD 117
REQD STEEL= 393.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	444.	4 - 12MM	0.	3102.	YES	NO

CRITICAL NEG MOMENT= 66.01 KN-MET AT 0.MM, LOAD 109
REQD STEEL= 410.MM2, RHO=0.0037, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 47. MMS
REQD. DEVELOPMENT LENGTH = 385. MMS

Cracked Moment of Inertia Iz at above location = 46484.9 cm^4

3	442.	2 - 16MM	2995.	6500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     61.56 KN-MET  AT  6500.MM, LOAD  108|
|   REQD STEEL=   394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

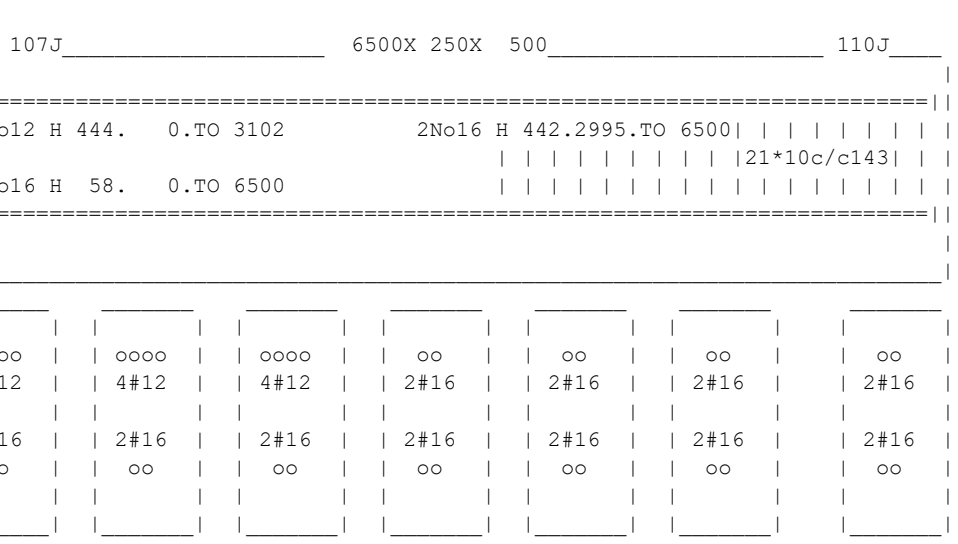
  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.         197./     418.          32./       66.      113/  109
  542.         182./     310.          29./       49.      113/  109
 1083.         162./     210.          26./       34.      113/  109
 1625.         155./     137.          25./       22.      108/  114
 2167.         196./      82.          31./       13.      117/  114
 2708.         252./      32.          40./        5.      117/  114
 3250.         300./       0.          48./        0.      117/    0
 3792.         252./      11.          40./        2.      117/  113
 4333.         197./      60.          32./       10.      117/  113
 4875.         139./     114.          22./       18.      109/  113
 5417.         146./     185.          24./       30.      114/  108
 5958.         168./     283.          27./       45.      114/  108
 6500.         184./     389.          30./       62.      114/  108
```

B E A M N O. 620 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    9.56 KNS  Vc=  113.40 KNS  Vs=    0.00 KNS
 Tu=    3.23 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   11.60 KNS  Vc=  107.48 KNS  Vs=    0.00 KNS
 Tu=    3.93 KN-MET  Tc=    3.9 KN-MET  Ts=    5.2 KN-MET  LOAD   101
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2810. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.27 SQ.CM.
```



ACI 318-11 BEAM NO. 621 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	167.	4146.	NO	NO
	CRITICAL POS MOMENT= 49.40 KN-MET AT 2250.MM, LOAD 117					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
2	442.	2 - 16MM	0.	1276.	YES	NO
	CRITICAL NEG MOMENT= 55.36 KN-MET AT 0.MM, LOAD 102					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
3	442.	2 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     59.66 KN-MET  AT  4500.MM, LOAD  102|
|   REQD STEEL=   394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4

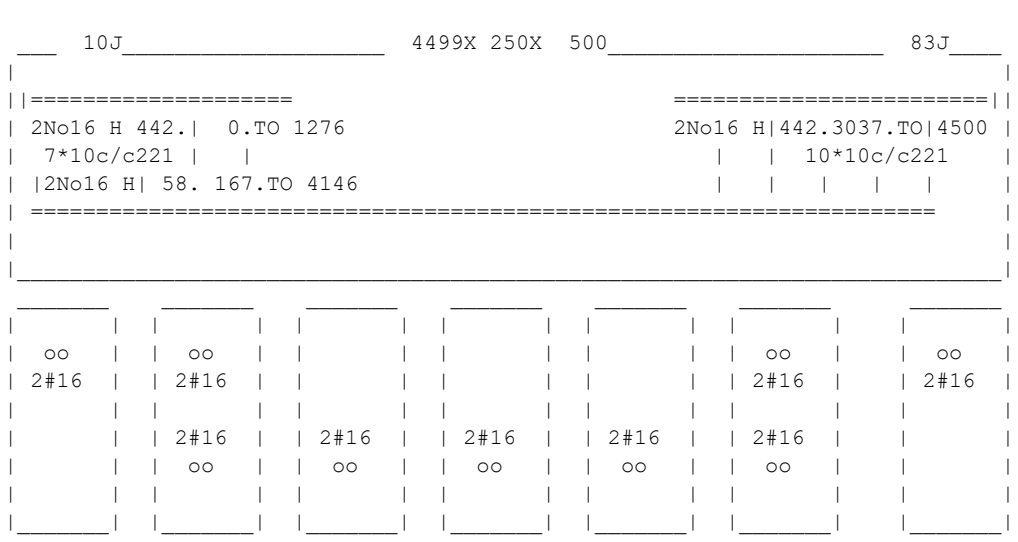

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     349.           0./       55.        0/  102
  375.           0./     183.           0./       29.        0/  102
  750.          19./      60.           3./       10.      117/  111
 1125.         136./       0.          22./        0.      117/    0
 1500.         224./       0.          36./        0.      117/    0
 1875.         283./       0.          45./        0.      117/    0
 2250.         311./       0.          49./        0.      117/    0
 2625.         308./       0.          49./        0.      117/    0
 3000.         275./       0.          44./        0.      117/    0
 3375.         211./       0.          34./        0.      117/    0
 3750.         119./      72.          19./       12.      117/  110
 4125.           0./     206.           0./       33.        0/  102
 4500.           0./     377.           0./       60.        0/  102
```

B E A M N O. 621 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   49.96 KNS  Vc=  116.15 KNS  Vs=    0.00 KNS
Tu=    0.05 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1310. MM

AT END   SUPPORT - Vu=   61.13 KNS  Vc=  110.14 KNS  Vs=    0.00 KNS
Tu=    0.09 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1810. MM
```



ACI 318-11 BEAM NO. 622 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	167.	4146.	NO	NO
	CRITICAL POS MOMENT= 49.40 KN-MET AT 2250.MM, LOAD 117					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
2	442.	2 - 16MM	0.	1276.	YES	NO
	CRITICAL NEG MOMENT= 55.66 KN-MET AT 0.MM, LOAD 102					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
3	442.	2 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     59.35 KN-MET  AT  4500.MM, LOAD  102|
|   REQD STEEL=   394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4

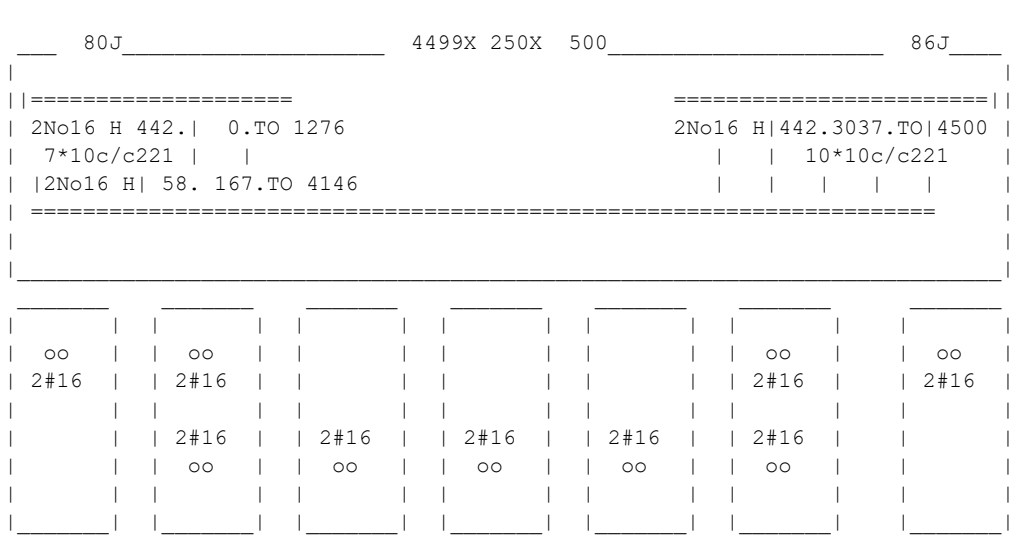

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )         (SQ.  MM )                (KNS-MET )

    0.           0./     351.           0./       56.        0/  102
  375.           0./     185.           0./       30.        0/  102
  750.          18./      60.           3./       10.      117/  111
 1125.         135./       0.          22./        0.      117/    0
 1500.         223./       0.          36./        0.      117/    0
 1875.         282./       0.          45./        0.      117/    0
 2250.         311./       0.          49./        0.      117/    0
 2625.         308./       0.          49./        0.      117/    0
 3000.         276./       0.          44./        0.      117/    0
 3375.         212./       0.          34./        0.      117/    0
 3750.         120./      73.          19./       12.      117/  110
 4125.           0./     204.           0./       33.        0/  102
 4500.           0./     375.           0./       59.        0/  102
```

B E A M N O. 622 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   50.08 KNS  Vc=  116.49 KNS  Vs=    0.00 KNS
Tu=    0.06 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1310. MM

AT END   SUPPORT - Vu=   61.13 KNS  Vc=  110.14 KNS  Vs=    0.00 KNS
Tu=    0.09 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1810. MM
```



ACI 318-11 BEAM NO. 624 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	542.	4500.	NO	YES

CRITICAL POS MOMENT= 50.45 KN-MET AT 2625.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	1276.	YES	NO

CRITICAL NEG MOMENT= 63.70 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 395.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 42068.0 cm^4

3	442.	2 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     44.59 KN-MET  AT  4500.MM, LOAD  108|
|   REQD STEEL=   367.MM2, RHO=0.0033, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4

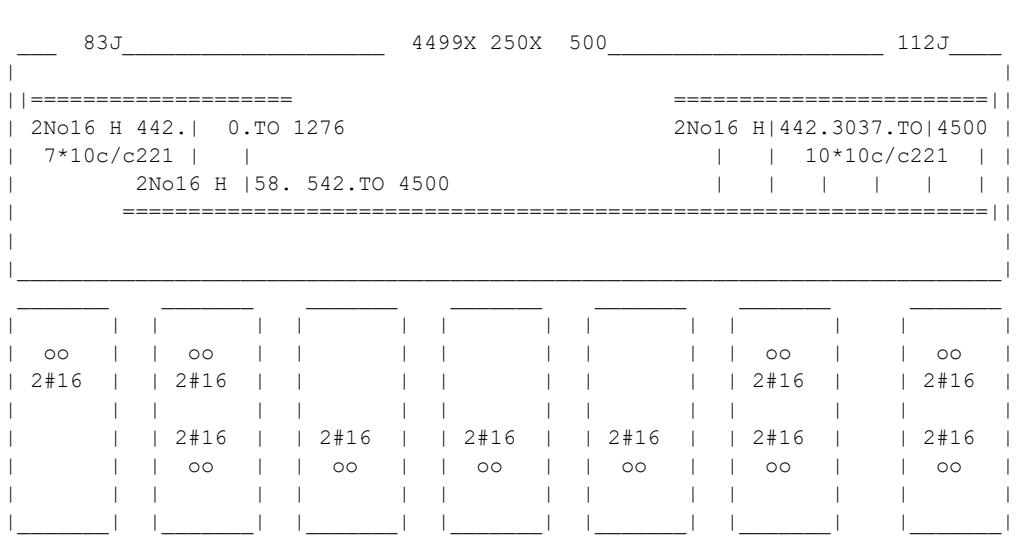

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     403.           0./       64.        0/  102
  375.           0./     223.           0./       36.        0/  102
  750.           0./      88.           0./       14.        0/  111
 1125.         117./       0.          19./        0.      117/    0
 1500.         212./       0.          34./        0.      117/    0
 1875.         277./       0.          44./        0.      117/    0
 2250.         313./       0.          50./        0.      117/    0
 2625.         317./       0.          50./        0.      117/    0
 3000.         291./       0.          46./        0.      117/    0
 3375.         235./       0.          38./        0.      117/    0
 3750.         149./      29.          24./        5.      117/  113
 4125.          34./     139.           5./       22.      117/  108
 4500.           0./     280.           0./       45.        0/  108
```

B E A M N O. 624 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   52.94 KNS  Vc=  122.94 KNS  Vs=    0.00 KNS
Tu=    0.34 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1310. MM

AT END   SUPPORT - Vu=   55.66 KNS  Vc=  117.29 KNS  Vs=    0.00 KNS
Tu=    0.78 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1810. MM
```



ACI 318-11 BEAM NO. 626 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	542.	4500.	NO	YES
	CRITICAL POS MOMENT= 50.36 KN-MET AT 2625.MM, LOAD 117					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
2	442.	2 - 16MM	0.	1276.	YES	NO
	CRITICAL NEG MOMENT= 64.12 KN-MET AT 0.MM, LOAD 102					
	REQD STEEL= 398.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 42068.0 cm^4					
3	442.	2 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     44.99 KN-MET  AT  4500.MM, LOAD  110|
|   REQD STEEL=   370.MM2, RHO=0.0033, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4

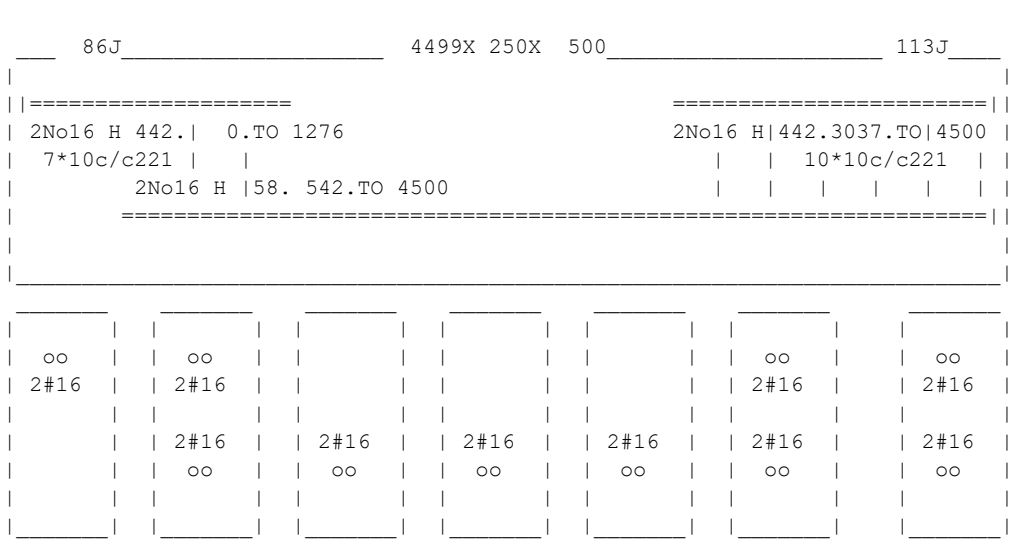

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     406.           0./       64.        0/  102
  375.           0./     225.           0./       36.        0/  102
  750.           0./      89.           0./       14.        0/  111
 1125.         115./       0.          19./        0.      117/    0
 1500.         210./       0.          34./        0.      117/    0
 1875.         276./       0.          44./        0.      117/    0
 2250.         312./       0.          50./        0.      117/    0
 2625.         317./       0.          50./        0.      117/    0
 3000.         291./       0.          46./        0.      117/    0
 3375.         235./       0.          38./        0.      117/    0
 3750.         149./      30.          24./        5.      117/  115
 4125.          34./     141.           6./       23.      117/  110
 4500.           0./     282.           0./       45.        0/  110
```

B E A M N O. 626 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   53.06 KNS  Vc=  124.24 KNS  Vs=    0.00 KNS
Tu=    0.19 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1310. MM

AT END   SUPPORT - Vu=   55.64 KNS  Vc=  117.35 KNS  Vs=    0.00 KNS
Tu=    0.73 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1810. MM
```



ACI 318-11 BEAM NO. 627 DESIGN RESULTS

LEN - 3000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	0.	3000.	YES	YES
	CRITICAL POS MOMENT= 38.46 KN-MET AT 1500.MM, LOAD 117					
	REQD STEEL= 314.MM2, RHO=0.0028, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					
2	444.	3 - 12MM	0.	1144.	YES	NO
	CRITICAL NEG MOMENT= 38.31 KN-MET AT 0.MM, LOAD 117					
	REQD STEEL= 313.MM2, RHO=0.0028, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					
3	442.	2 - 16MM	1599.	3000.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     43.27 KN-MET  AT  3000.MM, LOAD  102|
|   REQD STEEL=   356.MM2, RHO=0.0032, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4

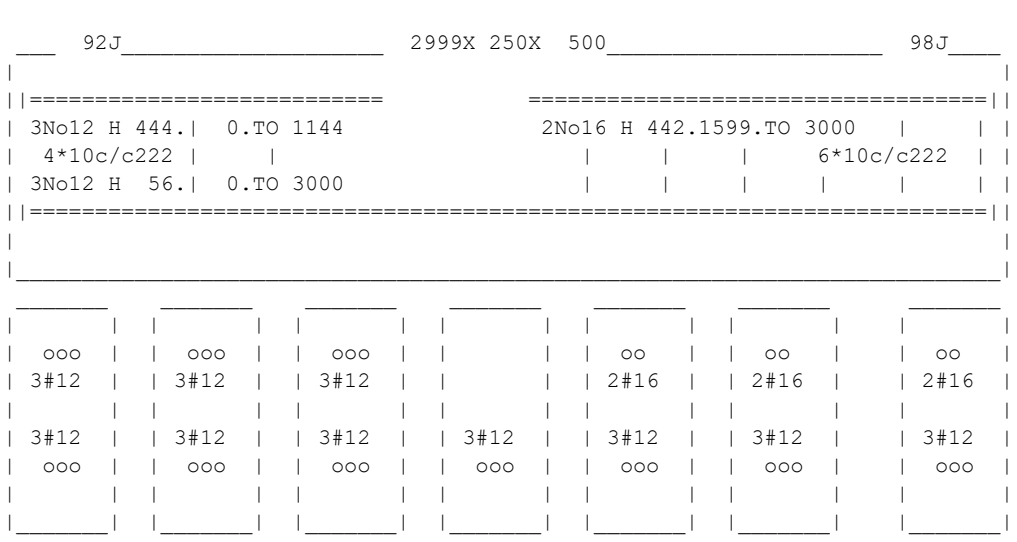

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )         (SQ.  MM )                (KNS-MET )

    0.           0./     240.           0./       38.        0/  117
  250.          20./     125.           3./       20.      115/  117
  500.          50./      48.           8./        8.      110/  116
  750.          84./      16.          14./        3.      110/  116
 1000.         133./       0.          21./        0.      117/    0
 1250.         193./       0.          31./        0.      117/    0
 1500.         241./       0.          38./        0.      117/    0
 1750.         234./       0.          37./        0.      117/    0
 2000.         214./       0.          34./        0.      117/    0
 2250.         181./      24.          29./        4.      117/  115
 2500.         134./      86.          22./       14.      117/  110
 2750.          74./     167.          12./       27.      117/  110
 3000.           2./     271.           0./       43.      117/  102
```

B E A M N O. 627 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   44.71 KNS  Vc=  109.47 KNS  Vs=    0.00 KNS
Tu=    1.16 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR  560. MM

AT END   SUPPORT - Vu=   55.72 KNS  Vc=  112.70 KNS  Vs=    0.00 KNS
Tu=    2.20 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   102
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 1060. MM
```



ACI 318-11 BEAM NO. 628 DESIGN RESULTS

LEN - 3000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	0.	3000.	YES	YES
	CRITICAL POS MOMENT= 28.90 KN-MET AT 2000.MM, LOAD 117					
	REQD STEEL= 235.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					
2	444.	3 - 12MM	0.	1144.	YES	NO
	CRITICAL NEG MOMENT= 33.51 KN-MET AT 0.MM, LOAD 117					
	REQD STEEL= 273.MM2, RHO=0.0025, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					
3	444.	3 - 12MM	1731.	3000.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     36.65 KN-MET  AT  3000.MM, LOAD  108|
|   REQD STEEL=   299.MM2, RHO=0.0027, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./   71. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                        |
|--------------------------------------------------------------|

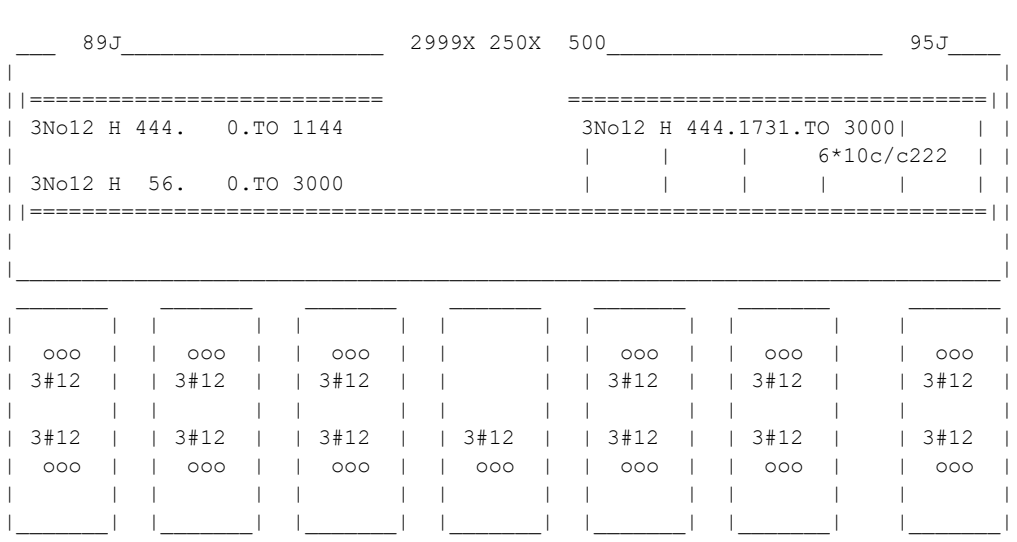
  Cracked Moment of Inertia Iz at above location =    36333.1 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           2./     209.           0./       34.      113/  117
  250.          32./     113.           5./       18.      113/  117
  500.          57./      54.           9./        9.      108/  116
  750.          80./      21.          13./        3.      108/  116
 1000.          91./       0.          15./        0.      108/    0
 1250.         133./       0.          21./        0.      117/    0
 1500.         162./       0.          26./        0.      117/    0
 1750.         178./       0.          29./        0.      117/    0
 2000.         180./       0.          29./        0.      117/    0
 2250.         169./      25.          27./        4.      117/  110
 2500.         144./      79.          23./       13.      117/  108
 2750.         107./     148.          17./       24.      117/  108
 3000.          56./     229.           9./       37.      117/  108
```

B E A M N O. 628 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   13.20 KNS  Vc=  106.56 KNS  Vs=    0.00 KNS
 Tu=    0.61 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   41.97 KNS  Vc=  109.24 KNS  Vs=    0.00 KNS
 Tu=    2.28 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   102
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 1060. MM
```



ACI 318-11 BEAM NO. 629 DESIGN RESULTS

LEN - 2250. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	261.	2250.	NO	YES

CRITICAL POS MOMENT= 51.00 KN-MET AT 2250.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	1276.	YES	NO

CRITICAL NEG MOMENT= 51.79 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

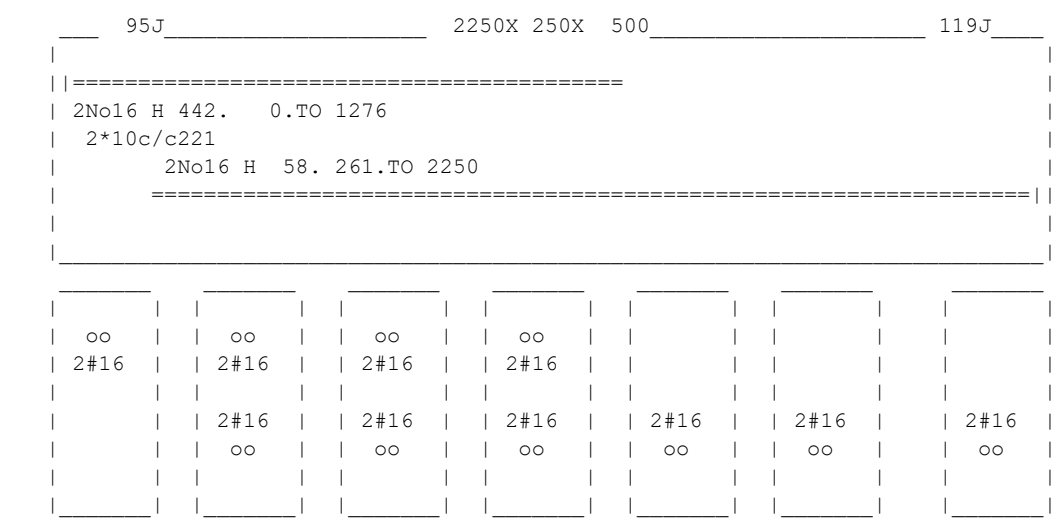
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 326.	0./ 52.	0/ 102
188.	0./ 240.	0./ 38.	0/ 102
375.	0./ 163.	0./ 26.	0/ 102
563.	0./ 98.	0./ 16.	0/ 111
750.	33./ 43.	5./ 7.	117/ 111
938.	94./ 0.	15./ 0.	117/ 0
1125.	149./ 0.	24./ 0.	117/ 0
1313.	196./ 0.	31./ 0.	117/ 0
1500.	236./ 0.	38./ 0.	117/ 0
1688.	269./ 0.	43./ 0.	117/ 0
1875.	294./ 0.	47./ 0.	117/ 0
2063.	311./ 0.	49./ 0.	117/ 0
2250.	321./ 0.	51./ 0.	117/ 0

B E A M N O. 629 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 49.55 KNS Vc= 113.58 KNS Vs= 0.00 KNS
Tu= 0.23 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 185. MM

AT END SUPPORT - Vu= 12.76 KNS Vc= 104.64 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 630 DESIGN RESULTS

LEN - 2250. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	448.	2250.	NO	YES

CRITICAL POS MOMENT= 55.58 KN-MET AT 2250.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	444.	4 - 12MM	0.	1332.	YES	NO

CRITICAL NEG MOMENT= 67.37 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 418.MM2, RHO=0.0038, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 47. MMS
REQD. DEVELOPMENT LENGTH = 385. MMS

Cracked Moment of Inertia Iz at above location = 46484.9 cm^4

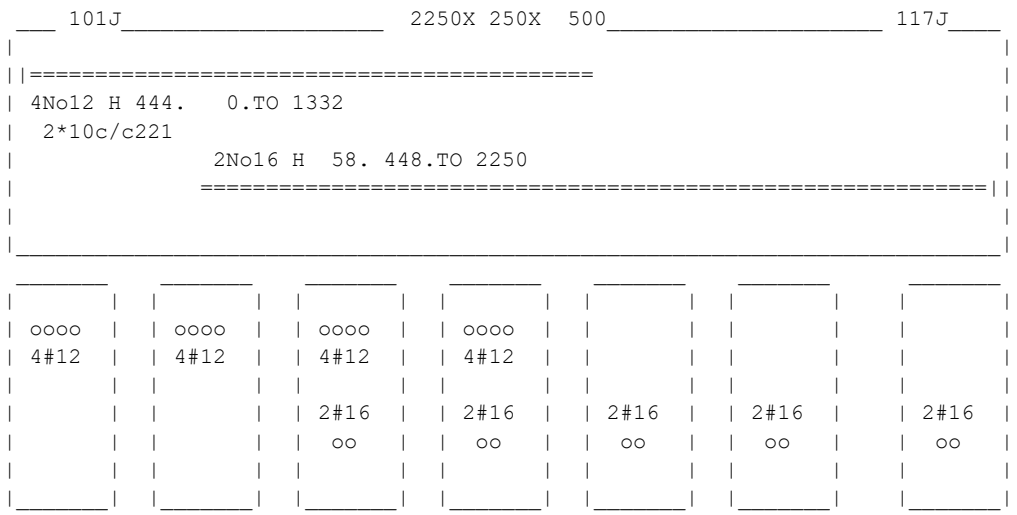
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 427.	0./ 67.	0/ 102
188.	0./ 328.	0./ 52.	0/ 102
375.	0./ 239.	0./ 38.	0/ 102
563.	0./ 158.	0./ 25.	0/ 102
750.	0./ 86.	0./ 14.	0/ 111
938.	36./ 31.	6./ 5.	117/ 111
1125.	103./ 0.	17./ 0.	117/ 0
1313.	162./ 0.	26./ 0.	117/ 0
1500.	215./ 0.	34./ 0.	117/ 0
1688.	260./ 0.	41./ 0.	117/ 0
1875.	298./ 0.	47./ 0.	117/ 0
2063.	328./ 0.	52./ 0.	117/ 0
2250.	350./ 0.	56./ 0.	117/ 0

B E A M N O. 630 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 45.31 KNS Vc= 107.75 KNS Vs= 0.00 KNS
Tu= 0.31 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 108
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 185. MM

AT END SUPPORT - Vu= 22.18 KNS Vc= 105.62 KNS Vs= 0.00 KNS
Tu= 0.76 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 631 DESIGN RESULTS

LEN - 2250. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	261.	2250.	NO	YES

CRITICAL POS MOMENT= 50.19 KN-MET AT 2250.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	1463.	YES	NO

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     54.37 KN-MET  AT     0.MM, LOAD  102|
|   REQD STEEL=   394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4

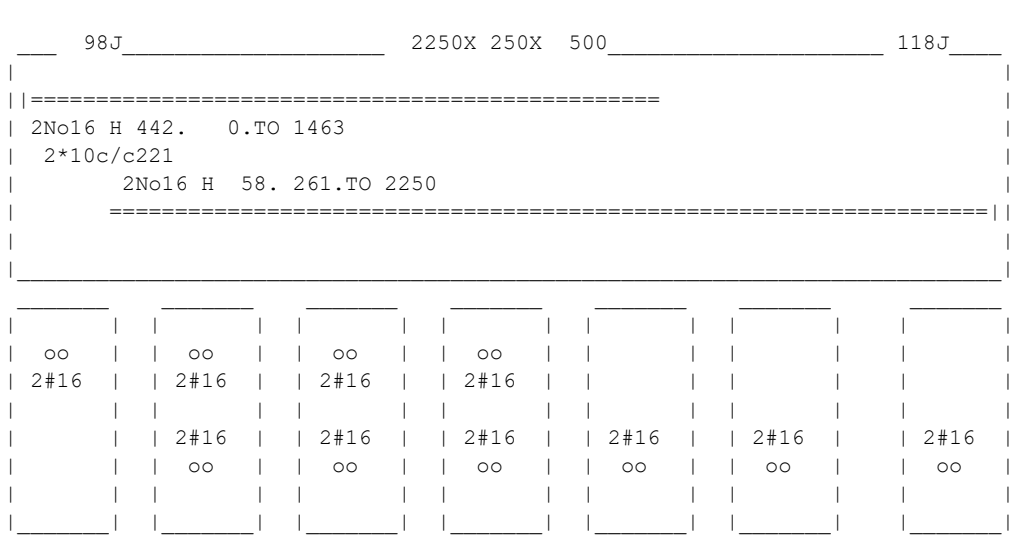

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     343.           0./       54.        0/  102
  188.           0./     256.           0./       41.        0/  102
  375.           0./     178.           0./       29.        0/  102
  563.           0./     108.           0./       17.        0/  102
  750.          23./      50.           4./        8.      117/  111
  938.          85./       1.          14./        0.      117/  111
 1125.         140./       0.          23./        0.      117/    0
 1313.         188./       0.          30./        0.      117/    0
 1500.         229./       0.          37./        0.      117/    0
 1688.         262./       0.          42./        0.      117/    0
 1875.         287./       0.          46./        0.      117/    0
 2063.         305./       0.          49./        0.      117/    0
 2250.         316./       0.          50./        0.      117/    0
```

B E A M N O. 631 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   50.03 KNS  Vc=  114.96 KNS  Vs=    0.00 KNS
 Tu=    0.60 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   117
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR  185. MM

 AT END   SUPPORT - Vu=   12.79 KNS  Vc=  104.65 KNS  Vs=    0.00 KNS
 Tu=    0.23 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 632 DESIGN RESULTS

LEN - 2250. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	448.	2250.	NO	YES

CRITICAL POS MOMENT= 55.63 KN-MET AT 2250.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	444.	4 - 12MM	0.	1332.	YES	NO

CRITICAL NEG MOMENT= 66.70 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 414.MM2, RHO=0.0037, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 47. MMS
REQD. DEVELOPMENT LENGTH = 385. MMS

Cracked Moment of Inertia Iz at above location = 46484.9 cm^4

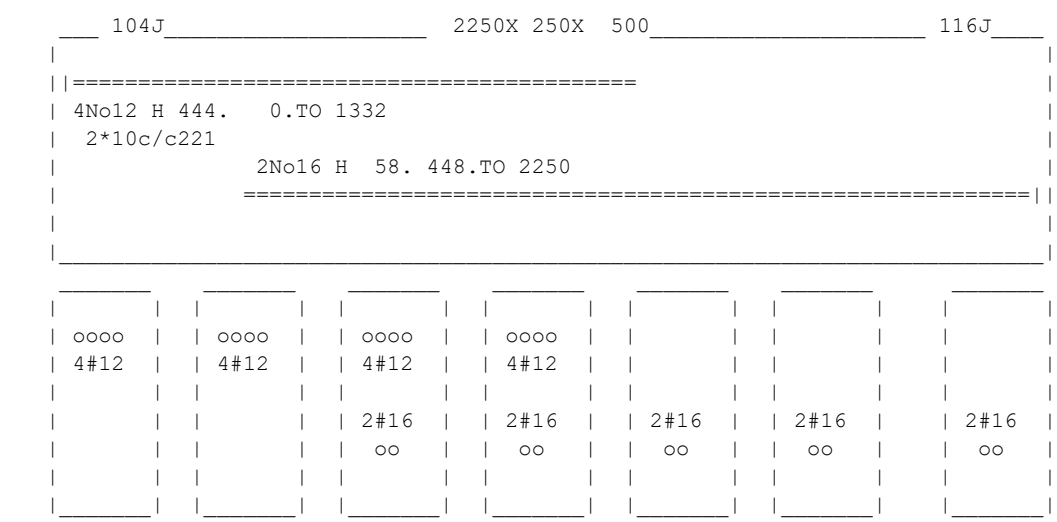
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 423.	0./ 67.	0/ 102
188.	0./ 324.	0./ 52.	0/ 102
375.	0./ 235.	0./ 38.	0/ 102
563.	0./ 155.	0./ 25.	0/ 101
750.	0./ 84.	0./ 14.	0/ 106
938.	38./ 29.	6./ 5.	117/ 111
1125.	104./ 0.	17./ 0.	117/ 0
1313.	164./ 0.	26./ 0.	117/ 0
1500.	216./ 0.	35./ 0.	117/ 0
1688.	261./ 0.	42./ 0.	117/ 0
1875.	298./ 0.	47./ 0.	117/ 0
2063.	328./ 0.	52./ 0.	117/ 0
2250.	351./ 0.	56./ 0.	117/ 0

B E A M N O. 632 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 46.41 KNS Vc= 107.24 KNS Vs= 0.00 KNS
Tu= 1.35 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 108
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 185. MM

AT END SUPPORT - Vu= 22.18 KNS Vc= 105.62 KNS Vs= 0.00 KNS
Tu= 0.77 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 633 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	367.	3570.	NO	NO
	CRITICAL POS MOMENT= 22.07 KN-MET AT 1875.MM, LOAD 117					
	REQD STEEL= 146.MM2, RHO=0.0011, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 39023.5 cm^4					
2	544.	3 - 12MM	0.	1144.	YES	NO
	CRITICAL NEG MOMENT= 42.71 KN-MET AT 0.MM, LOAD 111					
	REQD STEEL= 284.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 69. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 55974.3 cm^4					
3	542.	2 - 16MM	2662.	4500.	NO	YES
	CRITICAL NEG MOMENT= 57.42 KN-MET AT 4500.MM, LOAD 117					
	REQD STEEL= 384.MM2, RHO=0.0028, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 64443.6 cm^4					

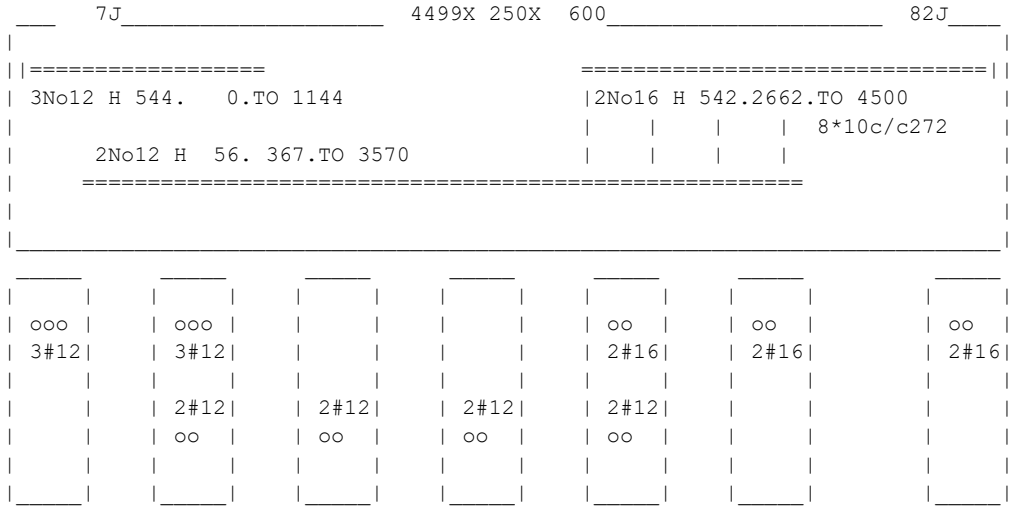
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 217.	0./ 43.	0/ 111
375.	0./ 125.	0./ 25.	0/ 111
750.	10./ 49.	2./ 10.	117/ 111
1125.	62./ 0.	12./ 0.	117/ 0
1500.	96./ 0.	19./ 0.	117/ 0
1875.	111./ 0.	22./ 0.	117/ 0
2250.	111./ 0.	22./ 0.	101/ 0
2625.	101./ 0.	20./ 0.	102/ 0

3000.	75./	0.	15./	0.	111/ 0
3375.	41./	8.	8./	2.	111/ 117
3750.	0./	84.	0./	17.	0/ 117
4125.	0./	178.	0./	35.	0/ 117
4500.	0./	292.	0./	57.	0/ 117

B E A M N O. 633 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 30.86 KNS Vc= 133.37 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 50.12 KNS Vc= 129.62 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 1710. MM



ACI 318-11 BEAM NO. 634 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	362.	3575.	NO	NO

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=     22.60 KN-MET  AT  2250.MM, LOAD  117|
|   REQD STEEL=   149.MM2, RHO=0.0011, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    39023.5 cm^4

  2        544.     3 - 12MM         0.         1519.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     45.07 KN-MET  AT     0.MM, LOAD  102|
|   REQD STEEL=   299.MM2, RHO=0.0022, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./   69. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    55974.3 cm^4

  3        542.     2 - 16MM      3037.         4500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     52.80 KN-MET  AT  4500.MM, LOAD  117|
|   REQD STEEL=   353.MM2, RHO=0.0026, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    64443.6 cm^4

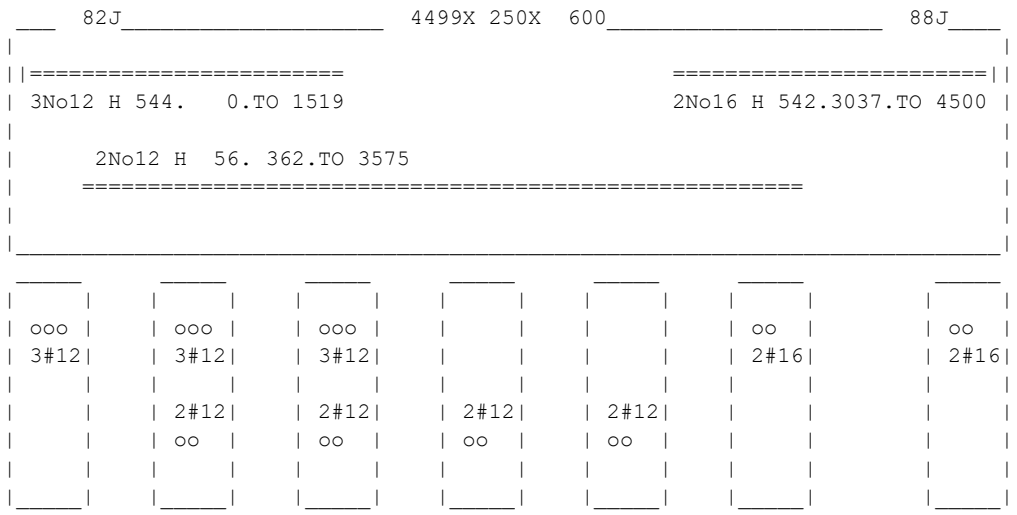

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     229.           0./       45.        0/  102
  375.           0./     133.           0./       26.        0/  111
  750.           3./      57.           1./       11.      117/  111
 1125.          58./       1.          12./        0.      117/  116
 1500.          95./       0.          19./        0.      117/    0
 1875.         114./       0.          23./        0.      117/    0
 2250.         114./       0.          23./        0.      117/    0
 2625.         102./       0.          20./        0.      101/    0
 3000.          76./       0.          15./        0.      102/    0
 3375.          40./       0.           8./        0.      111/    0
 3750.           0./      67.           0./       13.        0/  117
 4125.           0./     158.           0./       31.        0/  117
 4500.           0./     268.           0./       53.        0/  117
```

B E A M N O. 634 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   31.86 KNS  Vc=  132.98 KNS  Vs=    0.00 KNS
 Tu=    2.18 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```

AT END SUPPORT - Vu= 43.13 KNS Vc= 132.98 KNS Vs= 0.00 KNS
Tu= 2.18 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 635 DESIGN RESULTS

LEN - 3000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	244.	2381.	NO	NO

CRITICAL POS MOMENT= 14.85 KN-MET AT 1250.MM, LOAD 117
REQD STEEL= 98.MM2, RHO=0.0007, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

Cracked Moment of Inertia Iz at above location = 39023.5 cm^4

2	544.	2 - 12MM	0.	1144.	YES	NO

CRITICAL NEG MOMENT= 29.13 KN-MET AT 0.MM, LOAD 111
REQD STEEL= 193.MM2, RHO=0.0014, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

Cracked Moment of Inertia Iz at above location = 39023.5 cm^4

3	542.	2 - 16MM	1599.	3000.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     53.49 KN-MET  AT  3000.MM, LOAD  117|
|   REQD STEEL=   358.MM2, RHO=0.0026, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    64443.6 cm^4

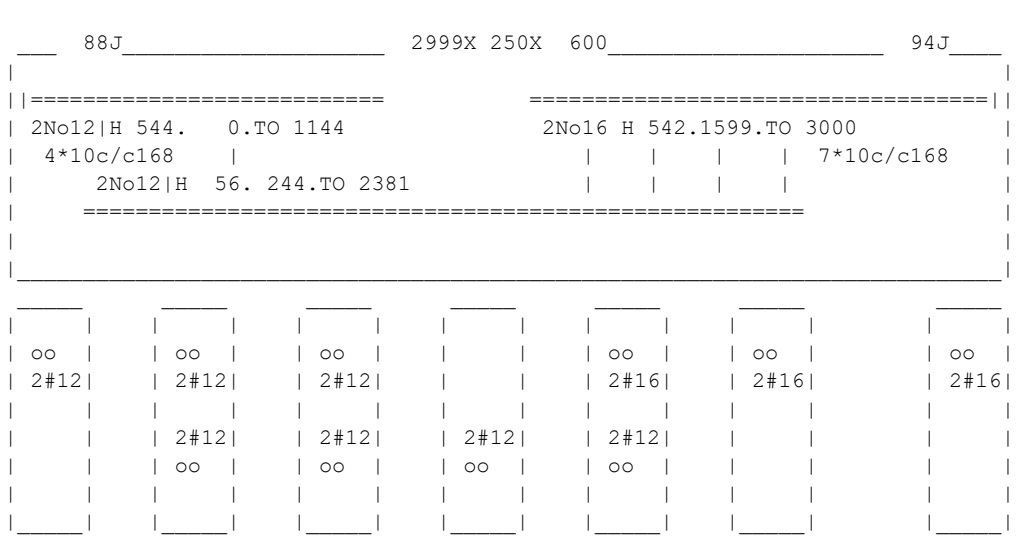

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     147.           0./       29.        0/  111
  250.           0./      86.           0./       17.        0/  111
  500.          35./      35.           7./        7.      117/  111
  750.          61./       1.          12./        0.      117/  116
 1000.          74./       0.          15./        0.      117/    0
 1250.          75./       0.          15./        0.      117/    0
 1500.          65./       0.          13./        0.      101/    0
 1750.          55./       0.          11./        0.      111/    0
 2000.          41./       0.           8./        0.      111/    0
 2250.          18./      47.           4./        9.      116/  117
 2500.           0./     109.           0./       22.        0/  117
 2750.           0./     184.           0./       36.        0/  117
 3000.           0./     272.           0./       53.        0/  117
```

B E A M N O. 635 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    7.83 KNS  Vc=  127.62 KNS  Vs=    0.00 KNS
Tu=    4.90 KN-MET  Tc=    4.9 KN-MET  Ts=    6.5 KN-MET  LOAD   110
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 168. MM  C/C FOR  460. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.53 SQ.CM.

AT END   SUPPORT - Vu=   40.46 KNS  Vc=  129.90 KNS  Vs=    0.00 KNS
Tu=    4.90 KN-MET  Tc=    4.9 KN-MET  Ts=    6.5 KN-MET  LOAD   110
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 168. MM  C/C FOR  960. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.53 SQ.CM.
```



ACI 318-11 BEAM NO. 636 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	731.	3582.	NO	NO
	CRITICAL POS MOMENT= 34.94 KN-MET AT 2250.MM, LOAD 102					
	REQD STEEL= 231.MM2, RHO=0.0017, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 69. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 55974.3 cm^4					
2	544.	4 - 12MM	0.	1144.	YES	NO
	CRITICAL NEG MOMENT= 63.35 KN-MET AT 0.MM, LOAD 101					
	REQD STEEL= 423.MM2, RHO=0.0031, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 46. MMS					
	REQD. DEVELOPMENT LENGTH = 395. MMS					
	Cracked Moment of Inertia Iz at above location = 71887.8 cm^4					
3	542.	3 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     80.00 KN-MET  AT  4500.MM, LOAD  117|
|   REQD STEEL=   483.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./   67. MMS         |
|   REQD. DEVELOPMENT LENGTH =   483. MMS                        |
|--------------------------------------------------------------|

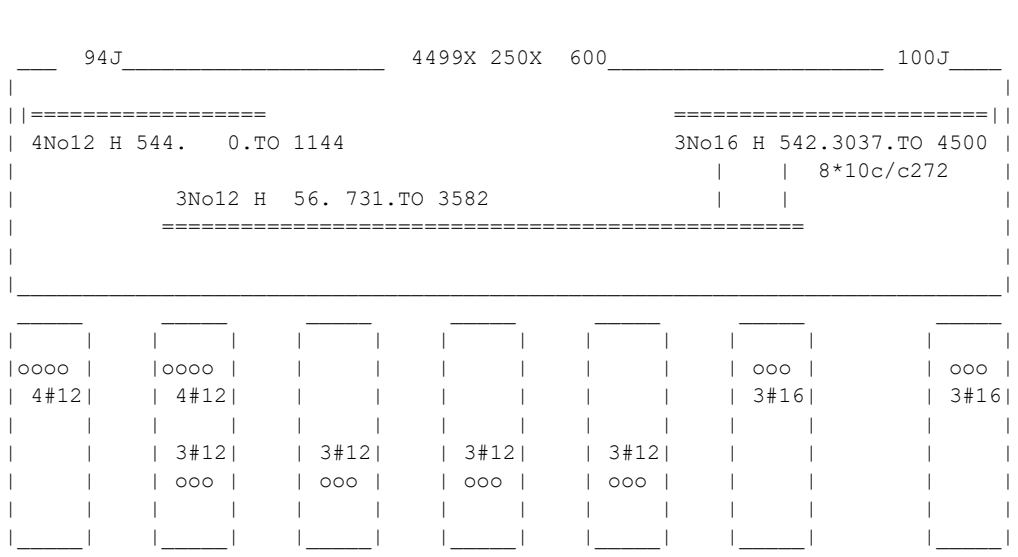
  Cracked Moment of Inertia Iz at above location =    91121.9 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     323.           0./       63.        0/  101
  375.           0./     171.           0./       34.        0/  111
  750.           0./      58.           0./       12.        0/  111
 1125.          83./       0.          17./        0.      117/    0
 1500.         142./       0.          28./        0.      117/    0
 1875.         173./       0.          34./        0.      117/    0
 2250.         177./       0.          35./        0.      102/    0
 2625.         161./       0.          32./        0.      101/    0
 3000.         116./       0.          23./        0.      101/    0
 3375.          53./       0.          11./        0.      111/    0
 3750.           0./      98.           0./       19.        0/  117
 4125.           0./     238.           0./       47.        0/  117
 4500.           0./     410.           0./       80.        0/  117
```

B E A M N O. 636 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   47.05 KNS  Vc=  152.80 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   68.81 KNS  Vc=  133.48 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 272. MM  C/C FOR 1710. MM
```



ACI 318-11 BEAM NO. 637 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	719.	3969.	NO	NO
	CRITICAL POS MOMENT= 36.93 KN-MET AT 2250.MM, LOAD 101					
	REQD STEEL= 245.MM2, RHO=0.0018, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 69. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 55974.3 cm^4					
2	542.	3 - 16MM	0.	1276.	YES	NO
	CRITICAL NEG MOMENT= 74.31 KN-MET AT 0.MM, LOAD 111					
	REQD STEEL= 483.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS					
	REQD. DEVELOPMENT LENGTH = 483. MMS					
	Cracked Moment of Inertia Iz at above location = 91121.9 cm^4					
3	542.	3 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     69.33 KN-MET  AT  4500.MM, LOAD  117|
|   REQD STEEL=   465.MM2, RHO=0.0034, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./   67. MMS         |
|   REQD. DEVELOPMENT LENGTH =   483. MMS                        |
|--------------------------------------------------------------|

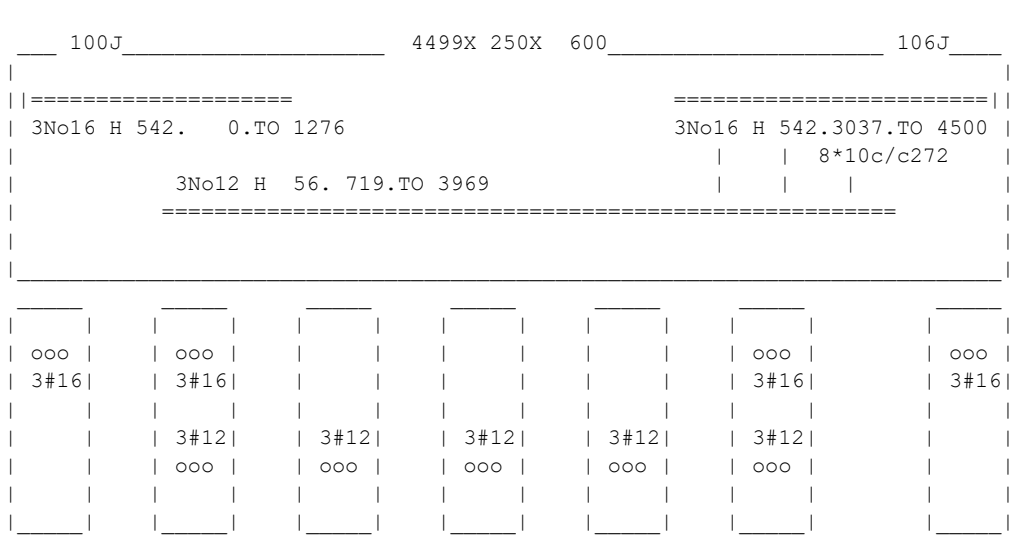
  Cracked Moment of Inertia Iz at above location =    91121.9 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     380.           0./       74.        0/  111
  375.           0./     225.           0./       44.        0/  111
  750.           0./      96.           0./       19.        0/  111
 1125.          67./       0.          13./        0.      117/    0
 1500.         134./       0.          27./        0.      117/    0
 1875.         172./       0.          34./        0.      117/    0
 2250.         187./       0.          37./        0.      101/    0
 2625.         182./       0.          36./        0.      101/    0
 3000.         163./       0.          32./        0.      111/    0
 3375.         121./       0.          24./        0.      111/    0
 3750.          56./      60.          11./       12.      116/  117
 4125.           0./     192.           0./       38.        0/  117
 4500.           0./     354.           0./       69.        0/  117
```

B E A M N O. 637 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   52.72 KNS  Vc=  236.22 KNS  Vs=    0.00 KNS
Tu=    2.51 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   63.15 KNS  Vc=  139.81 KNS  Vs=    0.00 KNS
Tu=    2.51 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 272. MM  C/C FOR 1710. MM
```



ACI 318-11 BEAM NO. 638 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	368.	3570.	NO	NO
	CRITICAL POS MOMENT= 22.02 KN-MET AT 1875.MM, LOAD 117					
	REQD STEEL= 145.MM2, RHO=0.0011, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 39023.5 cm^4					
2	544.	3 - 12MM	0.	1144.	YES	NO
	CRITICAL NEG MOMENT= 42.90 KN-MET AT 0.MM, LOAD 102					
	REQD STEEL= 285.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 69. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 55974.3 cm^4					
3	542.	2 - 16MM	2662.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     57.19 KN-MET  AT  4500.MM, LOAD  117|
|   REQD STEEL=   383.MM2, RHO=0.0028, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

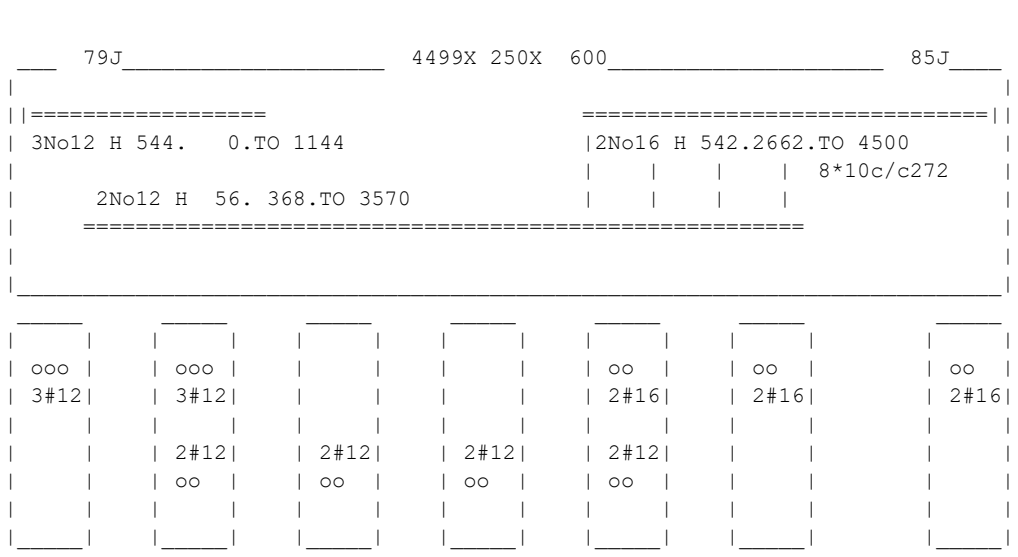
  Cracked Moment of Inertia Iz at above location =    64443.6 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     218.           0./       43.        0/  102
  375.           0./     124.           0./       25.        0/  111
  750.           9./      49.           2./       10.      117/  111
 1125.          61./       0.          12./        0.      117/    0
 1500.          95./       0.          19./        0.      117/    0
 1875.         111./       0.          22./        0.      117/    0
 2250.         111./       0.          22./        0.      101/    0
 2625.         102./       0.          20./        0.      102/    0
 3000.          74./       0.          15./        0.      111/    0
 3375.          40./       8.           8./        2.      111/  117
 3750.           0./      83.           0./       17.        0/  117
 4125.           0./     177.           0./       35.        0/  117
 4500.           0./     291.           0./       57.        0/  117
```

B E A M N O. 638 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   30.86 KNS  Vc=  133.36 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   50.01 KNS  Vc=  129.61 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 272. MM  C/C FOR 1710. MM
```



ACI 318-11 BEAM NO. 639 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	362.	3575.	NO	NO
	CRITICAL POS MOMENT= 22.64 KN-MET AT 2250.MM, LOAD 117					
	REQD STEEL= 150.MM2, RHO=0.0011, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 39023.5 cm^4					
2	544.	3 - 12MM	0.	1519.	YES	NO
	CRITICAL NEG MOMENT= 45.57 KN-MET AT 0.MM, LOAD 102					
	REQD STEEL= 303.MM2, RHO=0.0022, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 69. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 55974.3 cm^4					
3	542.	2 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     52.25 KN-MET  AT  4500.MM, LOAD  117|
|   REQD STEEL=   349.MM2, RHO=0.0026, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

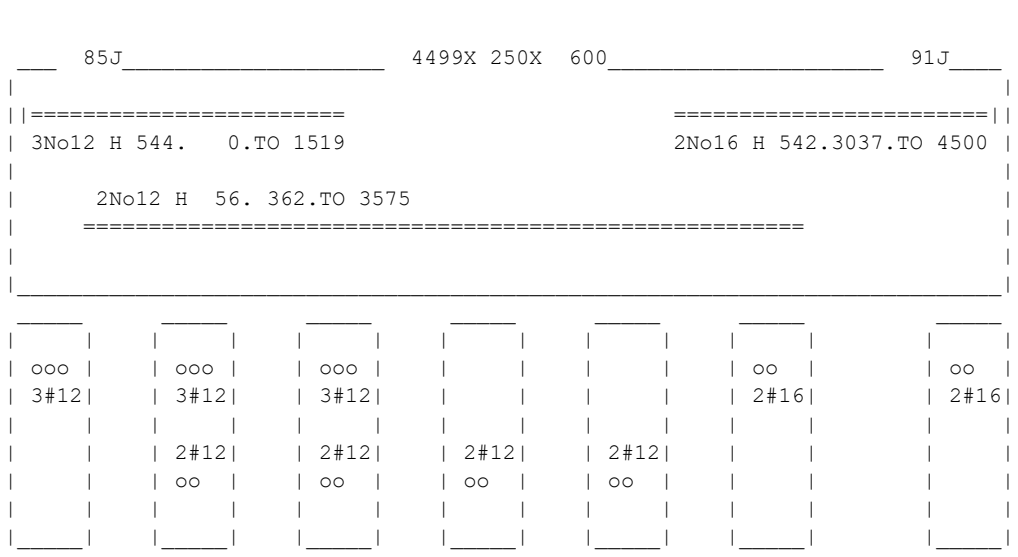
  Cracked Moment of Inertia Iz at above location =    64443.6 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     231.           0./       46.        0/  102
  375.           0./     134.           0./       27.        0/  111
  750.           2./      57.           0./       11.      117/  111
 1125.          57./       1.          11./        0.      117/  116
 1500.          94./       0.          19./        0.      117/    0
 1875.         113./       0.          22./        0.      117/    0
 2250.         114./       0.          23./        0.      117/    0
 2625.         103./       0.          20./        0.      102/    0
 3000.          77./       0.          15./        0.      102/    0
 3375.          40./       0.           8./        0.      111/    0
 3750.           0./      65.           0./       13.        0/  117
 4125.           0./     155.           0./       31.        0/  117
 4500.           0./     266.           0./       52.        0/  117
```

B E A M N O. 639 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   31.87 KNS  Vc=  132.96 KNS  Vs=    0.00 KNS
 Tu=    2.18 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   43.11 KNS  Vc=  132.96 KNS  Vs=    0.00 KNS
 Tu=    2.18 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 640 DESIGN RESULTS

LEN - 3000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	246.	2379.	NO	NO
	CRITICAL POS MOMENT= 14.58 KN-MET AT 1250.MM, LOAD 117					
	REQD STEEL= 96.MM2, RHO=0.0007, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 39023.5 cm^4					
2	544.	2 - 12MM	0.	1144.	YES	NO
	CRITICAL NEG MOMENT= 28.84 KN-MET AT 0.MM, LOAD 111					
	REQD STEEL= 191.MM2, RHO=0.0014, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 39023.5 cm^4					
3	542.	2 - 16MM	1599.	3000.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     54.03 KN-MET  AT  3000.MM, LOAD  117|
|   REQD STEEL=   361.MM2, RHO=0.0027, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    64443.6 cm^4

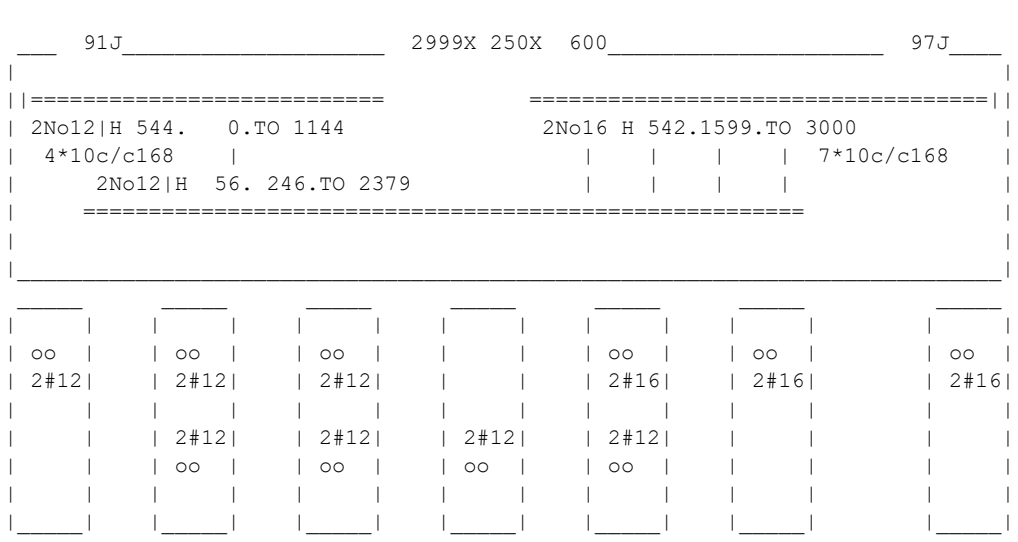

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     146.           0./       29.        0/  111
  250.           0./      85.           0./       17.        0/  111
  500.          34./      35.           7./        7.      117/  111
  750.          60./       0.          12./        0.      117/  116
 1000.          73./       0.          14./        0.      117/    0
 1250.          73./       0.          15./        0.      117/    0
 1500.          65./       0.          13./        0.      101/    0
 1750.          53./       0.          11./        0.      111/    0
 2000.          39./       0.           8./        0.      111/    0
 2250.          17./      49.           3./       10.      116/  117
 2500.           0./     111.           0./       22.        0/  117
 2750.           0./     186.           0./       37.        0/  117
 3000.           0./     275.           0./       54.        0/  117
```

B E A M N O. 640 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   11.12 KNS  Vc=  130.92 KNS  Vs=    0.00 KNS
Tu=    5.76 KN-MET  Tc=    4.8 KN-MET  Ts=    7.7 KN-MET  LOAD   108
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 168. MM  C/C FOR  460. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.79 SQ.CM.

AT END   SUPPORT - Vu=   37.17 KNS  Vc=  130.92 KNS  Vs=    0.00 KNS
Tu=    5.76 KN-MET  Tc=    4.8 KN-MET  Ts=    7.7 KN-MET  LOAD   108
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 168. MM  C/C FOR  960. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.79 SQ.CM.
```



ACI 318-11 BEAM NO. 641 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	730.	3583.	NO	NO
	CRITICAL POS MOMENT= 35.09 KN-MET AT 2250.MM, LOAD 102					
	REQD STEEL= 233.MM2, RHO=0.0017, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 69. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 55974.3 cm^4					
2	544.	4 - 12MM	0.	1144.	YES	NO
	CRITICAL NEG MOMENT= 63.29 KN-MET AT 0.MM, LOAD 101					
	REQD STEEL= 422.MM2, RHO=0.0031, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 46. MMS					
	REQD. DEVELOPMENT LENGTH = 395. MMS					
	Cracked Moment of Inertia Iz at above location = 71887.8 cm^4					
3	542.	3 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     80.56 KN-MET  AT  4500.MM, LOAD  117|
|   REQD STEEL=   483.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./   67. MMS         |
|   REQD. DEVELOPMENT LENGTH =   483. MMS                        |
|--------------------------------------------------------------|

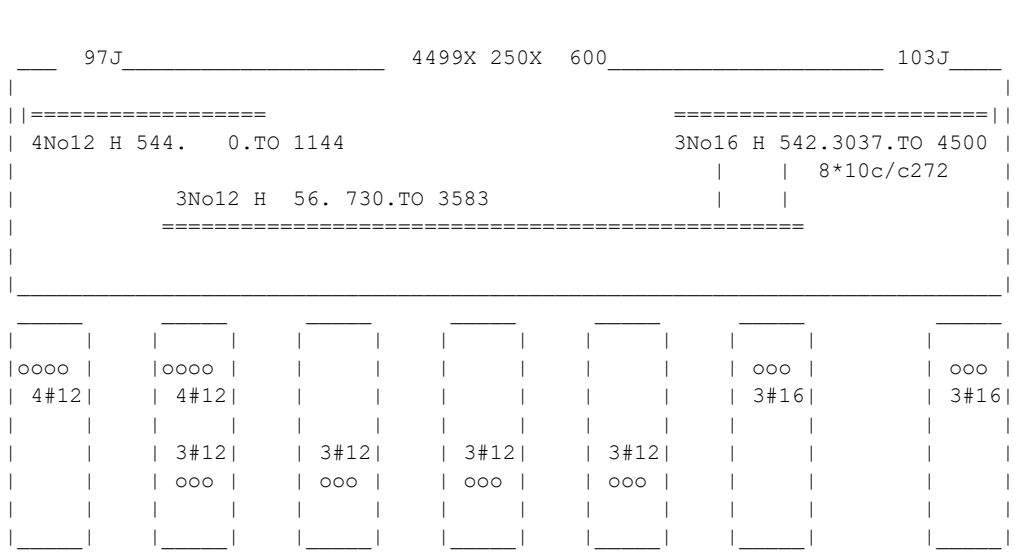
  Cracked Moment of Inertia Iz at above location =    91121.9 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     323.           0./       63.        0/  101
  375.           0./     167.           0./       33.        0/  111
  750.           0./      55.           0./       11.        0/  111
 1125.          86./       0.          17./        0.      117/    0
 1500.         144./       0.          29./        0.      117/    0
 1875.         174./       0.          34./        0.      117/    0
 2250.         178./       0.          35./        0.      102/    0
 2625.         161./       0.          32./        0.      102/    0
 3000.         116./       0.          23./        0.      101/    0
 3375.          52./       0.          10./        0.      111/    0
 3750.           0./     100.           0./       20.        0/  117
 4125.           0./     241.           0./       47.        0/  117
 4500.           0./     413.           0./       81.        0/  117
```

B E A M N O. 641 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   47.03 KNS  Vc=  152.68 KNS  Vs=    0.00 KNS
Tu=    0.03 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   68.84 KNS  Vc=  133.51 KNS  Vs=    0.00 KNS
Tu=    0.03 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 272. MM  C/C FOR 1710. MM
```



ACI 318-11 BEAM NO. 642 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	719.	3969.	NO	NO
	CRITICAL POS MOMENT= 36.93 KN-MET AT 2250.MM, LOAD 101					
	REQD STEEL= 245.MM2, RHO=0.0018, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 69. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 55974.3 cm^4					
2	542.	3 - 16MM	0.	1276.	YES	NO
	CRITICAL NEG MOMENT= 74.09 KN-MET AT 0.MM, LOAD 101					
	REQD STEEL= 483.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS					
	REQD. DEVELOPMENT LENGTH = 483. MMS					
	Cracked Moment of Inertia Iz at above location = 91121.9 cm^4					
3	542.	3 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     69.61 KN-MET  AT  4500.MM, LOAD  117|
|   REQD STEEL=   467.MM2, RHO=0.0034, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./   67. MMS         |
|   REQD. DEVELOPMENT LENGTH =   483. MMS                        |
|--------------------------------------------------------------|

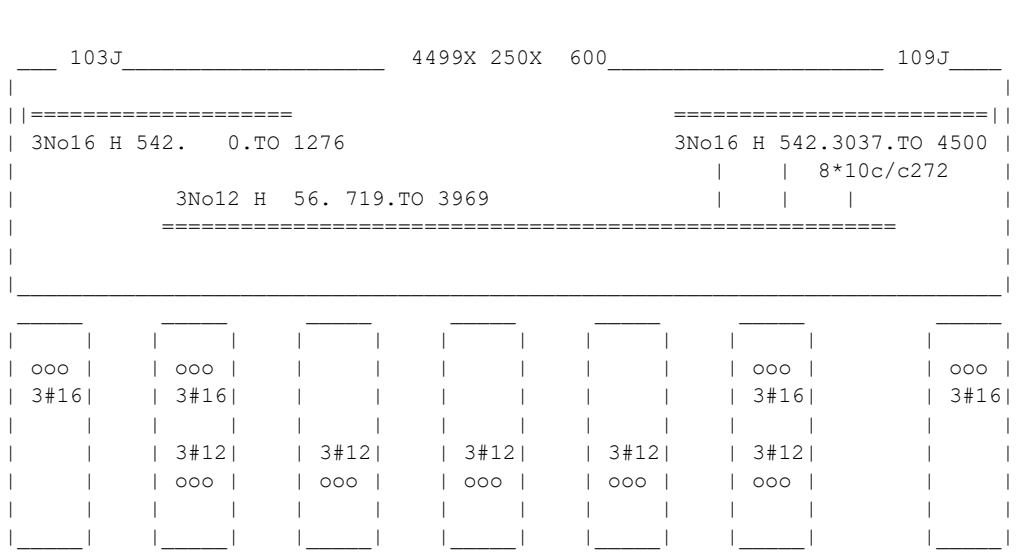
  Cracked Moment of Inertia Iz at above location =    91121.9 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     379.           0./       74.        0/  101
  375.           0./     223.           0./       44.        0/  111
  750.           0./      94.           0./       19.        0/  111
 1125.          68./       0.          13./        0.      117/    0
 1500.         134./       0.          27./        0.      117/    0
 1875.         172./       0.          34./        0.      117/    0
 2250.         187./       0.          37./        0.      101/    0
 2625.         182./       0.          36./        0.      101/    0
 3000.         162./       0.          32./        0.      111/    0
 3375.         119./       0.          24./        0.      111/    0
 3750.          55./      61.          11./       12.      116/  117
 4125.           0./     193.           0./       38.        0/  117
 4500.           0./     356.           0./       70.        0/  117
```

B E A M N O. 642 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   52.72 KNS  Vc=  236.22 KNS  Vs=    0.00 KNS
 Tu=    2.50 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   63.15 KNS  Vc=  139.81 KNS  Vs=    0.00 KNS
 Tu=    2.50 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 272. MM  C/C FOR 1710. MM
```



ACI 318-11 BEAM NO. 643 DESIGN RESULTS

LEN - 6500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	0.	6500.	YES	YES
	CRITICAL POS MOMENT= 28.41 KN-MET AT 6500.MM, LOAD 114					
	REQD STEEL= 231.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					
2	442.	2 - 16MM	0.	3234.	YES	NO
	CRITICAL NEG MOMENT= 52.85 KN-MET AT 0.MM, LOAD 109					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
3	442.	2 - 16MM	2995.	6500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     48.15 KN-MET  AT  6500.MM, LOAD  108|
|   REQD STEEL=   394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

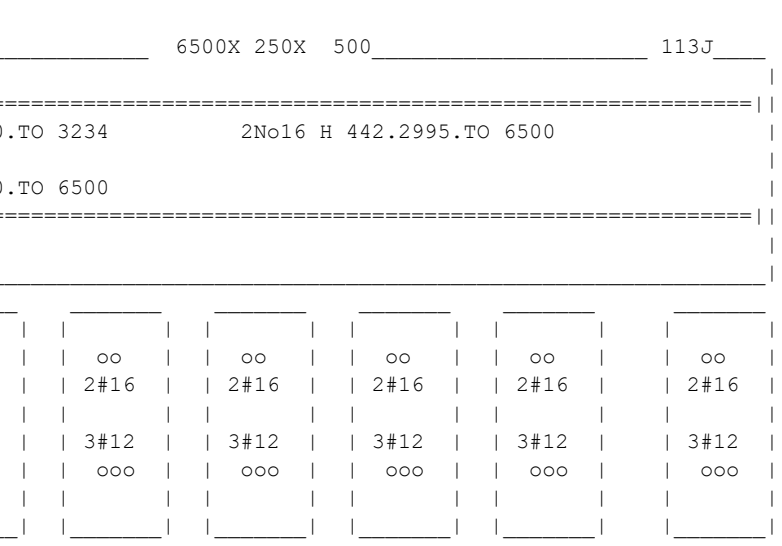
  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.         157./     333.          25./       53.      113/  109
  542.         147./     254.          24./       41.      113/  109
 1083.         132./     183.          21./       29.      113/  109
 1625.         116./     119.          19./       19.      117/  109
 2167.         136./      65.          22./       11.      117/  114
 2708.         149./      18.          24./        3.      117/  114
 3250.         154./       0.          25./        0.      117/    0
 3792.         152./      14.          24./        2.      117/  113
 4333.         142./      58.          23./        9.      117/  113
 4875.         125./     106.          20./       17.      117/  113
 5417.         145./     163.          23./       26.      114/  108
 5958.         164./     229.          26./       37.      114/  108
 6500.         177./     303.          28./       48.      114/  108
```

B E A M N O. 643 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    9.67 KNS  Vc=  108.19 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   11.49 KNS  Vc=  106.04 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 644 DESIGN RESULTS

LEN - 1750. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	0.	1750.	YES	YES

CRITICAL POS MOMENT= 35.84 KN-MET AT 0.MM, LOAD 117
REQD STEEL= 292.MM2, RHO=0.0026, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

Cracked Moment of Inertia Iz at above location = 36333.1 cm^4

2	444.	3 - 12MM	0.	1750.	YES	YES

CRITICAL NEG MOMENT= 40.89 KN-MET AT 1750.MM, LOAD 108
REQD STEEL= 336.MM2, RHO=0.0030, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

Cracked Moment of Inertia Iz at above location = 35984.5 cm^4

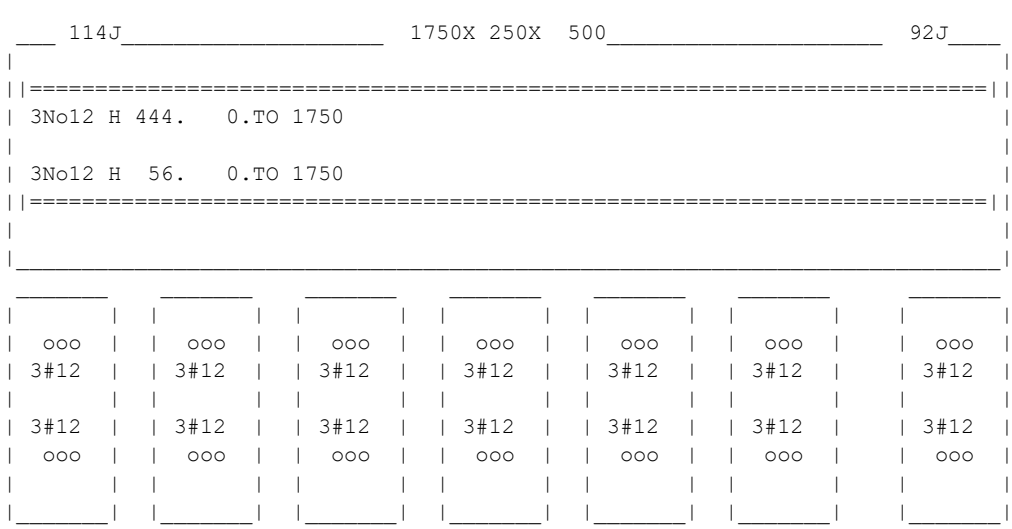
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	224./ 31.	36./ 5.	117/ 113
146.	204./ 43.	33./ 7.	117/ 113
292.	182./ 55.	29./ 9.	117/ 113
438.	161./ 68.	26./ 11.	117/ 113
583.	139./ 81.	22./ 13.	117/ 113
729.	116./ 94.	19./ 15.	117/ 113
875.	93./ 113.	15./ 18.	114/ 108
1021.	91./ 135.	15./ 22.	114/ 108
1167.	89./ 158.	14./ 25.	114/ 108
1313.	87./ 182.	14./ 29.	114/ 108
1458.	84./ 206.	14./ 33.	114/ 108
1604.	81./ 231.	13./ 37.	114/ 108
1750.	77./ 256.	12./ 41.	114/ 108

B E A M N O. 644 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 644 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

AT END SUPPORT - Vu= 15.65 KNS Vc= 108.09 KNS Vs= 0.00 KNS
Tu= 0.65 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 645 DESIGN RESULTS

LEN - 1750. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	3 - 12MM	0.	1750.	YES	YES
	CRITICAL POS MOMENT= 39.62 KN-MET AT 0.MM, LOAD 117					
	REQD STEEL= 324.MM2, RHO=0.0029, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					
2	442.	2 - 16MM	0.	1750.	YES	YES
	CRITICAL NEG MOMENT= 50.23 KN-MET AT 1750.MM, LOAD 108					
	REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					

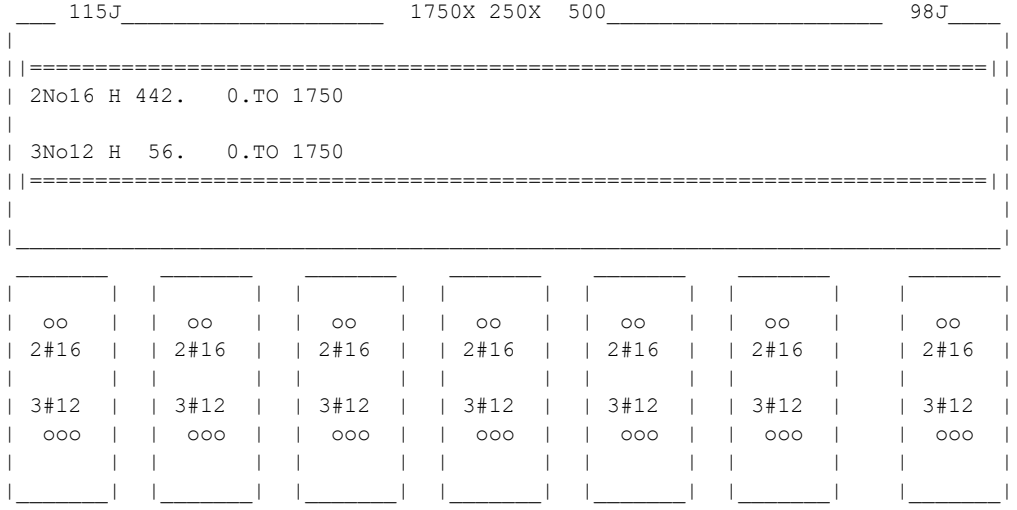
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	248./	24.	40./	4.	117/	113
146.	219./	37.	35./	6.	117/	113
292.	190./	51.	30./	8.	117/	113
438.	160./	66.	26./	11.	117/	113
583.	130./	80.	21./	13.	117/	113
729.	105./	96.	17./	15.	109/	113
875.	103./	124.	17./	20.	114/	108
1021.	102./	155.	16./	25.	114/	108
1167.	101./	186.	16./	30.	114/	108
1313.	99./	217.	16./	35.	114/	108
1458.	97./	250.	16./	40.	114/	108
1604.	95./	283.	15./	45.	114/	108
1750.	92./	316.	15./	50.	114/	108

B E A M N O. 645 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 645 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

AT END SUPPORT - Vu= 16.07 KNS Vc= 109.50 KNS Vs= 0.00 KNS
Tu= 1.12 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 646 DESIGN RESULTS

LEN - 3000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	0.	3000.	YES	YES
	CRITICAL POS MOMENT= 23.45 KN-MET AT 1500.MM, LOAD 117					
	REQD STEEL= 215.MM2, RHO=0.0022, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 142. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 19768.7 cm^4					
2	394.	2 - 12MM	0.	894.	YES	NO

```
|------------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      5.71 KN-MET  AT     0.MM, LOAD  102|
|   REQD STEEL=    52.MM2, RHO=0.0005, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./  142. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                        |
|------------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    19768.7 cm^4

  3        394.     2 - 12MM      2231.         3000.        NO   YES
|------------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      1.72 KN-MET  AT  3000.MM, LOAD  109|
|   REQD STEEL=    16.MM2, RHO=0.0002, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./  142. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                        |
|------------------------------------------------------------------|

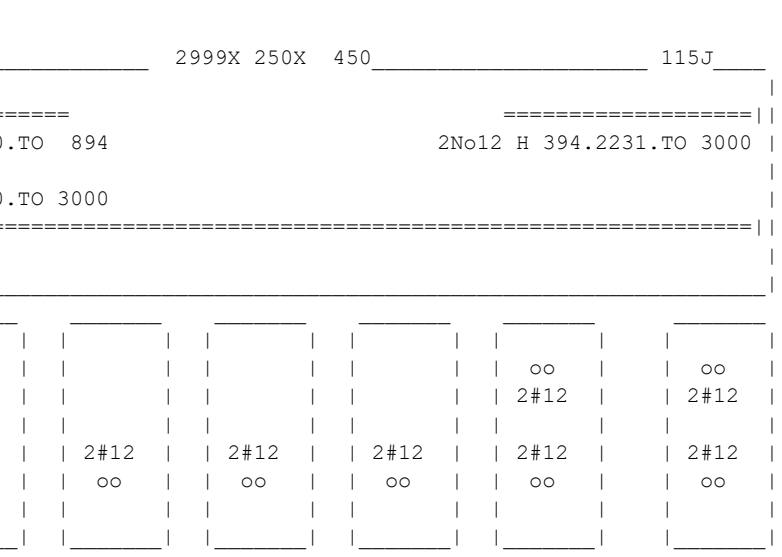
  Cracked Moment of Inertia Iz at above location =    19768.7 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           4./      40.           1./        6.      113/  102
  250.           8./      15.           1./        2.      113/  111
  500.          42./       4.           6./        1.      117/  116
  750.          75./       0.          11./        0.      117/    0
 1000.         107./       0.          15./        0.      117/    0
 1250.         137./       0.          19./        0.      117/    0
 1500.         165./       0.          23./        0.      117/    0
 1750.         147./       0.          21./        0.      117/    0
 2000.         127./       0.          18./        0.      117/    0
 2250.         105./       0.          15./        0.      117/    0
 2500.          82./       0.          12./        0.      117/    0
 2750.          57./       3.           8./        0.      117/  114
 3000.          30./      12.           4./        2.      117/  109
```

B E A M N O. 646 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    2.06 KNS  Vc=   96.77 KNS  Vs=    0.00 KNS
 Tu=    0.23 KN-MET  Tc=    3.3 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    4.34 KNS  Vc=   96.77 KNS  Vs=    0.00 KNS
 Tu=    0.23 KN-MET  Tc=    3.3 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 647 DESIGN RESULTS

LEN - 2250. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	2250.	YES	YES

CRITICAL POS MOMENT= 58.98 KN-MET AT 375.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	444.	3 - 12MM	1200.	2250.	NO	YES

CRITICAL NEG MOMENT= 33.55 KN-MET AT 2250.MM, LOAD 102
REQD STEEL= 273.MM2, RHO=0.0025, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

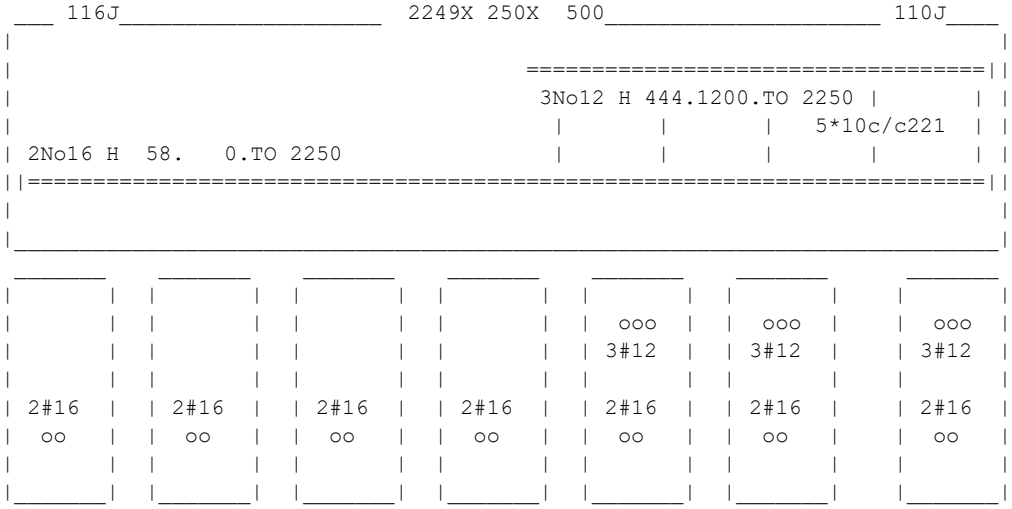
Cracked Moment of Inertia Iz at above location = 36333.1 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	351./ 0.	56./ 0.	117/ 0
187.	365./ 0.	58./ 0.	117/ 0
375.	372./ 0.	59./ 0.	117/ 0
562.	372./ 0.	59./ 0.	117/ 0
750.	363./ 0.	58./ 0.	117/ 0
937.	347./ 0.	55./ 0.	117/ 0
1125.	323./ 0.	51./ 0.	117/ 0
1312.	291./ 0.	46./ 0.	117/ 0
1500.	252./ 0.	40./ 0.	117/ 0
1687.	206./ 20.	33./ 3.	117/ 111
1875.	152./ 74.	24./ 12.	117/ 111
2062.	92./ 134.	15./ 22.	117/ 111
2250.	24./ 210.	4./ 34.	117/ 102

B E A M N O. 647 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.71 KNS Vc= 105.88 KNS Vs= 0.00 KNS
Tu= 0.77 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 54.29 KNS Vc= 137.97 KNS Vs= 0.00 KNS
Tu= 0.91 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 102
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 685. MM



ACI 318-11 BEAM NO. 648 DESIGN RESULTS

LEN - 2250. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	2250.	YES	YES

CRITICAL POS MOMENT= 59.02 KN-MET AT 375.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	444.	3 - 12MM	1200.	2250.	NO	YES

CRITICAL NEG MOMENT= 33.03 KN-MET AT 2250.MM, LOAD 102
REQD STEEL= 269.MM2, RHO=0.0024, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 71. MMS
REQD. DEVELOPMENT LENGTH = 305. MMS

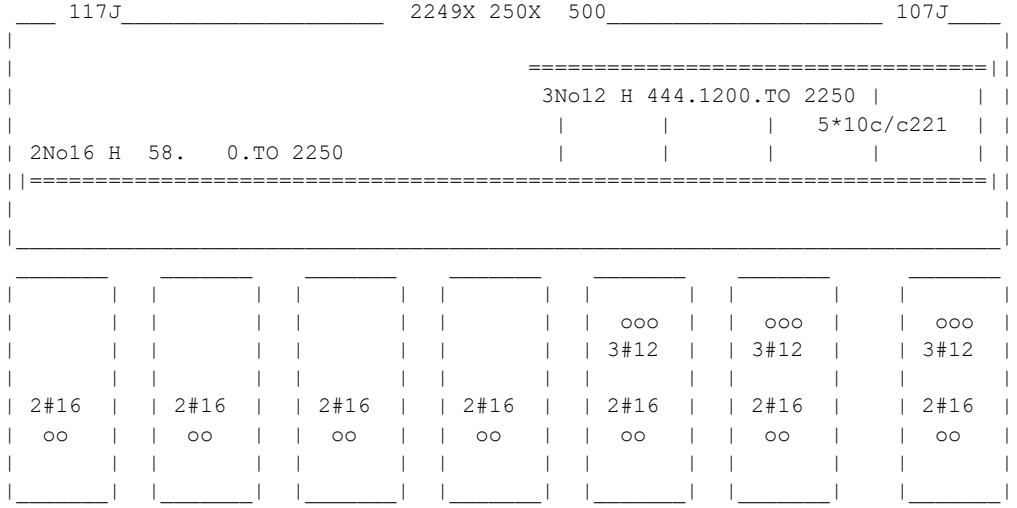
Cracked Moment of Inertia Iz at above location = 36333.1 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	350./	0.	56./	0.	117/	0
187.	365./	0.	58./	0.	117/	0
375.	373./	0.	59./	0.	117/	0
562.	372./	0.	59./	0.	117/	0
750.	364./	0.	58./	0.	117/	0
937.	348./	0.	55./	0.	117/	0
1125.	324./	0.	52./	0.	117/	0
1312.	293./	0.	47./	0.	117/	0
1500.	254./	0.	41./	0.	117/	0
1687.	208./	17.	33./	3.	117/	111
1875.	155./	70.	25./	11.	117/	111
2062.	94./	131.	15./	21.	117/	111
2250.	27./	206.	4./	33.	117/	102

B E A M N O. 648 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.70 KNS Vc= 105.88 KNS Vs= 0.00 KNS
Tu= 0.76 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 54.03 KNS Vc= 140.57 KNS Vs= 0.00 KNS
Tu= 0.87 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 102
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 685. MM



ACI 318-11 BEAM NO. 650 DESIGN RESULTS

LEN - 2250. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	2250.	YES	YES

CRITICAL POS MOMENT= 50.60 KN-MET AT 187.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	693.	2250.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     60.18 KN-MET  AT  2250.MM, LOAD  102|
|   REQD STEEL=   394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

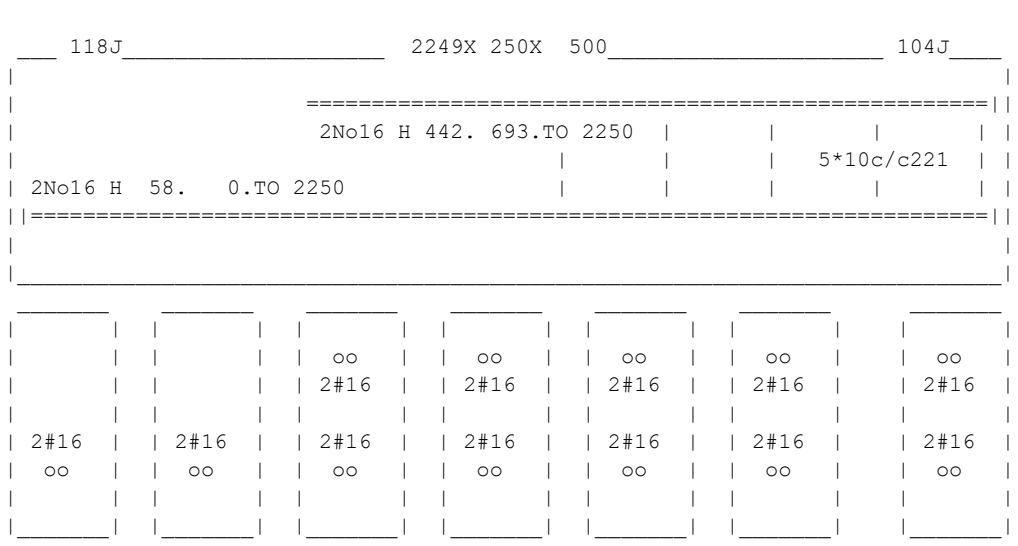
  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.         316./       0.          50./        0.      117/    0
  187.         318./       0.          51./        0.      117/    0
  375.         313./       0.          50./        0.      117/    0
  562.         301./       0.          48./        0.      117/    0
  750.         280./       0.          45./        0.      117/    0
  937.         252./       0.          40./        0.      117/    0
 1125.         217./       0.          35./        0.      117/    0
 1312.         174./      29.          28./        5.      117/  110
 1500.         124./      81.          20./       13.      117/  110
 1687.          67./     139.          11./       22.      117/  110
 1875.           3./     208.           0./       33.      117/  102
 2062.           0./     290.           0./       46.        0/  102
 2250.           0./     380.           0./       60.        0/  102
```

B E A M N O. 650 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   33.10 KNS  Vc=  111.84 KNS  Vs=    0.00 KNS
 Tu=    0.23 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   62.03 KNS  Vc=  109.95 KNS  Vs=    0.00 KNS
 Tu=    0.23 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR  685. MM
```



ACI 318-11 BEAM NO. 651 DESIGN RESULTS

LEN - 2250. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	2250.	YES	YES

CRITICAL POS MOMENT= 51.33 KN-MET AT 187.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	693.	2250.	NO	YES

CRITICAL NEG MOMENT= 60.67 KN-MET AT 2250.MM, LOAD 102
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 138. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

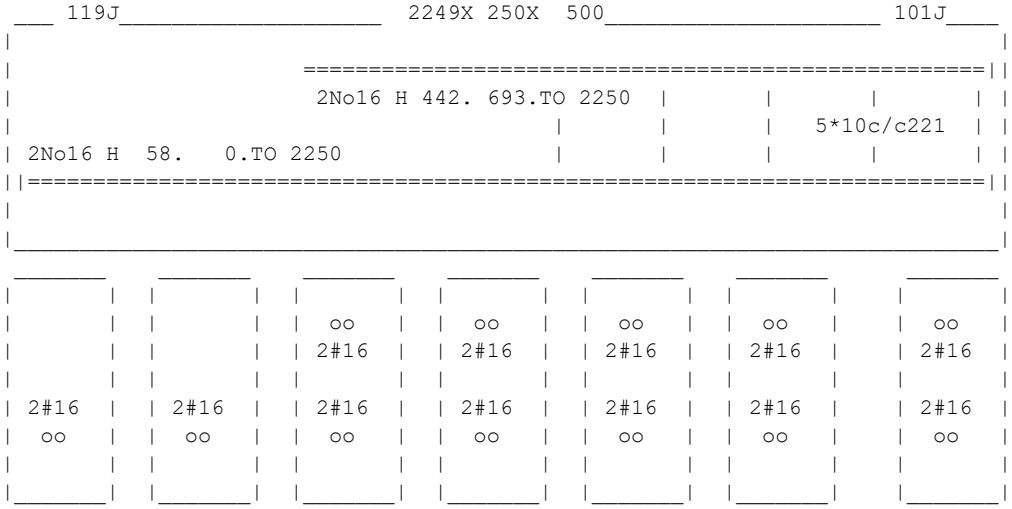
Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	321./ 0.	51./ 0.	117/ 0
187.	323./ 0.	51./ 0.	117/ 0
375.	317./ 0.	50./ 0.	117/ 0
562.	304./ 0.	48./ 0.	117/ 0
750.	283./ 0.	45./ 0.	117/ 0
937.	255./ 0.	41./ 0.	117/ 0
1125.	219./ 0.	35./ 0.	117/ 0
1312.	175./ 26.	28./ 4.	117/ 110
1500.	125./ 78.	20./ 13.	117/ 110
1687.	67./ 137.	11./ 22.	117/ 110
1875.	2./ 210.	0./ 34.	117/ 102
2062.	0./ 292.	0./ 47.	0/ 102
2250.	0./ 383.	0./ 61.	0/ 102

B E A M N O. 651 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 33.13 KNS Vc= 111.81 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 62.06 KNS Vc= 109.95 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 685. MM



**********END OF BEAM DESIGN**********

235. DESIGN COLUMN 3 4 8 31 32 34 583 585 591 TO 594 597 TO 600 603 TO 606 609 -
236. 610 TO 612 615 TO 618 623 625

ACI 318-11 COLUMN NO. 3 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

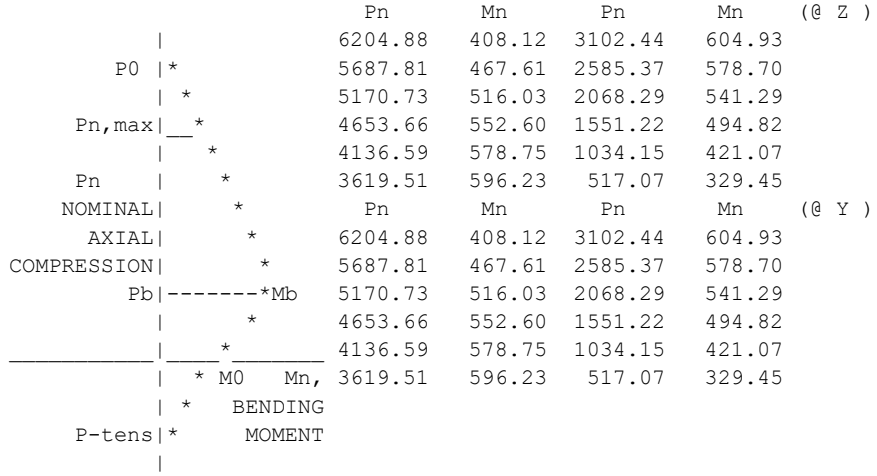
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	115.76	19.46	0.04004

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	115.76	42.09	0.08659



ACI 318-11 COLUMN NO. 4 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

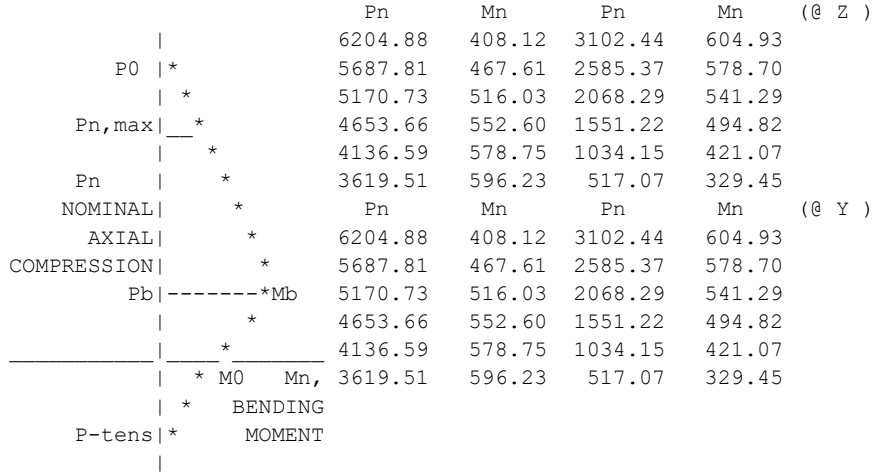
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	115.77	19.46	0.04002

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	115.77	42.13	0.08665



ACI 318-11 COLUMN NO. 8 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

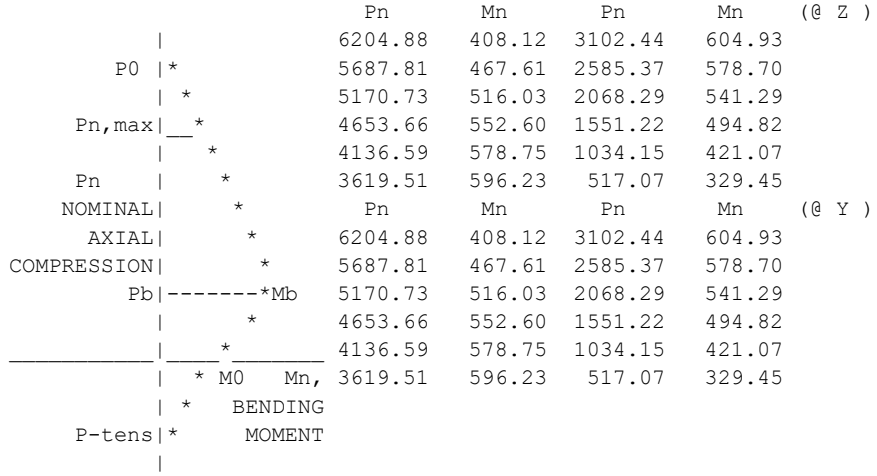
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	243.85	14.12	0.01379

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	243.85	5.15	0.00503



ACI 318-11 COLUMN NO. 31 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

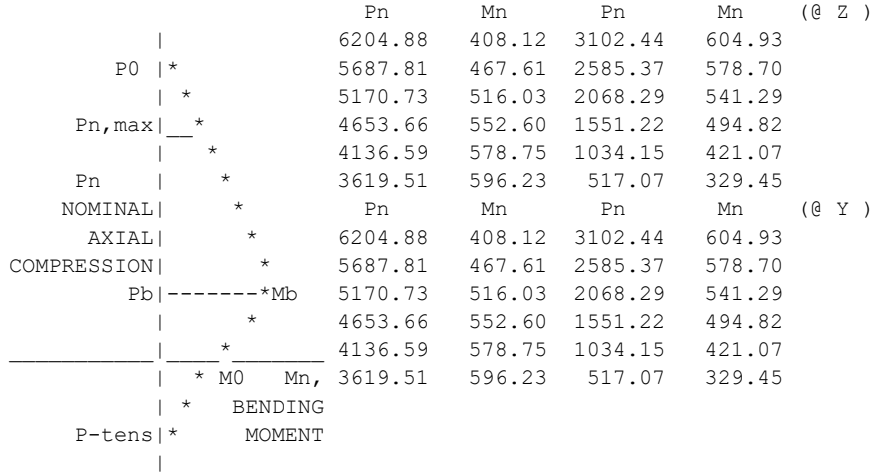
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	398.95	31.70	0.03973

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	398.95	8.68	0.01088



ACI 318-11 COLUMN NO. 32 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

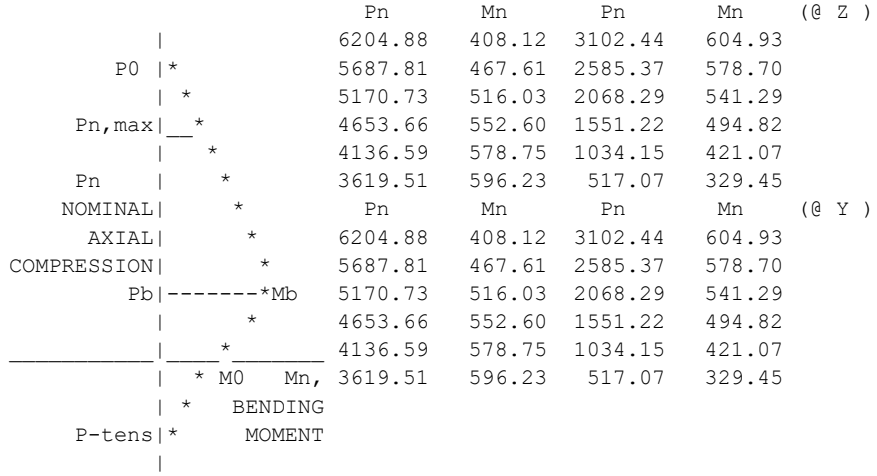
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	398.97	31.70	0.03972

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	398.97	8.69	0.01090



ACI 318-11 COLUMN NO. 34 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

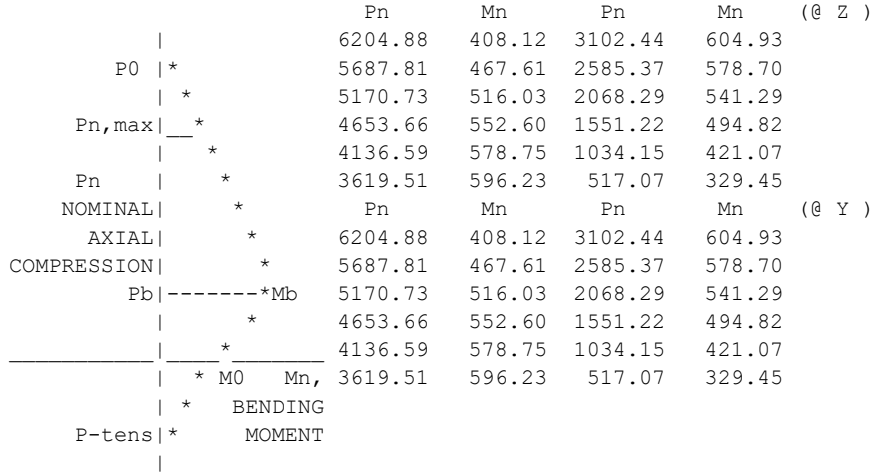
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	493.32	6.23	0.00631

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	493.32	2.31	0.00234



ACI 318-11 COLUMN NO. 583 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

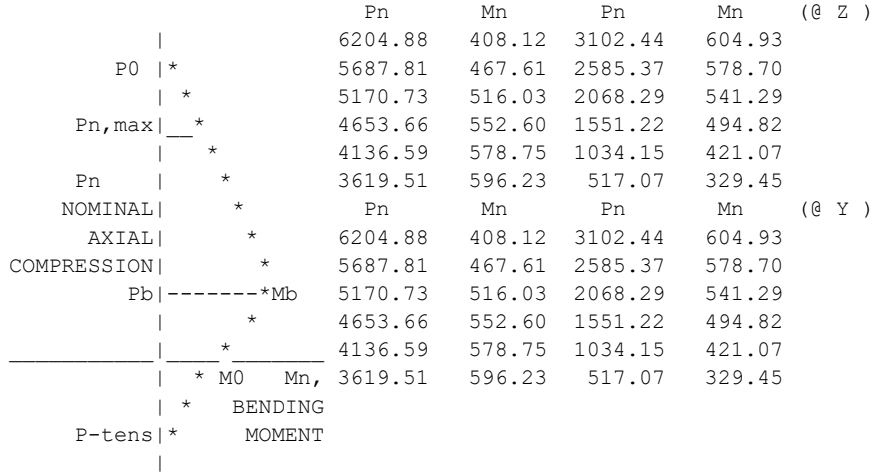
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	243.84	14.11	0.01378

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	243.84	5.20	0.00508



ACI 318-11 COLUMN NO. 585 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

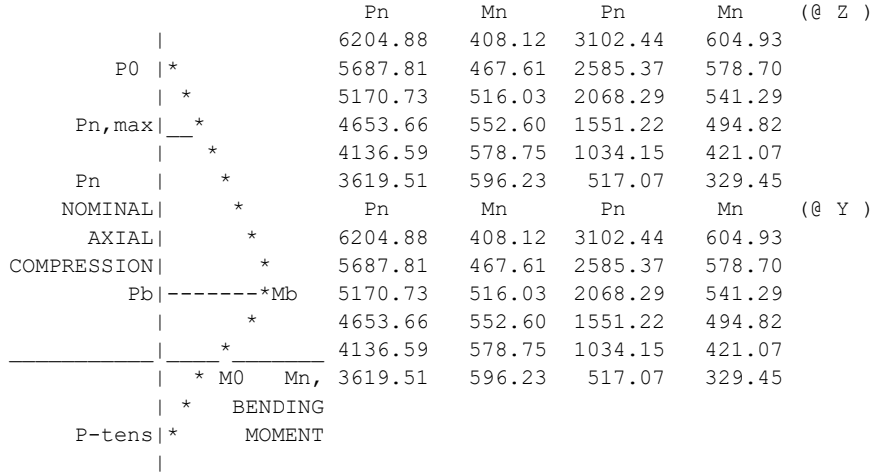
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	493.30	6.20	0.00629

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	493.30	2.32	0.00235



ACI 318-11 COLUMN NO. 591 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

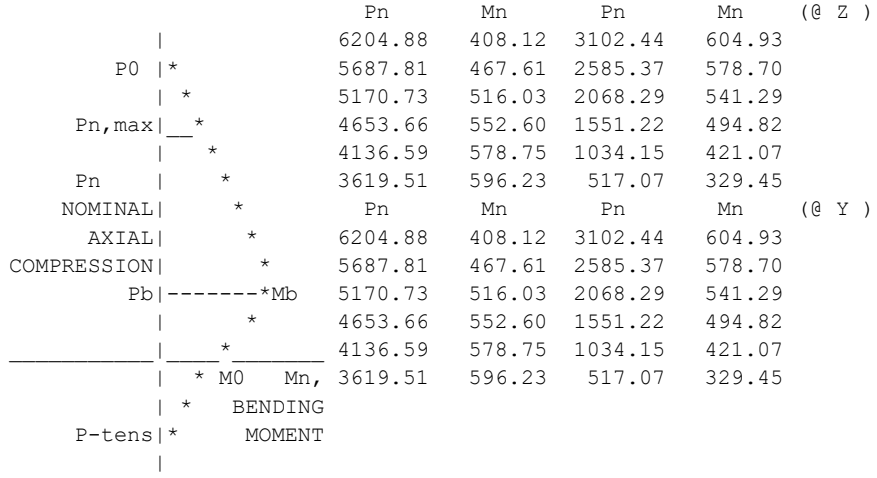
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	259.33	17.25	0.01584

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	259.33	5.92	0.00544



ACI 318-11 COLUMN NO. 592 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

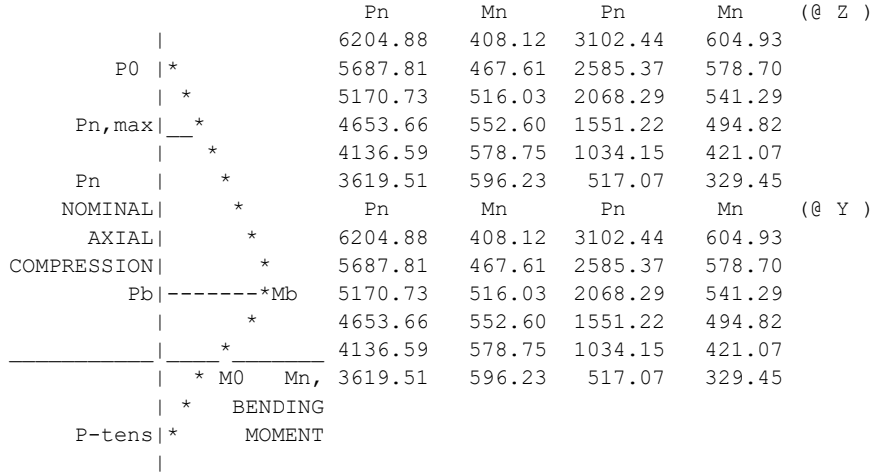
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	517.48	7.49	0.00723

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	517.48	0.05	0.00005



ACI 318-11 COLUMN NO. 593 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

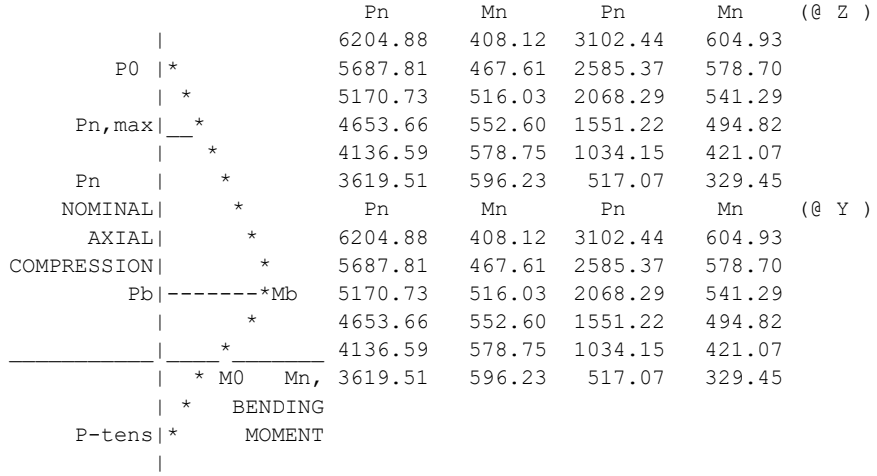
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	259.31	17.17	0.01576

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	259.31	6.00	0.00551



ACI 318-11 COLUMN NO. 594 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

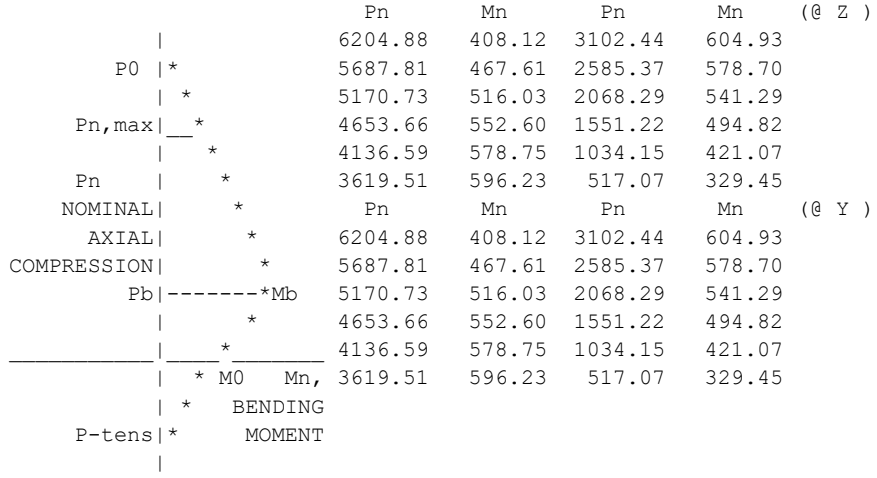
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	517.47	7.30	0.00705

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	517.47	0.05	0.00004



ACI 318-11 COLUMN NO. 597 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

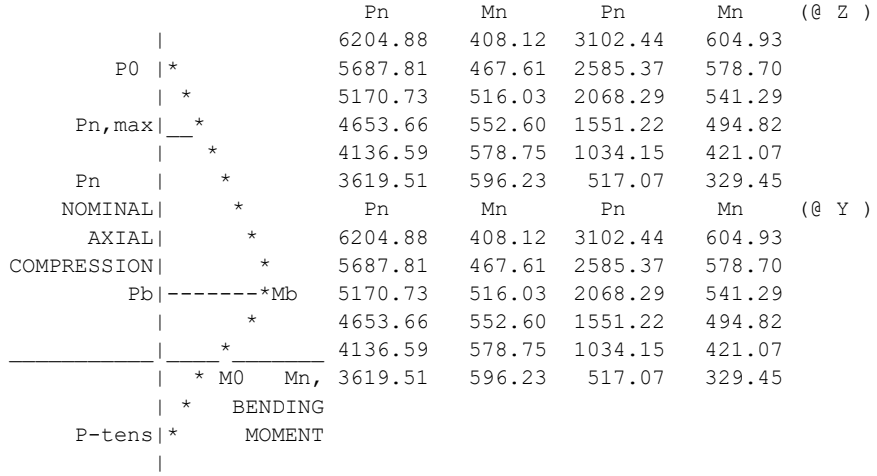
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	251.88	25.10	0.02373

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	251.88	46.21	0.04369



ACI 318-11 COLUMN NO. 598 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

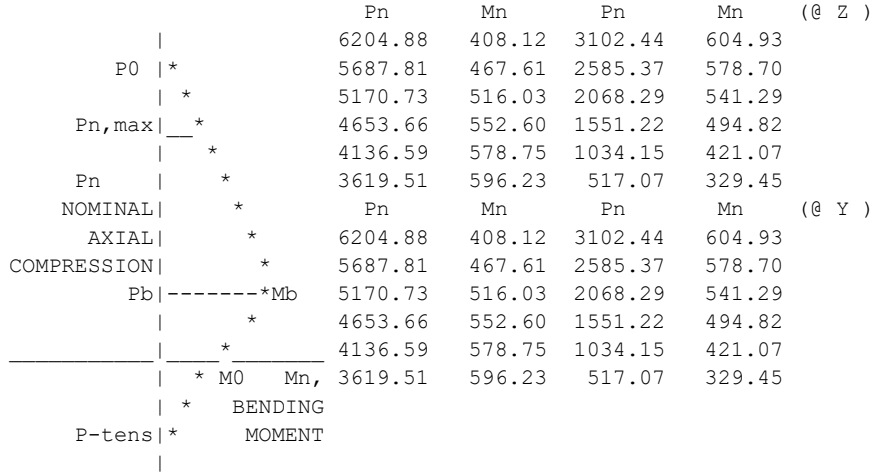
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	701.97	48.40	0.03448

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	701.97	1.33	0.00095



ACI 318-11 COLUMN NO. 599 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

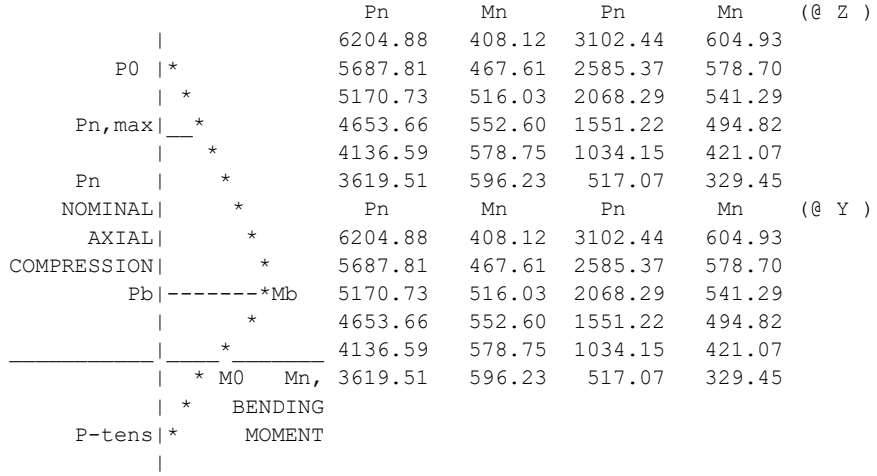
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	255.71	25.55	0.02380

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	255.71	46.10	0.04293



ACI 318-11 COLUMN NO. 600 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

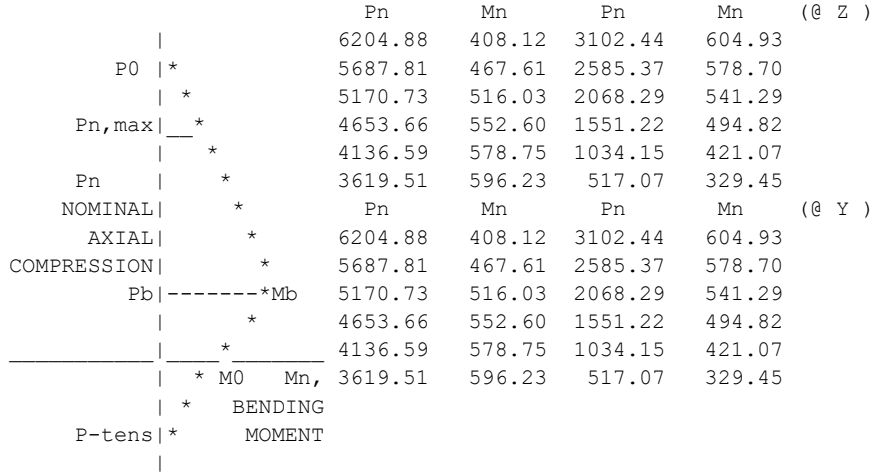
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	705.66	48.16	0.03412

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	705.66	1.34	0.00095



ACI 318-11 COLUMN NO. 603 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

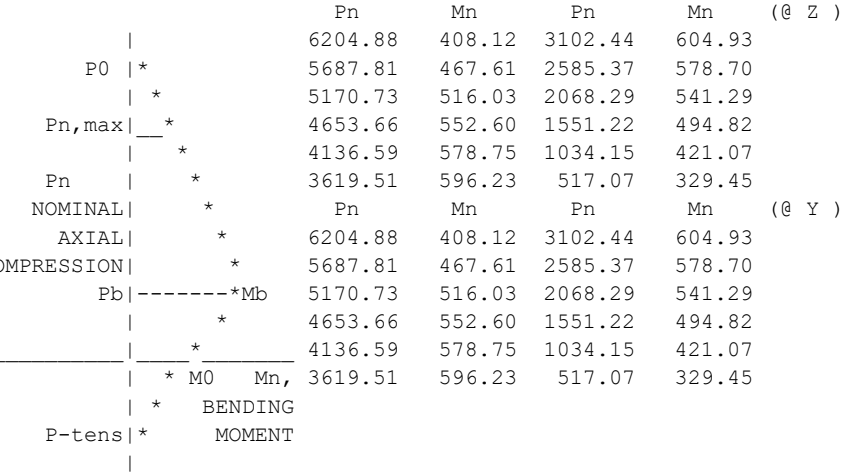
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	220.02	19.66	0.01334

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	220.02	24.05	0.01632



ACI 318-11 COLUMN NO. 604 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

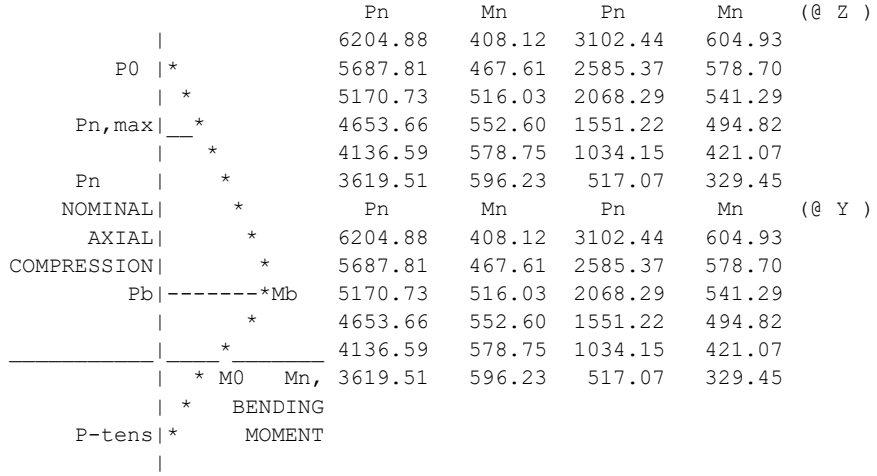
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	564.53	6.96	0.00617

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	564.53	8.49	0.00752



ACI 318-11 COLUMN NO. 605 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

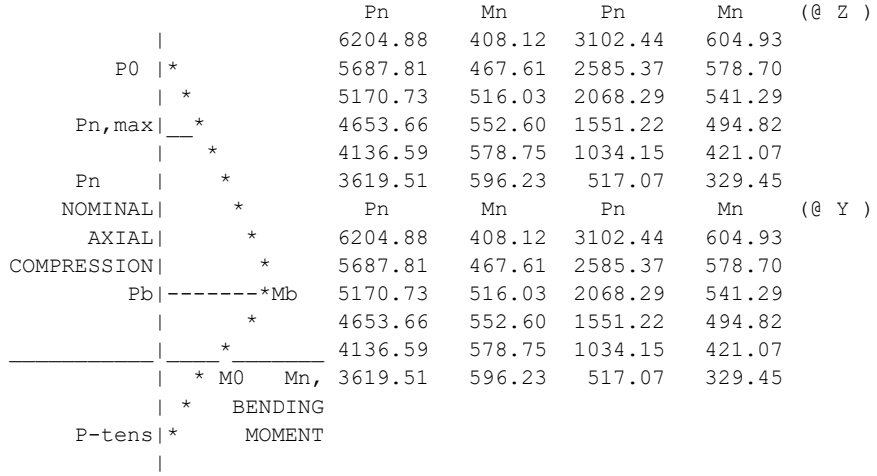
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	224.80	21.35	0.01417

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	224.80	23.56	0.01565



ACI 318-11 COLUMN NO. 606 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

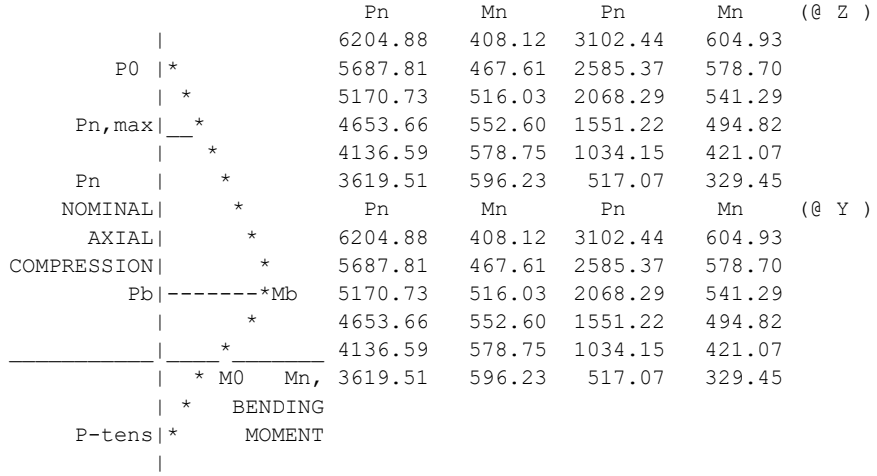
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	569.29	6.95	0.00611

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	569.29	8.53	0.00749



ACI 318-11 COLUMN NO. 609 DESIGN RESULTS

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FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

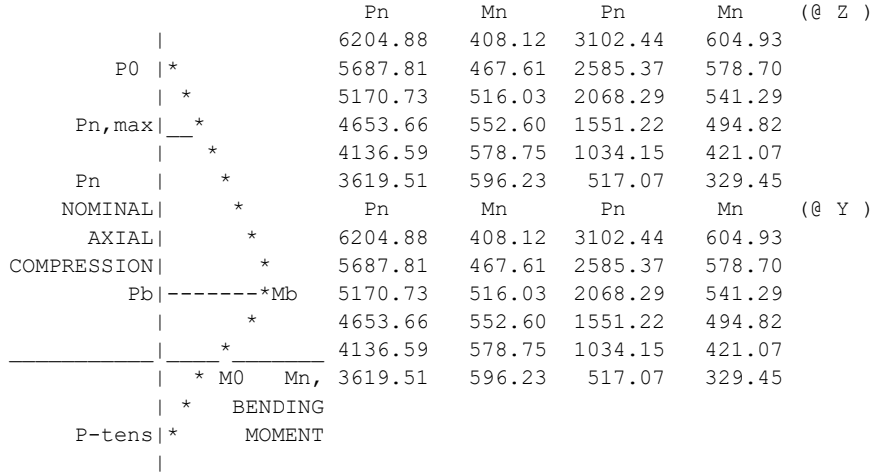
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	265.44	16.47	0.00926

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	265.44	10.53	0.00592



ACI 318-11 COLUMN NO. 610 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

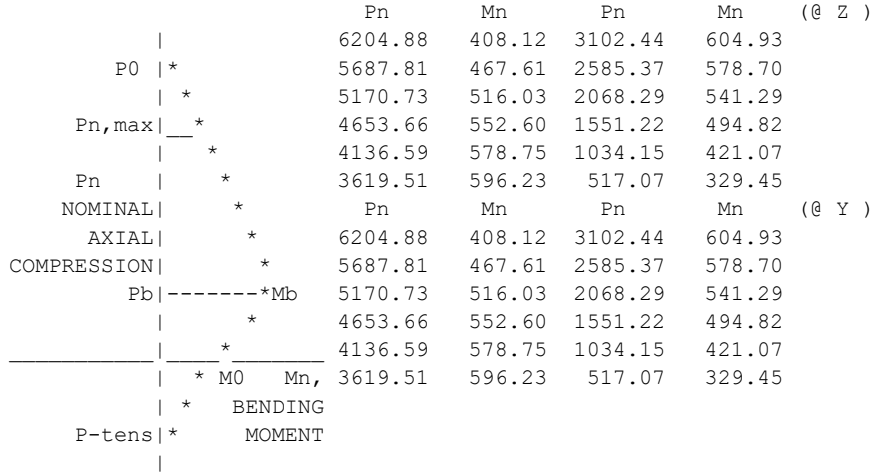
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	659.49	6.55	0.00496

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	659.49	2.92	0.00221



ACI 318-11 COLUMN NO. 611 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

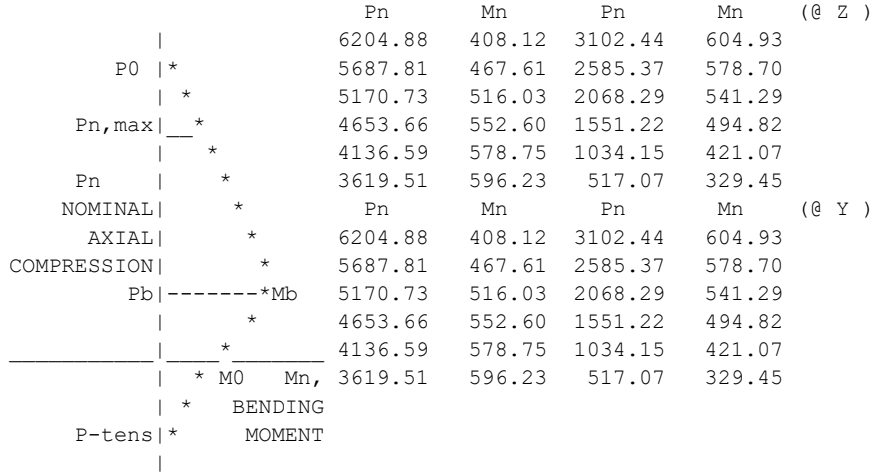
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	265.25	16.22	0.00913

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	265.25	10.52	0.00592



ACI 318-11 COLUMN NO. 612 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

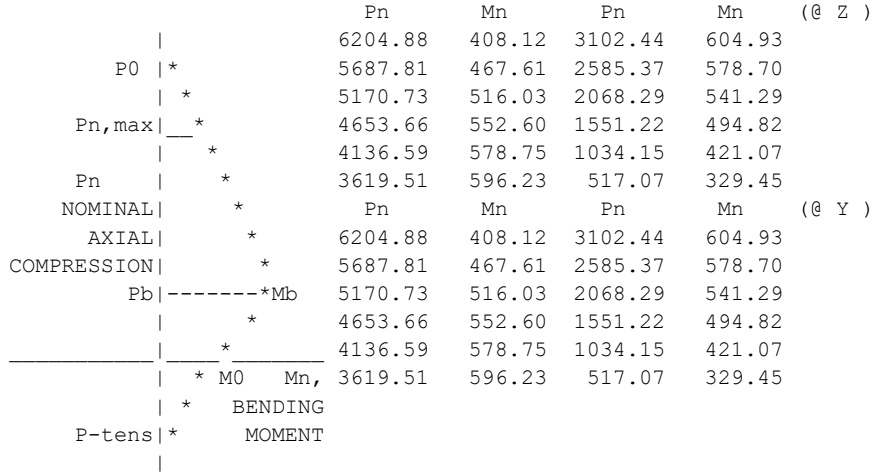
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	659.33	6.13	0.00465

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	659.33	2.89	0.00220



ACI 318-11 COLUMN NO. 615 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

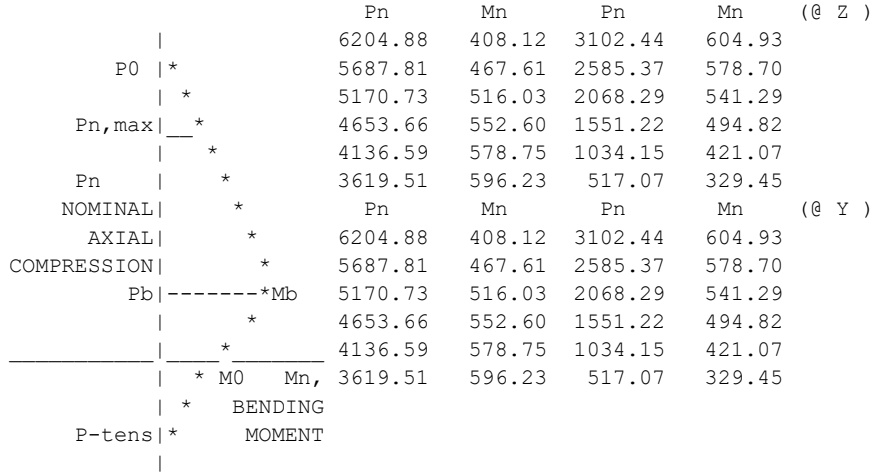
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	124.17	24.06	0.02893

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	124.17	56.15	0.06750



ACI 318-11 COLUMN NO. 616 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

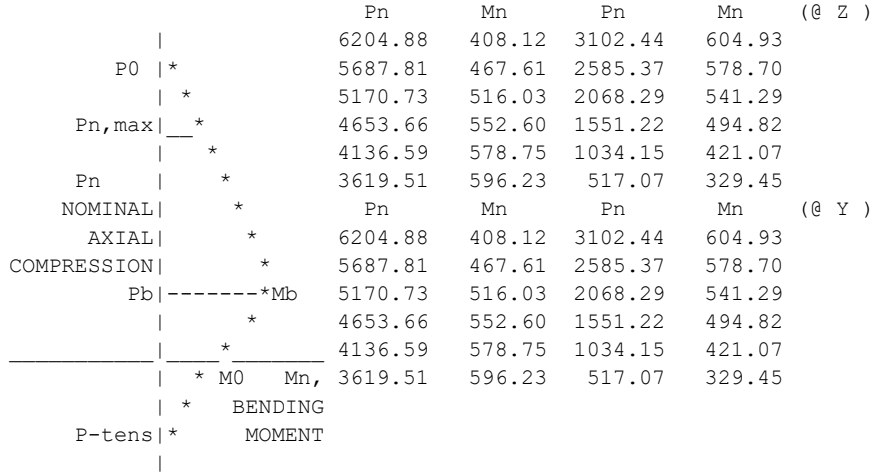
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	563.32	53.82	0.04777

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	563.32	15.39	0.01366



ACI 318-11 COLUMN NO. 617 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

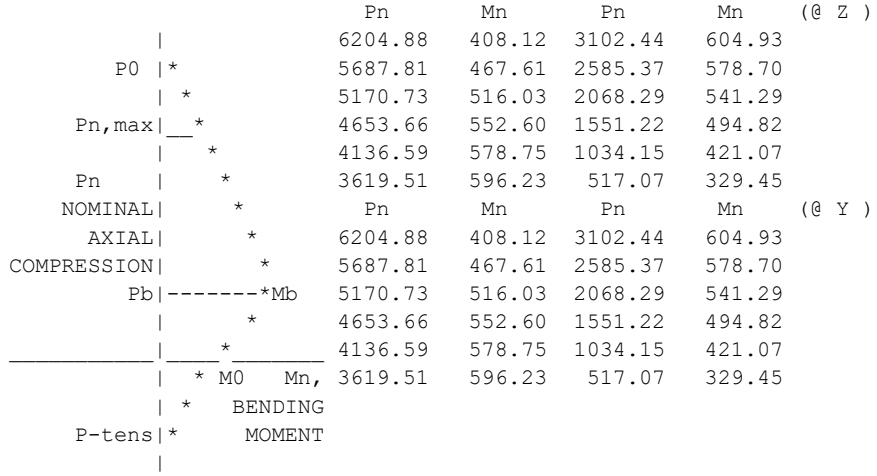
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	124.14	23.97	0.02882

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	124.14	56.16	0.06753



ACI 318-11 COLUMN NO. 618 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

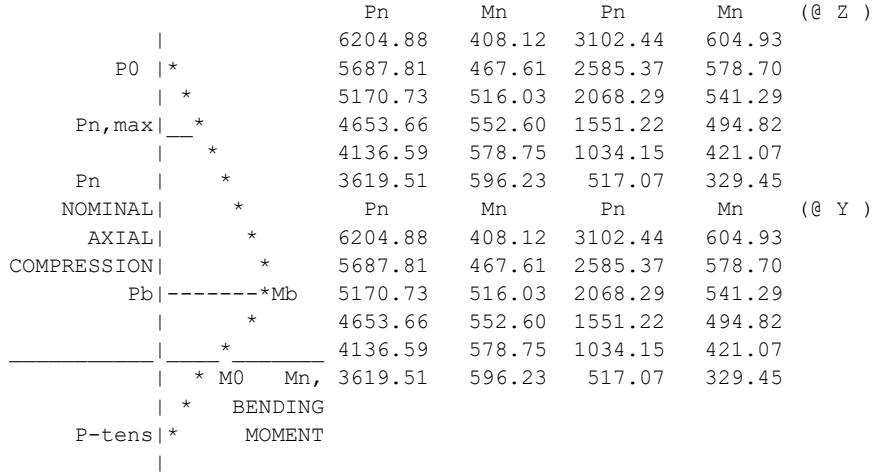
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	563.26	53.71	0.04768

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
226.86	-1039.71	563.26	15.40	0.01367



ACI 318-11 COLUMN NO. 623 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

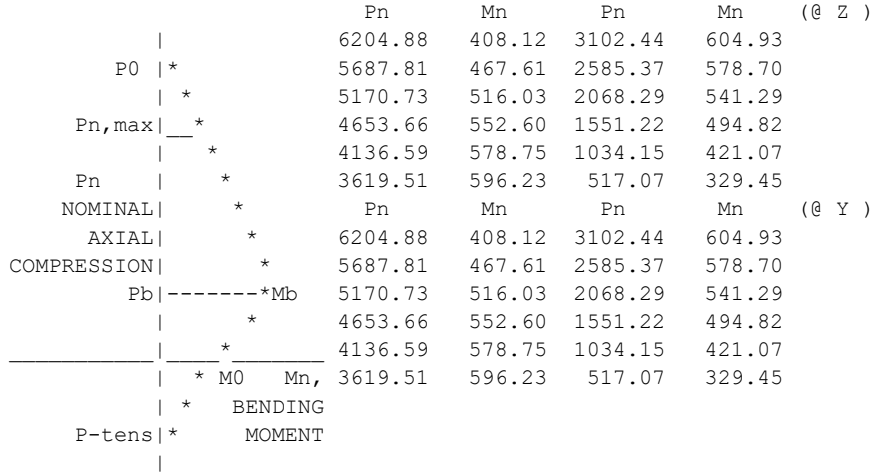
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	91.09	24.57	0.10787

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	91.09	28.60	0.12556



ACI 318-11 COLUMN NO. 625 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

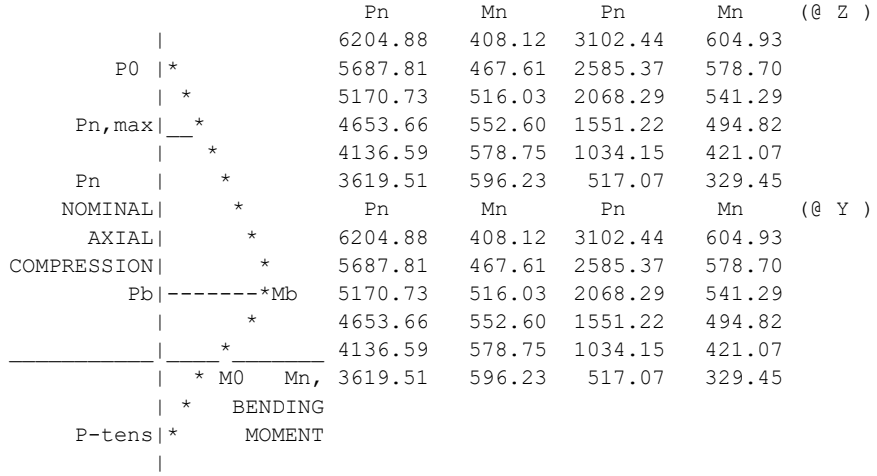
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	95.35	26.13	0.10963

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3145.35	606.41	192.8
M0	P-tens.	Des.Pn	Des.Mn	e/h
226.86	-1039.71	95.35	29.08	0.12198



**********END OF COLUMN DESIGN RESULTS**********

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237. CONCRETE TAKE
238. END CONCRETE DESIGN
```

********** CONCRETE TAKE OFF **********
(FOR BEAMS, COLUMNS AND PLATES DESIGNED ABOVE)

NOTE: CONCRETE QUANTITY REPRESENTS VOLUME OF CONCRETE IN BEAMS, COLUMNS, AND PLATES DESIGNED ABOVE.
REINFORCING STEEL QUANTITY REPRESENTS REINFORCING STEEL IN BEAMS AND COLUMNS DESIGNED ABOVE.
REINFORCING STEEL IN PLATES IS NOT INCLUDED IN THE REPORTED QUANTITY.

TOTAL VOLUME OF CONCRETE = 50.2 CU.METER

BAR SIZE NUMBER	WEIGHT (in lbs)
10	725
12	656
16	1100
20	4799
*** TOTAL=	7280

239. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 15,2020 TIME= 16: 1:15 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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